Lecture 1 Three Classes of PDEs.

1.i) Introduction + Syllabus

1.ii) Definitions

{Single Variable Calc. $f(x)$, $\frac{d}{dx}$} ⇝ Ordinary Differential Equations (ODEs)

{Multivariable Calc $f(x,y,z)$, $\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z}$} ⇝ Partial Differential Equations (PDEs)

Def 1.1: An ODE is an equation relating an unknown function $f(x)$ of a single variable to its own derivatives.

Ex 1.2: Some example ODEs

$$\frac{dP}{dt} = kP \qquad \frac{dT}{dt} = -k(T - T_c) \qquad \frac{d^2\theta}{dt^2} + \frac{g}{\ell}\theta = 0$$

(P = population, t = time) (T = temp, t = time; Newton's law of cooling) (θ = angle, t = time; Pendulum motion)

Motivation 1.3: Many (in fact most) quantities in nature depend on multiple variables. Eg heat obviously depends on both time and space

⇝ $u(t,x,y,z)$.

Def 1.4: A PDE is an equation relating an unknown function $u(t,x,y,z)$ of many variables to its own partial derivatives.

The order of the equation is the highest derivative that appears.

More generally $u(x_1, x_2, \ldots x_n)$

Ex 1.5: Some sample PDEs

i) $\frac{\partial u}{\partial t} + \frac{\partial u}{\partial x} = 0$ (first order)

ii) $3\frac{\partial^2 u}{\partial x^2} + 4\frac{\partial^2 u}{\partial x \partial y} - \frac{\partial^2 u}{\partial y \partial z} + 6\frac{\partial^2 u}{\partial z^2} - 14\frac{\partial u}{\partial y} + 7u = 0$ (second order const. coeff)

iii) $a(t,x)\frac{\partial^2 u}{\partial x^2} + b(t,x)\frac{\partial^2 u}{\partial x \partial y} + \cdots f(t,x)\cdot u = 0.$ (second order non-const coeff)

1.iii) Three classes of Linear PDEs of Second Order

Def 1.6 : A PDE is linear if each term is a (fixed, known) function times a partial derivative i.e. it depends linearly on the derivative.

linear ✓
$\cos(t+x)\frac{\partial^2 u}{\partial y^2}$ ✓
$3x\frac{\partial u}{\partial z}$ ✓
$e^{y+z}\frac{\partial^{(4)} u}{\partial x^3 \partial y^2 \partial z}$

nonlinear ✗
$\left(\frac{\partial u}{\partial x}\right)^2$ ✗
$\frac{\partial u}{\partial x}\cdot\frac{\partial u}{\partial y}$ ✗
$e^u \frac{\partial u}{\partial z}$

Bizarre Fact of Nature 1.7 : Effectively every equation coming from physics, chemistry, engineering, economics, ... is a first or second order equation. ← third derivatives just do not appear!

Notation 1.8 : $u(t,x,y,z)$ we denote $\frac{\partial}{\partial t} = \partial_0$

$\frac{\partial}{\partial x} = \partial_1$ $\frac{\partial}{\partial y} = \partial_2$ $\frac{\partial}{\partial z} = \partial_3$

then a second-order linear PDE can be written

(I) $$\sum_{i,j=0}^{3} A_{ij}(t,x,y,z)\partial_i\partial_j u + \sum_{k=0}^{3} B_k(t,x,y,z)\partial_k u + C(t,x,y,z)u = 0.$$

1.iii a : Elliptic Equations

Prototypical Example 1.9 : The Laplace equation

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} = 0$$

Shorthand notation → $$\Delta u = \nabla\cdot\nabla u = \nabla^2 u$$

← $\left(\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z}\right)\cdot\left(\frac{\partial u}{\partial x}, \dots \frac{\partial u}{\partial z}\right)$ divergence of gradient.

Applications 1.10
- The electromagnetic potential $\varphi(x,y,z)$ at equilibrium $\Delta\varphi(x,y,z) = 0$.
- The (Newtonian) gravitational potential $U(x,y,z)$

Def 1.11 : An elliptic 2nd order linear PDE is an equation (independent of t) $u(x,y,z)$

no t! → $$\sum_{i,j=1}^{3} a_{ij}(x,y,z)\partial_i\partial_j u + \sum_{k=1}^{3} b_k(x,y,z)\partial_k u + c(x,y,z)u = 0$$

such that $\begin{pmatrix} a_{11} & a_{12} & \\ & \ddots & a_{33} \end{pmatrix}$ is invertible.

Ex 1.12 (other elliptic equations)

- $\Delta u + \lambda u = 0 \qquad -\infty < \lambda < \infty$ (Helmholtz Equation)
- $-\Delta u + V(x,y,z)u = 0$ (time-independent Schrödinger Eq.)
  - for $V = \frac{1}{4\pi\varepsilon_0} \cdot \frac{e}{\sqrt{x^2+y^2+z^2}}$ solutions are electron orbitals from chemistry

1.iii b : Parabolic Equations

Prototypical Example 1.13 : the Heat equation $u(t,x,y,z)$

$$\frac{\partial u}{\partial t} - \Delta u = 0$$

Applications / Poincaré Principle 1.14 : u = concentration of solute in e.g. water

$\frac{\partial u}{\partial t} - \Delta u = 0$ is also the diffusion equation

➡ Intuition from one setting can be applied in the other!

Def 1.15 : A **parabolic** 2nd order linear PDE is a time-dependent equation

$$\frac{\partial u}{\partial t} - \left\{ \begin{array}{l} \text{any time-independent} \\ 2^{\text{nd}} \text{ order linear elliptic PDE} \end{array} \right\} = 0.$$

1.iii c : Hyperbolic Equations

Prototypical Example 1.16 : the Wave equation

$$\frac{\partial^2 u}{\partial t^2} - c^2 \Delta u = 0$$

(c ← propagation speed.)

Applications 1.17 : E&M, acoustics, seismic, fluid dynamics, gravitational, ...

Def 1.18 : A **hyperbolic** 2nd order linear PDE is a time-dependent equation

$$\frac{\partial^2 u}{\partial t^2} - \left\{ \begin{array}{l} \text{any time-independent} \\ 2^{\text{nd}} \text{ order linear elliptic PDE} \end{array} \right\} = 0.$$

Etymology 1.19 : In 2d replace $\frac{\partial}{\partial x} \rightsquigarrow X$ $\frac{\partial}{\partial t} \rightsquigarrow T$

- $X^2 + Y^2 = 0$ (ellipse)
- $T - X^2 = 0$ (parabola)
- $T^2 - X^2 = 0$ (hyperbola)

Lecture 2: ODEs I: solving techniques

Recall a first-order ODE is an equation

$$\frac{du}{dt} = F(u,t)$$

for an unknown function $u:(a,b) \longrightarrow \mathbb{R}$ (we allow $a=-\infty, b=\infty$).

2.i) Solving first-order ODEs

Method 2.1 (Direct integration)

If $F(u,t) = f(t)$ is independent of u,

$$u'(t) = \frac{du}{dt} = f(t)$$

$$\int_0^t u'(t_1)\,dt_1 = \int_0^t f(t_1)\,dt_1$$

FToC

$$u(t) = C + \int_0^t f(t_1)\,dt_1.$$

Ex 2.2: $F(u,t) = \lambda u$. $\quad u'(t) = \lambda u$.

Method 2.3 You know this one, the solution is just

$$u(t) = Ce^{\lambda t}$$

Verify by plugging in: if u is as above,

$$u'(t) = \frac{d}{dt}\left(Ce^{\lambda t}\right) = C\lambda e^{\lambda t} = \lambda\left(Ce^{\lambda t}\right) = \lambda u \quad \checkmark.$$

This is the "guess and check" method.

Def 2.4: An ODE $u'(t) = F(u,t)$ is said to be separable if

$$F(u,t) = g(u)f(t).$$

Method 2.5 (Separation of variables for ODEs)

$$\frac{du}{dt} = g(u)f(t).$$

↘ pretend it's a fraction. Formally $\int \frac{1}{g(u)}\frac{du}{dt}\,dt = \int f(t)\,dt$

$$\frac{du}{g(u)} = f(t)\,dt$$

$\int \frac{1}{g(u)}\,du$

$$\int \frac{du}{g(u)} = \int f(t)\,dt$$

by u-substitution.

Ex 2.6: Solve $\frac{dy}{dt} = 3t^2(y^2-1)$

$$\frac{dy}{y^2-1} = 3t^2\,dt$$

$$\frac{1}{2}\ln\left|\frac{y-1}{y+1}\right| = t^3 + C$$

$$\frac{y-1}{y+1} = De^{2t^3} \quad (D = e^C)$$

$$y(t) = \frac{1+De^{2t^3}}{1-De^{2t^3}}$$

→ Recall: partial fraction decomp.

$$\frac{1}{y^2-1} = \frac{1}{2}\left[\frac{1}{y-1} - \frac{1}{y+1}\right]$$

$$\int \frac{1}{y^2-1}\,dy = \frac{1}{2}\left[\ln|y-1| - \ln|y+1|\right] = \frac{1}{2}\ln\left|\frac{y-1}{y+1}\right|$$

Remark 2.7: It is very easy to write down an ODE resulting in an integral w/ no anti-derivative in terms of standard functions:

$$\frac{dy}{dt} = e^{-t^2}\cdot y$$

$$\ln|y| = \int e^{-t^2}\,dt$$



→ No anti-derivative in terms of $x, x^2, \sin x, \cos x$ $e^x, \ln x, \ldots$

Upsetting Fact of life 2.8: Not all ODEs have closed form solutions in terms of functions we have names for. Solutions still exist, but must be understood qualitatively or numerically.

2.ii) Second-order (or higher) linear ODEs + Characteristic polynomials

Consider

$$u''(t) + au'(t) + bu(t) = 0.$$

for $a, b \in \mathbb{R}$.

Method 2.9 (Characteristic polynomials). Take the guess or ansatz

$$u(t) = Ce^{\lambda t}.$$

Plug in: $u''(t) = C\lambda^2 e^{\lambda t}$ $\quad u'(t) = C\lambda e^{\lambda t}$

$$0 = u'' + au' + bu = \lambda^2 \cdot Ce^{\lambda t} + a\lambda Ce^{\lambda t} + bCe^{\lambda t}$$

$$= (\lambda^2 + a\lambda + b)\,Ce^{\lambda t}$$

$\Rightarrow$ (true if and only if) $0 = (\lambda^2 + a\lambda + b)$

$\lambda =$ use quadratic formula. λ_1, λ_2.

$$u = C_1 e^{\lambda_1 t} + C_2 e^{\lambda_2 t}.$$

Def 2.10: An ODE is said to be

$$a u^{(k)} + b u^{(k-1)} + \dots + \omega u = 0 \quad \text{(homogeneous)}$$

$$a u^{(k)} + \qquad\qquad + \omega u = f(t) \quad \text{(inhomogeneous)}$$

depending on whether there are terms that don't involve u. The same holds for non-const coeff or nonlinear terms.

Interpretation 2.11: Inhomogeneous terms are interpretted as external sources or driving forces.

Ex 2.12:

$$\frac{dP}{dt} = \underbrace{\alpha P}_{\text{birth rate}} + \underbrace{I(t)}_{\text{immigration/ emmigration rate}}$$

P = population

$$\frac{dT}{dt} = \underbrace{k(T - T_0)}_{\text{Newton's law}} + \underbrace{s(t)}_{\text{external heat source}}$$

T = temp

2.iii) Initial Value Problems

Recall

Thm 2.13: An ODE of order n has a solution with precisely n arbitrary constants.

Def 2.14: An **initial value** or **boundary value** problem imposes n ($\mathbb{R}$-valued) contraints at $t = a$ or $t = a$ and b respectively.

Ex 2.15

$$\begin{cases} u''(t) + b u'(t) + c u(t) = f(t) \\ u(0) = 3 \\ u'(0) = 4 \end{cases}$$

or on t in $[0,1]$ for boundary value.

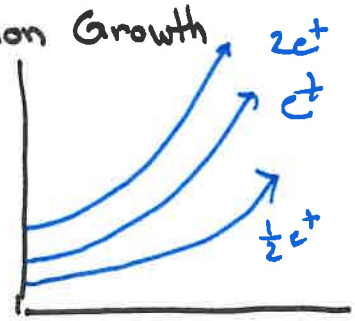
$u(0) = 3$

$u(1) = 4$

More generally we can impose k constraints at $t_1, t_2 \dots, t_k$ and have $n-k$ arbitrary constants in the solution.

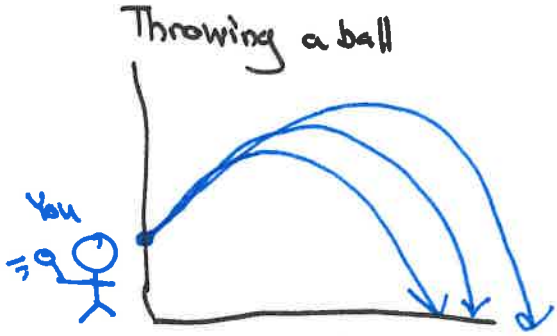
Interpretation 2.16 :

Population Growth



Initial value $P(0) = __$ gives state of system at time 0. This information is required for the solution to be determined completely

Throwing a ball



Both initial position $h(0)$ and speed $h'(0)$ are required to know full trajectory.

2.iv) Systems of ODEs

Def 2.17 A **system of ODEs** or **vector-valued ODE** is an equation

$$\frac{d}{dt}\begin{bmatrix} u_1 \\ \vdots \\ u_n \end{bmatrix} = \vec{F}\left(\begin{bmatrix} u_1 \\ \vdots \\ u_n \end{bmatrix}, t\right) = \begin{pmatrix} F_1(u_1 \dots u_n, t) \\ \vdots \\ F_n(u_1 \dots u_n, t) \end{pmatrix}$$

for an unknown function $u : \mathbb{R}(a,b) \longrightarrow \mathbb{R}^n$.

Ex 2.18 : If $F_i = \lambda_i u_i$ then

$$\frac{d}{dt}\begin{bmatrix} u_1 \\ \vdots \\ u_n \end{bmatrix} = \begin{pmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{pmatrix}\begin{bmatrix} u_1 \\ \vdots \\ u_n \end{bmatrix}$$

↳ matrix form.

$$\begin{bmatrix} u_1 \\ \vdots \\ u_n \end{bmatrix} = \begin{bmatrix} C_1 e^{\lambda_1 t} \\ \vdots \\ C_n e^{\lambda_n t} \end{bmatrix}$$

For linear systems $\frac{d}{dt}\begin{bmatrix} u_1 \\ \vdots \\ u_n \end{bmatrix} = \begin{pmatrix} & A & \end{pmatrix}\begin{pmatrix} u_1 \\ \vdots \\ u_n \end{pmatrix}$

Solve via eigenvectors + eigenvalues ← reviewed in a few lectures.

3.i) Motivation

Motivating Example 3.1 (Bessel)

The equation $u(r)$

$$r^2 \frac{d^2u}{dr^2} + r\frac{du}{dr} + (r^2 \pm \alpha^2)u = 0 \qquad \alpha \in \mathbb{R}.$$

→ The two solutions are, by definition, the Bessel functions $J_\alpha(r)$, $K_\alpha(r)$

will appear later when studying vibrations of circular drums + other areas. It is the Bessel ODE. It is not solvable by any standard methods, and (the solution has no closed-form expression in terms of other functions.)

Question 3.2 : In what sense can we "know" the solution? Calculate values? qualitative behavior? Good discrete approximation?

Motivating Example 3.3 Consider now

$$r^2 \frac{d^2u}{dr^2} + r\frac{du}{dr} + (r^2 \pm \alpha^2)u + r^3\cos(r)u = 0$$
$$r \in [0, \tfrac{\pi}{2}].$$

Question 3.4 : Not solvable by standard methods, or even in terms of "special functions" Does it even have a solution? How much should the solution "look like" the Bessel function.

Question 3.5 : If we use numerical methods (Lec 20+21) to approximate a solution on a computer, can we guarantee correctness + accuracy?

Motivating Example 3.6 : In some applications, (eg heat, stresses, etc) one is interested in the maximum heat/stress part of a device should withstand.

⇒ We want to know the ratio

$$\frac{\max_{(x,y,z,t)} |u(x,y,z,t)|}{\int |u(x,y,z,t)|\,dV} \leq ?$$

of max to total heat, we don't need full function u in every possible situation.

⇒ Maximum principles, Harnack inequalities, regularity theory, + qualitative study.

3.ii) Three levels of structure on $\mathbb{R}^n$

Recall $\mathbb{R}^n = \{(x_1, \dots, x_n) \mid x_i \in \mathbb{R}\}$

is a vector space of dimension n. This gives it algebraic operations

$v+w =$ (addition)

$cv =$ (scalar multiplication)

And we can talk about linear maps $f: V \to W$ ← vector spaces obeying

$$f(v+w) = f(v) + f(w)$$
$$f(cv) = cf(v).$$

Def 3.7: the Euclidean norm is the function $\|\cdot\| : \mathbb{R}^n \to \mathbb{R}^{\geq 0}$

$$\|x\| = \sqrt{x_1^2 + \dots + x_n^2}$$

so $\|x-y\| = \text{dist}(x,y)$.

⇒ gives $\mathbb{R}^n$ a notion of geometry (distance).
+ topology (open + closed sets)

Def 3.8: A sequence $x_k \in \mathbb{R}^n$ has limit x if

$$\|x_k - x\| \longrightarrow 0 \quad \text{(in } \mathbb{R}\text{)}$$

Def 3.9: A set $\Omega \subseteq \mathbb{R}^n$ is

- Open if for any $x \in \Omega$, there is $r > 0$ so the ball $\{\|x-y\| < r\} \subseteq \Omega$

⇒ contains none of its boundary ← fails at end



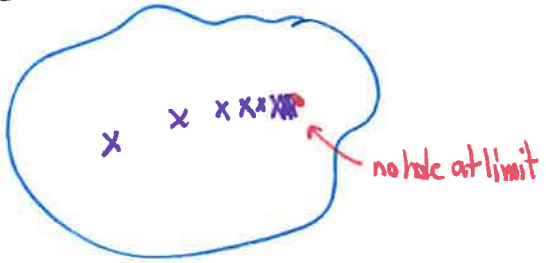


- Closed if it contains all its limit points (or if Ω^c is open).

Fact 3.10: $\mathbb{R}^n$ is complete ie contains all limit points, ie

if $x_k \longrightarrow x$ then $x \in \mathbb{R}$

(not true for $\mathbb{Q}^n$ rationals).



3.iii) Function Spaces

Define $C^0([a,b];\mathbb{R}^k) = \{f:[a,b]\to\mathbb{R}^k \mid f \text{ is continuous}\}$

with norm

$$\|f\|_{C^0} = \max_{x\in[a,b]} |f(x)|.$$

and

$$C^1([a,b];\mathbb{R}^k) = \left\{f:[a,b]\to\mathbb{R}^k \;\middle|\; \begin{array}{l} f \text{ is continuous} \\ \frac{df}{dx} \text{ is continuous} \end{array}\right\}.$$

$$\|f\|_{C^1} = \max_{[a,b]} |f(x)| + \max_{[a,b]} |f'(x)|.$$

Fact 3.11: If f is cont, g is cont then $f+g$ is cont.
" $c\in\mathbb{R}$ then cf is cont.
$\Rightarrow$ $C^0([a,b];\mathbb{R}^k)$, $C^1([a,b];\mathbb{R}^k)$ are vector spaces.

Thm 3.12: C^0, C^1, C^2, ... are complete ie. if
$f_k(x) \longrightarrow f(x)$ i.e. $\|f_k(x)-f(x)\|_{C^0} = \max_x |f_k(x)-f(x)| \to 0$.
then $f(x)$ in C^0, C^1, .. also.

3.iv) The existence and Uniqueness Theorem

Let $u'(t) = F(u,t)$ be an ODE.

Thm 3.13: Suppose that F is C^1, so there is an $M = \max F'$
so $|F(x,t)-F(y,s)| \le M(|x-y|+|t-s|)$.

then $\forall t_0\in\mathbb{R}$, $\exists T>0$ so that there is a solution u
on $u:[t_0,t_0+T]\longrightarrow\mathbb{R}^k$.
and there is a unique solution w/ $u(t_0) = w_0$ (initial value problem).

Rem 3.14
Same is true for higher-order

Abstract Lemma 3.15: Suppose $T: V \to V$ is a map of complete, vector spaces w/ norms and there's an $\varepsilon > 0$ so that

$$\|Tx - Ty\| \leq (1-\varepsilon)\|x-y\|.$$

Then, there is a unique $x \in V$ so that $x = Tx$ ← called a fixed point.

Proof: 1) Existence: take x_0 and point. and consider

$$x_0 = x_0$$
$$x_1 = Tx_0$$
$$x_2 = T(T(x_0))$$
$$x_3 = T(T\ldots(x_0)).$$

$\alpha^k \to 0$ if $\alpha < 1$.

then $\|x_{k+1} - x_k\| = \|T(x_k) - T(x_{k-1})\| \leq (1-\varepsilon)\|x_k - x_{k-1}\|$

$$\leq (1-\varepsilon)^k \|x_1 - x_0\|.$$



so $x_k \to x$ by completeness. (exists by completeness)

2) If $x = Tx$ and $y = Ty$ then $x = y$ (uniqueness).

$\|Tx - Ty\| = \|x - y\|$ but $\|Tx - Ty\| \leq (1-\varepsilon)\|x-y\|$.

$\|x-y\| \leq (1-\varepsilon)\|x-y\|$ so $\|x-y\| = 0$ and $x = y$. □

Proof of Theorem: We define a map $T: e^1 \to e$ whose fixed point is the solution.

Define $T(u) = w_0 + \int_{t_0}^{t} F(u(s), s)\, ds$.

1) If $u = T(u) = w_0 + \int_{t_0}^{t} \ldots ds$ then $u(t_0) = w_0$ and $u' = F(u(t), t)$ by FToC.

2) $\|Tu - Tv\|_e = \max_{[t_0, t_0+\tau]} \left| \left(w_0 + \int_{t_0}^{t} F(u(s), s)\, ds - w_0 - \int_{t_0}^{t} F(v(s), s)\, ds \right) \right|$

$$= \max \left| \left(\int_{t_0}^{t} F(u, s) - F(v, s)\, ds \right) \right|$$

$$\leq \max_t \int_{t_0}^{t} M \|u(s) - v(s)\| \, ds$$

$$\leq M \cdot \tau \cdot \|u - v\|_{C^0}. \quad \text{so if } \tau < \tfrac{1}{2M} \text{ then w/ } \varepsilon = \tfrac{1}{2}.$$

$\square$

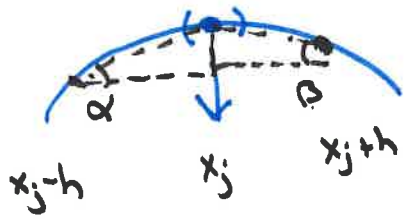
Lecture 4 | Model PDEs I: 1d heat and waves.

4.i) Derivation of the wave equation

Consider a rope with fixed ends, and call the height displacement from flat $u(t,x)$.



Assume that the rope has density ρ. An infinitessimal segment with length h has mass ρh, at $x = x_j$



restoring vertical force

$$F = T\sin\alpha_j + T\sin\beta_j$$

$$\approx T\tan\alpha_j + T\tan\beta_j$$

$$\approx T\left(\frac{u(x_{j-1},t) - u(x_j,t)}{h} + \frac{u(x_{j+1},t) - u(x_j,t)}{h}\right) \approx T\left[u'(x_j,t) - u'(x_{j-1},t)\right]$$

(T: tension)

Newton's law

$$ma = F(x,t)$$

$$\rho h \cdot \frac{d^2u}{dt^2} = \{\text{vertical restoring force}\}$$

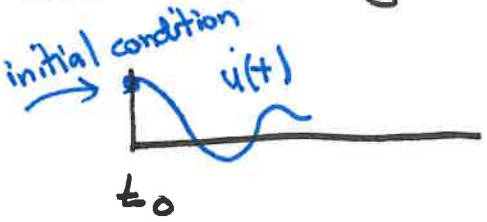
$$\frac{\partial^2 u}{\partial t^2} = \frac{T}{\rho}\left[\frac{u'(x_j,t) - u'(x_{j-1},t)}{h}\right] = \frac{T}{\rho}\left[u''(x_j,t)\right] \quad \checkmark \text{ w/ } c^2 = \frac{T}{\rho}$$

Thus the wave equation in 1d reads

$$\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}.$$

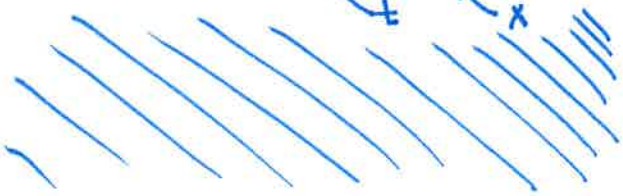
4.ii) Boundary + Initial Value Problems + [redacted]

Recall the paradigm for solving ODEs.



1) Apply a method of solving to obtain a general solution (w/ arbitrary constants)
2) Plug in knowledge of specific situation eg initial conditions to get specific solution.

$\Omega = [0,\infty) \times \mathbb{R}$ (t vertical, x horizontal)



$u(t,x)$ has value at each point (t,x) in Ω.

$(-\infty, \infty) = \mathbb{R}$

$\Rightarrow$ We have to specify initial value + velocity (Second order) at <u>every</u> value of x.

Def 4.1: The <u>initial value problem</u> for the wave equation is to solve

$$\begin{cases} \frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2} \\ u(0,x) = g(x) \\ \partial_t u(0,x) = h(x). \end{cases}$$

for given functions $g(x), h(x)$.

Theme 4.2: Often to adapt concepts from ODEs to PDEs we "promote" values to function.

$u(t)$ is a number (ODE)
$u(t,-)$ is a function (PDE)

The initial condition similarly promotes $u(0) \rightsquigarrow u(0,-)$.

Ex 4.3 (traveling waves) The solution of the IVP

$$\begin{cases} \frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2} \\ u(0,x) = g(x) \\ \partial_t u(0,x) = -c g'(x) \end{cases}$$



is $u(t,x) = g(x-ct)$.

Proof:

$$\frac{\partial^2 u}{\partial t^2} - c^2 \frac{\partial^2 u}{\partial x^2} = \left(\frac{\partial}{\partial t}\right)^2 g(x-ct) - c^2 \left(\frac{\partial}{\partial x}\right)^2 g(x-ct)$$

$$= \frac{\partial}{\partial t}\left(-c\, g'(x-ct)\right) - c^2 \frac{\partial}{\partial x}\left(g'(x-ct)\right)$$

$= (-c)^2 g''(x-ct) - c^2 g''(x-ct) = 0$ ✓

and $g(x-ct)\big|_{t=0} = g(x)$ ✓

$\frac{\partial}{\partial t} g(x-ct) = g'(x-ct)\cdot -c\big|_{t=0}$ ✓

$= -cg'(x)$

Thm 4.4 (general traveling waves)

The solution of the IVP

$$\begin{cases} \frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2} \\ u(0,x) = g(x) \\ \partial_t u(0,x) = h(x) \end{cases}$$

is

$$u(t,x) = \frac{1}{2}\left[g(x-ct) + g(x+ct)\right] + \frac{1}{2c}\int_{x-ct}^{x+ct} h(s)\,ds.$$

Proof: Use linearity to apply to each term. For remember FToC $\frac{d}{dz}\int_a^z f(s)\,ds = f(z)$ → $\frac{d}{dz}\int_z^a f(s)\,ds = -f(z)$.

Finish on HW.

Def 4.5: The initial value problem for the heat equation is

$$\begin{cases} \frac{\partial u}{\partial t} = \alpha \frac{\partial^2 u}{\partial x^2} \\ u(0,x) = g(x) \end{cases}$$

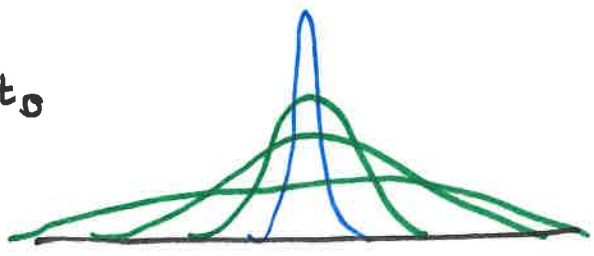
← 1st order in t, so one initial condition just like ODEs.

Ex 4.6 (heat kernel) The solution of the IVP

$$\begin{cases} \frac{\partial u}{\partial t} = \alpha \frac{\partial^2 u}{\partial x^2} \\ u(t_0,x) = \frac{1}{\sqrt{4\pi t_0}} e^{-x^2/4t_0} \end{cases}$$

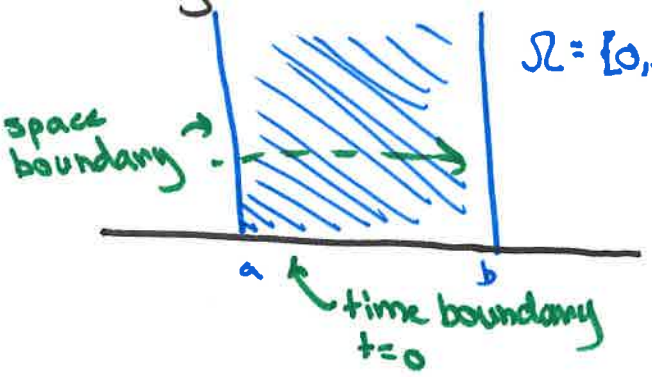
is

$$u(t,x) = \frac{1}{\sqrt{4\pi t}} e^{-x^2/4t}.$$



4.iii) Boundary Value Problems

Strings / rooms are not infinite in x-direction. In this case we need to specify function at both time + spatial boundaries.



Def 4.7: The IVP for the wave (resp. heat) eq w/ Dirichlet Boundary Conditions at $x=a, x=b$ is

$$\begin{cases} \frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2} \\ u(0,x) = g(x) \\ \partial_t u(0,x) = h(x) \end{cases} \quad \text{And} \quad u(t,a) = u(t,b) = 0$$

w/ Neumann Boundary Conditions at $x = a, b$

$$\begin{cases} \text{same} \end{cases} \quad \text{And} \quad \partial_x u(t,a) = \partial_x u(t,b) = 0.$$

Ex 4.8: $[a,b] = [0,\pi]$ then sol of $\begin{cases} \frac{\partial^2 u}{\partial t^2} = c^2 \Delta u \\ u(0,x) = \sin(x) \\ \partial_t u(0,x) = 0 \end{cases}$ w/ Dirichlet is $u(t,x) = \sin(x)\cos(ct)$

or more generally $= \sin(nx)\cos(cnt)$ $n = 1, 2, 3, \ldots$

Demonstration 4.9: Standing + traveling waves.

5.i) Separation of variables

Consider a) heat eq. $\partial_t u - \alpha \Delta u = 0$
b) wave eq. $\partial_t^2 u - c^2 \Delta u = 0$
on $\Omega = [0,\infty) \times [0, L]$

Today we will see a general solving method.

Ansatz 5.1 : Assume that the solution has the form

$$u(t,x) = V(t)\varphi(x)$$

i.e. it is a product of functions of a single variable.

Prop 5.2 : If $u(t,x)$ satisfies the heat equation, then $V(t), \varphi(x)$ individually solve

$$V'(t) = -\lambda V(t)$$
$$\varphi''(x) = -\lambda \varphi(x)$$

for some $\lambda \in \mathbb{R}$.

Proof/derivation : Plug in

$$\partial_t u = \partial_t (V(t)\varphi(x)) = V'(t)\varphi(x)$$
$$\Delta u = \partial_x^2 (V(t)\varphi(x)) = V(t)\varphi''(x)$$

so u solves ➡ $V'(t)\varphi(x) = V(t)\varphi''(x)$ or $\frac{V'(t)}{V(t)} = \frac{\varphi''(x)}{\varphi(x)}$

← ignore places where $V(t)$ or $\varphi(x) = 0$.

Fact about functions 5.3 : If two functions $f(x) = g(y)$ are equal for all values of x, y, then both are constant.

$$g(y) = f(0) \text{ for all } y$$
$$f(x) = g(0) = f(0) \text{ for all } x.$$

$$\Rightarrow \frac{V'(t)}{V(t)} = -\lambda \quad \text{and} \quad \frac{\varphi''(x)}{\varphi(x)} = -\lambda.$$

□

Prop 5.3 : For the wave equation,

$$V''(t) = -\lambda V(t)$$
$$\varphi''(x) = -\lambda \varphi(x)$$

for $\lambda \in \mathbb{R}$.

Ex 5.4 (Solving the t equation)

$$\dot{V}(t) = -\lambda V(t)$$

then $V(t) = V(0)e^{-\lambda t}$ since $V(0) = Ce^{\lambda \cdot 0} = C$ in general sol.

For the wave equation

$$V''(t) = -\lambda V(t)$$

$V(t) = C_1 \sin(\sqrt{\lambda}t) + C_2 \cos(\sqrt{\lambda}t)$ provided $\lambda \geq 0$.

5.ii) The Helmholtz Equation

Def 5.5 the Helmholtz equation is the (elliptic) PDE

$$\Delta \varphi = -\lambda \varphi$$

In 1-d this is an ODE $\left(\frac{\partial}{\partial x}\right)^2 \varphi = -\lambda \varphi$.

Ex 5.6 (Solving the φ equation)
With Dirichlet B.C.

$$\begin{cases} \varphi''(x) = -\lambda \varphi(x) \\ \varphi(0) = \varphi(L) = 0 \end{cases}$$

$\varphi(x) = A\sin(\sqrt{\lambda}x) + B\cos(\sqrt{\lambda}x)$

Plug in $0 = \varphi(0) = A \cdot 0 + B \cdot 1$
$\Rightarrow B = 0$
$0 = \varphi(L) = A\sin(\sqrt{\lambda}L)$
so $\sqrt{\lambda}L = n\pi$ $n = 0, 1, -1, 2, \ldots$
$\lambda = \frac{\pi^2 n^2}{L^2}$

Thm 5.7 A solution of the BVP (Dirichlet B.C.)

$$\begin{cases} \partial_t u = \Delta u \\ u(t,0) = u(t,L) = 0 \\ u(0,x) = \sin\left(\frac{n\pi}{L}x\right) \cdot A_n \end{cases}$$

is $u = V(t)\varphi(x) = A_n e^{-\frac{n^2\pi^2}{L^2}t} \sin\left(\frac{n\pi}{L}x\right)$. for $n = 1, 2, 3, \ldots$

Thm 5.8 (Heat Eq., Neumann Boundary Conditions)

A solution of the BVP

$$\begin{cases} \partial_t u = \Delta u \\ \partial_x u(t,0) = \partial_x u(t,L) = 0 \\ u(0,x) = \beta_n \cos\left(\frac{n\pi}{L}x\right) \end{cases} \qquad n = 1,2,3,\ldots$$

is

$$u(x,t) = \beta_n e^{-\frac{n^2\pi^2}{L^2}t} \cos\left(\frac{n\pi}{L}x\right)$$

Proof: Same but cosines.

Thm 5.9 (Positive Semi-definiteness)

Suppose that φ is a solution of the Helmholtz equation

$$\Delta\varphi = -\lambda\varphi$$

on $[0,L]$ with either Dirichlet or Neumann boundary conditions.

Then $\lambda \geq 0$.

Proof: Consider

$$\int_0^L \varphi \Delta\varphi \, dx = \int_0^L \varphi\left(\frac{\partial^2}{\partial x^2}\varphi\right) dx$$

↓ Integrate by parts

$$= -\int_0^L \left|\frac{\partial\varphi}{\partial x}\right|^2 dx + \underbrace{\varphi\frac{\partial\varphi}{\partial x}\Big|_{x=0}^{L}}_{=0 \text{ by B.C.}}$$

and on the other hand,

$$\int_0^L \varphi\Delta\varphi\, dx = -\lambda\int_0^L |\varphi|^2$$

so

$$\lambda = \frac{\int_0^L \left|\frac{\partial\varphi}{\partial x}\right|^2 dx}{\int_0^L |\varphi|^2 dx} \geq 0.$$

Ex/Notation 5.10: We've used the "adult" version of integration by parts

$$\int f\left(\frac{d}{dx}g\right) dx = -\int \left(\frac{d}{dx}f\right) g \, dx + fg\Big|_{x=0}^{L}$$

flip derivative, add negative sign

evaluate on the boundary.

More generally, the divergence theorem says

$$\int_\Omega \nabla \cdot (f \nabla g)\, dV = \int_{\partial\Omega} f \nabla g \cdot \vec{n}\, d\sigma$$

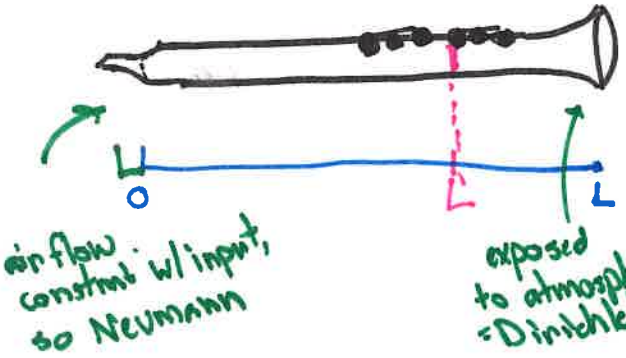


so and $\nabla \cdot (f \nabla g) = \nabla f \nabla g + f \underbrace{\nabla^2 g}_{\Delta g}$.

$$\int_\Omega \nabla f \nabla g\, dV = -\int_\Omega f \Delta g\, dV + \int_{\partial R} f \underbrace{\nabla g \cdot \vec{n}}_{= \partial_\nu g} d\sigma$$

← derivative in normal direction

Ex 5.11 : (The Clarinet + Overtones)



$u(t,x)$ pressure (ie sound) at (t,x).

mixed bd

$$\begin{cases} \frac{\partial^2 u}{\partial t^2} = \Delta u \\ \partial_x u(t,0) = 0, \quad u(t,L) = 0 \\ u(0,x) = ? \\ \partial_t u(0,x) = ? \end{cases}$$

$$u(t,x) = \left[\begin{matrix} C_1 \sin(\sqrt{\lambda} t) \\ + C_2 \cos(\sqrt{\lambda} t) \end{matrix}\right]\left[A\cos(\sqrt{\lambda} x) + B \sin(\sqrt{\lambda} x)\right]$$

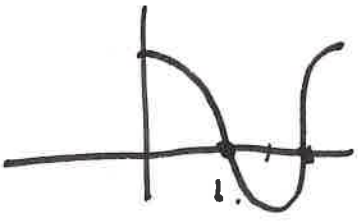
$$\partial_x u\Big|_{x=0} = A \cdot \sqrt{\lambda} \cdot -\sin(\sqrt{\lambda} x) + B\sqrt{\lambda}\cos(\sqrt{\lambda} x) = 0 \quad \Rightarrow \quad B = 0.$$

$$u(0,L) = A\cos(\sqrt{\lambda} x) = 0$$

$$\Rightarrow \sqrt{\lambda} x = (n \pm 1/2)\pi$$

$$\lambda = \frac{\pi^2}{L^2}(n - 1/2)^2$$

$n = 1, 2, 3, \ldots$



$\Rightarrow$ Playing the clarinet by changing the length L by opening/closing air columns produces different notes.

Model PDEs III : the principle of superposition + operator viewpoint.

6.1) Superposition

Recall a 2nd order linear equation has the form

$$\sum_{i=0}^{3}\sum_{j=0}^{3} A_{ij}(t,x,y,z)\partial_i\partial_j u + \sum_{k=0}^{3} B_k(t,x,y,z)\partial_k u + C(t,x,y,z)u = 0.$$

Theorem 6.1 (version 1) (superposition): If u, v are solutions to a linear equation as above, then

- $u+v$
- $cu \quad \forall c \in \mathbb{R}$

are also solutions. In particular, the solutions form a vector space.

Proof: u, v solutions

$$\sum_{i,j} A_{ij}(\dots)\partial_i\partial_j(u+v) + \sum_k B_k \partial_k(u+v) + C(u+v)$$

(✓ derivatives distribute)

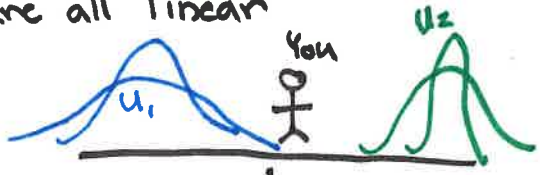
$$= \sum_{i,j} A_{ij}\partial_i\partial_j u + \sum_k B_k\partial_k u + Cu \quad = 0 + 0 = 0 \ \checkmark$$

$$+ \sum_{i,j} A_{ij}\partial_i\partial_j v + \sum_k B_k\partial_k v + Cv$$

and similarly for $c \cdot u$. □

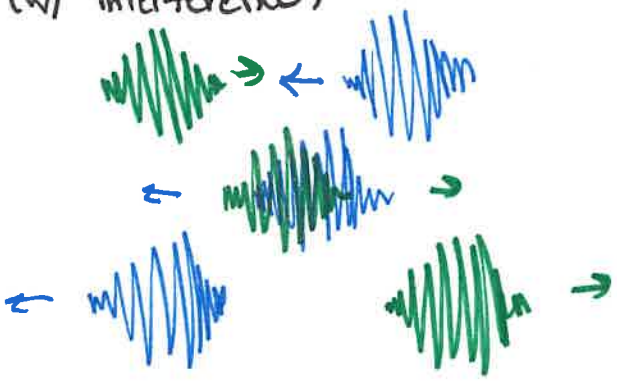
Interpretation 6.2: Laplace, heat, wave are all linear

- heat does not interact with itself!



heat felt is just sum of the two sources.

- Waves pass through each other (w/ interference)



- electric or gravitational force from two planets or charges is just the sum.

Sidenote: Einstein's equations for general relativity are highly nonlinear

Pop Sci 6.3 : (Schrödinger and the cat)

The full time-dependent Schrödinger equation is

$$i\frac{\partial \psi}{\partial t} = -\Delta \psi + V(t,x,y,z)\psi$$

$\psi : \Omega \to \mathbb{C}$ is complex valued.

↖ potential energy

This is linear! So if ψ_1, ψ_2 are solutions so is $\frac{1}{\sqrt{2}}(\psi_1 + \psi_2) = \Psi$

In particular, if ψ_1 = spin up, ψ_2 = spin down, an electron in Ψ is in a "superposition" of the two.

Similarly if ψ_1 = cat alive, ψ_2 = cat dead then the superposition is a solution!

(this thought experiment shows Quantum Mechanics breaks down at macroscopic scales)

6.ii) The operator viewpoint

Recall that a function

$$F : \mathbb{R}^n \to \mathbb{R}^m$$
$$x = (x_1, \dots x_n) \to (F_1(x_1, \dots x_n), \dots, F_m(x_1, \dots x_n)) = F(x)$$

is an operation that "eats" an $x \in \mathbb{R}^n$ and "spits out" a $y = F(x) \in \mathbb{R}^m$.

Recall 6.4 : Recall from linear algebra

$$A : \mathbb{R}^n \to \mathbb{R}^m$$

is linear if $A(\alpha x + \beta y) = \alpha A(x) + \beta A(y)$.

Similarly, we can think of a PDE as an operation that "eats" a function and "spits out" a function.

Def 6.5 : The differential operator associated to a PDE is the operator

$$L : C^2(\Omega; \mathbb{R}) \to C^0(\Omega; \mathbb{R})$$

given by

$$L = \left[\sum_{ij} A_{ij}(t,x,y,z)\partial_i\partial_j + \sum_k B_k(t,x,y,z)\partial_k + C(t,x,y,z)\right]$$

so that the PDE may be written

$$Lu = 0.$$

Terminology 6.6 : L is a "function from functions to functions" (L) ($u(t,x,y,z)$ $Lu(x,y,z)$) functions defined on spaces like C^2, C^0 are called operators.

Thm 6.7 (Superposition Version 2) : A PDE is linear $\Longleftrightarrow$ L is a linear operator

Proof : Each term $A_{ij}\partial_i\partial_j(u+v)$ distributes just as in the proof before. (Any nonlinear term, there will be c or u so that this doesn't work).

Corollary 6.8 : The solutions of a linear PDE are the vector (subspace)

$$\operatorname{Ker} L \subseteq C^2(\Omega : \mathbb{R}).$$

Rem 6.9 : The inhomogeneous equation $\sum A_{ij}\partial_i\partial_j u + \sum B_k\partial_k u + C_u = f$ becomes the linear system

$$Lu = f.$$

Later, we will see that thinking of this as infinite vectors/matrices makes sense and it's useful!

6.iii) Applications of Superposition

Ex 6.10 : (Nonzero Dirichlet Boundary Conditions)

Solve $\begin{cases} \partial_t u = \Delta u \\ u(0,t) = a \quad u(L,t) = b \\ u(0,x) = g(x) \end{cases}$ $(L = \partial_t - \Delta)$

$g(0) = a \quad g(L) = b.$

Consider 

u_0 is t-independent and linear so

$$(\partial_t - \Delta)u_0 = 0.$$

Take u_1 = solution of Dirichlet = 0 boundary value problem w/ $\tilde{g} = g - u_0$. [6.4]

Then $u = u_0 + u_1$ is the desired solution, by superposition.

Ex 6.11 : Solve the IVP for the heat equation

$$\begin{cases} (\partial_t - \Delta)u = 0 \\ u(t,0) = u(t,L) = 0 \\ u(0,x) = \underbrace{3\sin\left(\frac{\pi}{L}x\right) + 7\sin\left(\frac{5\pi}{L}x\right)}_{g(x)} \end{cases}$$

We know $u_n = A_n e^{-\left(\frac{n\pi}{L}\right)^2 t} \sin\left(\frac{n\pi}{L}x\right)$ is a solution w/

$u_n(0,x) = A_n \sin\left(\frac{n\pi}{L}x\right)$.

By superposition, $u_1(0,x) = 3\sin\left(\frac{\pi x}{L}\right)$ $A_1 = 3$

$u_5(0,x) = 7\sin\left(\frac{5\pi}{L}x\right)$ $A_5 = 7$.

then $u = u_1 + u_5$ solves eq. and $u(0,x) = g(x)$.

Ex 6.13 : Solve the IVP for the wave equation w/ Neumann B.C.

$$\begin{cases} (\partial_t^2 - \Delta)u = 0 \\ \partial_x u(t,0) = \partial_x u(t,L) = 0 \\ u(0,x) = -4\cos\left(\frac{3\pi}{L}x\right) + 2\cos\left(\frac{119\pi}{L}x\right) \\ \partial_t u(0,x) = 7\cos\left(\frac{3\pi}{L}x\right) - 3\cos\left(\frac{119\pi}{L}x\right). \end{cases}$$

Recall the general solution is

$$u(t,x) = \left[C_n \sin\left(\frac{n\pi}{L}t\right) + D_n \cos\left(\frac{n\pi}{L}t\right)\right]\cos\left(\frac{\pi n}{L}x\right)$$

$\Rightarrow$ For each n we have an ODE IVP. (solve each n independently)

For $n=3$ term,

$$-4\cos\left(\frac{3\pi}{L}x\right) = u(0,x) = \left[C_3\sin(0) + D_3\cos(0)\right]\cos\left(\frac{3\pi}{L}x\right)$$

$$\Rightarrow D_{n=3} = -4$$

$$7\cos\left(\frac{3\pi}{L}x\right) = \partial_t u(0,x) = \left[\frac{3\pi}{L}C_3\cos + \frac{3\pi}{L}\cdot -D_3\sin(\ )\right]\cos\left(\frac{3\pi}{L}x\right) \Rightarrow 7 = \frac{3\pi}{L}C_3$$

$$C_3 = \frac{7L}{3\pi}$$

Lecture 7 The heat equation + Separation of variables in 2d.

Reading 6.1, 6.2, 5.2.

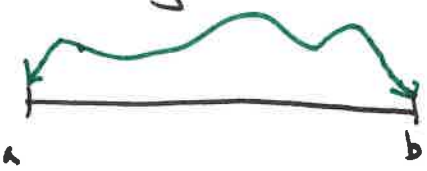
7.i) Derivation of the heat equation

Choose a region $D \subseteq \Omega \subseteq \mathbb{R}^n$. Let H be the amount of heat energy

$$H(t) = \int_D c\rho\, u(t,x)\, dV$$

(c: specific heat, ρ: density, u: temperature)

The amount of heat leaving D at any given time is proportional to the derivative at the boundary. In 1d



$$\frac{dH}{dt} = K \frac{du}{dx}\Big|_{x=b} - K \frac{du}{dx}\Big|_{x=a}$$

K = heat conductivity

More generally

$$\frac{dH}{dt} = \int_{\partial D} K \nabla u \cdot n \, d\omega$$

↓ divergence thm

$$= \int_D \nabla \cdot (K \nabla u)\, dV$$

but

$$\frac{dH}{dt} = \frac{d}{dt} \int_D c\rho\, u(t,x)\, dV = \int_D c\rho\, \partial_t u(t,x)\, dV$$

so

$$\int_D c\rho\, \partial_t u(t,x)\, dV = \int_D \nabla \cdot (K \nabla u)\, dV$$

for every region D. $D \to$ infinitessimal,

$$c\rho \frac{\partial u}{\partial t} = \nabla \cdot (K \nabla u)\, dV$$

$$\frac{\partial u}{\partial t} = \alpha \Delta u$$

↙ if K is constant. $\alpha = \frac{K}{c\rho}$

7.ii) The heat + wave equations in higher dimensions (IVP)

Let $\Omega \subseteq \mathbb{R}^n$ $n = 2, 3$ be a closed, connected region with boundary $\partial\Omega$.

Def 7.1: The IVP for the heat equation is

$$\begin{cases} \frac{\partial u}{\partial t} = \alpha \Delta u + f(t,x) \\ u(0,x) = g(x) \\ u|_{\partial\Omega} = 0 \end{cases}$$

← heat source/sink

(vanishing Dirichlet)

$(\partial_\nu u|_{\partial\Omega} = 0$ vanishing Neumann)

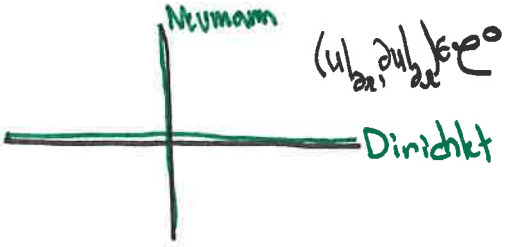
Interpretation 7.2: Dirichlet = perfect infinite heat sink touching boundary
Neumann = perfectly insulating boundary.
$f(t,x)$ = internal heat source or sink.

More generally non-vanishing Dirichlet/Neumann boundary conditions

$$u|_{\partial\Omega} = v(x)$$

or $$\partial_\nu u|_{\partial\Omega} = w(x)$$

Rem 7.3:



$(u|_{\partial\Omega}, \partial_\nu u|_{\partial\Omega}) \in C^0(\partial\Omega; \mathbb{R}) \times C^1(\partial\Omega; \mathbb{R})$

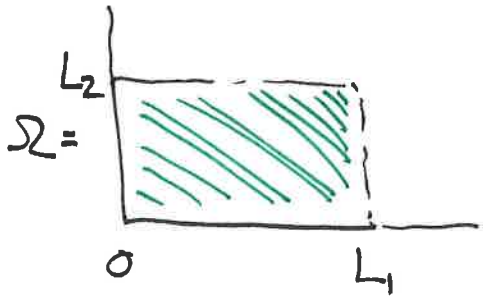
More generally any "Lagrangian" subspace gives a valid boundary condition, e.g.

$$a u|_{\partial\Omega} + b \partial_\nu u|_{\partial\Omega} = 0 \qquad a+b=1 \quad \text{(Robbin boundary condition)}$$

Def 7.4: Same for wave equation.

7.iii) Separation of variables, Again

Assume



Ansatz 7.5: Assume $u(t,x,y) = T(t)X(x)Y(y)$.

$$0 = \left(\frac{\partial}{\partial t} - \frac{\partial^2}{\partial x^2} - \frac{\partial^2}{\partial y^2}\right)u = T'(t)X(x)Y(y) - X''(x)T(t)Y(y) - Y''(y)X(x)T(t).$$

$$\frac{T'(t)}{T(t)} = \frac{X''(x)}{X(x)} + \frac{Y''(y)}{Y(y)} = -\Lambda \quad \text{must be const.}$$

Solving the T ODE: Same as before

$$T'(t) = -\Lambda T(t)$$

$$\Rightarrow T(t) = e^{-\Lambda t}$$

Solving the X, Y ODEs:

$$\frac{X''}{X} = -\lambda_1 \qquad \frac{Y''(y)}{Y(y)} = -\lambda_2 \qquad \Lambda = \lambda_1 + \lambda_2$$

are both individually constant.

$$X(x) = \sin\left(\frac{n\pi}{L_1}x\right) \qquad \lambda_1 = \frac{n^2\pi^2}{L_1^2}$$

$$Y(y) = \sin\left(\frac{m\pi}{L_2}y\right) \qquad \lambda_2 = \frac{m^2\pi^2}{L_2^2}.$$

Thm 7.6: The solution of the IVP on the rectangle w/ Dirichlet boundary conditions

$$\begin{cases} \frac{\partial u}{\partial t} - \Delta u = 0 \\ u(0,x) = \sum_{n,m} A_{nm}\sin\left(\frac{n\pi}{L_1}x\right)\sin\left(\frac{m\pi}{L_2}y\right) \\ u|_{\partial\Omega} = 0 \end{cases}$$

is

$$u(t,x) = \sum_{n,m=1}^{\infty} A_{nm}\sin\left(\frac{n\pi}{L_1}x\right)\sin\left(\frac{m\pi}{L_2}y\right)e^{-\left[\left(\frac{n\pi}{L_1}\right)^2 + \left(\frac{m\pi}{L_2}\right)^2\right]t}.$$

Rem 7.7 : Same for 3d.
Same for Neumann boundary conditions except w/ cos.

7.iv)

Eigenfunctions of the Laplacian

A fancier version of separation of variables.

$$u(t,x) = T(t)\,\Phi(x,y,z,\dots)$$

$$\frac{T'(t)}{T(t)} = -\lambda \qquad \text{and} \qquad \Delta\Phi = -\lambda\Phi$$

i.e. Φ solves the Helmholtz equation.

Def 7.8 : A function $\varphi : \Omega \to \mathbb{R}$ is called an (Dirichlet) eigenfunction of Δ on Ω w/ eigenvalue λ if it obeys

$$\begin{cases} \Delta\varphi = -\lambda\varphi \\ \varphi|_{\partial\Omega} = 0. \end{cases}$$

Then if $\varphi_1, \varphi_2, \dots$ are all eigenfunctions of the Laplacian.
$\lambda_1 \dots$

$$u(t,x) = \sum_{j=1}^{\infty} e^{-\lambda_j t}\varphi_j \qquad \text{solves} \qquad \begin{cases} \partial_t u - \Delta u = 0 \\ u|_{\partial\Omega} = 0 \\ u(0,x) = \sum_j \varphi_j. \end{cases}$$

➡ Solving the heat equation reduces to "diagonalizing" the Laplacian on Ω, i.e. finding eigenvectors + eigenvalues.

Lecture 8 Elliptic PDEs I: Laplaces Equation.

Reading: Strauss Ch. 6.

Recall the Laplacian is

$$\Delta u = \sum_{i=1}^{n} \partial_i^2 u = \nabla \cdot \nabla u$$

Today we focus on the Poisson equation w/ Dirichlet BC

$$\begin{cases} -\Delta u = f \\ u|_{\partial\Omega} = g. \end{cases}$$

8.i) The time-independent heat equation

Recall that a stationary value of an ODE

$$u'(t) = F(u(t), t))$$

is one such that $F(y,t) = 0$ $\forall t$. Then $u(t) = y$ is a steady-state solution.

Lemma 8.1: If $\begin{cases} -\Delta u = f \\ u|_{\partial\Omega} = g \end{cases}$ then u is a steady state solution of the heat equation.

Proof: Obvious: $\partial_t u = 0$ so $\partial_t u - \Delta u = f$

Lemma 8.2: If u solves $\begin{cases} -\Delta u = 0 \\ u|_{\partial\Omega} = g \end{cases}$ and v solves $\begin{cases} -\Delta v = f \\ u|_{\partial\Omega} = 0 \end{cases}$

then $u+v$ solves the full Poisson equation.

Proof: Both $-\Delta$, $|_{\partial\Omega}$ are linear operations. □

Lemma 8.3: If $-\Delta\phi_j = \lambda_j \phi_j$ are eigenfunctions of the Laplacian on Ω satisfying Dirichlet or Neumann (or more general boundary) conditions,

and $u = \sum_{j=0}^{n} A_j e^{-\lambda_j t} \phi_j$ solves $\begin{cases} (\partial_t - \Delta)u = 0 \\ u|_{\partial\Omega} = .. \\ u(0,x) = g(x) \end{cases}$

then $\lim_{t\to\infty} u(t,x) = \sum A_{\phi_K} \phi_{\phi_K}$, where ϕ_{ϕ_K} solves

$$\begin{cases} -\Delta u = 0 \\ u|_{\partial\Omega} = \text{⌊} \end{cases}$$

is a sum over the $\lambda = 0$ eigenspace.

Proof: unless $\lambda_j = 0$ then

$$\lim_{t\to\infty} A_j e^{-\lambda_j t}\phi_j(x) = 0. \quad \forall x.$$

□.

8.ii) Some solutions of the Laplace Equation

Ex 8.4 :(the linear algebra solution) If (ϕ_j, λ_j) are the eigenfunctions w/ eigenvalues λ_j satisfying (linear) boundary condition $B(\phi_j) = 0$. And

$$f = \sum_j \alpha_j \phi_j$$

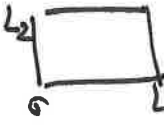
then the solution of

$$\begin{cases} -\Delta u = f \\ B(u) = 0 \end{cases}$$

is

$$u = \sum_j \frac{1}{\lambda_j}\alpha_j \phi_j.$$

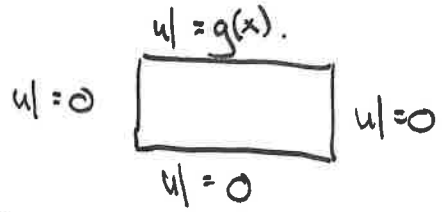
Proof: $\Delta\phi_j = \lambda_j\phi_j$ so $\Delta_g u = \sum_j \frac{1}{\lambda_j}\alpha_j \Delta\phi_j = \sum_j \alpha_j\phi_j = f.$

Ex 8.5 (rectangle eigenfunctions) we know $\phi_{mn} = \sin(\frac{n\pi}{L_1}x)\sin(\frac{m\pi}{L_2}y)$ are eigenfunctions w Dirichlet BC on 

, so we

can solve for $f = \sum_{m,n} A_{mn}\sin(\ x)\sin(\ y)$. Same for cos.

Ex 8.6 (Non-zero Dirichlet)



Suppose that $u = X(x)Y(y)$. Then

$$\frac{X''}{X} + \frac{Y''}{Y} = 0 \Rightarrow \frac{X''}{X} = \lambda$$
$$\frac{Y''}{Y} = -\lambda.$$

$X'' = \lambda X \quad \Rightarrow \quad X(x) = \sin\left(\frac{n\pi}{L_1}x\right) \qquad \lambda_n = \left(\frac{n\pi}{L_1}\right)^2$

$Y'' = -\lambda Y, \qquad Y(y) = A_n \sinh\left(\frac{n\pi}{L_1}y\right) + B_n \cosh\left(\frac{n\pi}{L_1}y\right)$

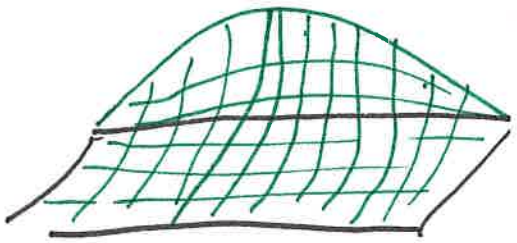
(Recall $\sinh = \frac{e^x - e^{-x}}{2}$ $\cosh = \frac{e^x + e^{-x}}{2}$)

boundary conditions say that

$$Y(0) = 0 \quad \Rightarrow \quad B_n = 0 \quad \forall n.$$

and $$u(x,y) = \sum_{n=1}^{\infty} A_n \sinh\left(\frac{\pi n}{L_1}y\right) \cdot \sin\left(\frac{\pi n}{L_1}x\right).$$

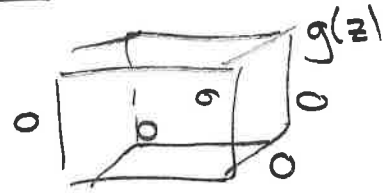
when $g(x) = \sum A_n \sin\left(\frac{\pi n}{L_1}x\right)$.



← $\sin(x)$.

↗ $\sim \sinh \sim e^{-y} + e^{y}$

Ex 8.7 (three-dim)



$$u(x,y,z) = \sum_{m,n=0}^{\infty} A_{mn} \sin\left(\frac{n\pi}{L_1}x\right) \sin\left(\frac{m\pi}{L_2}y\right) \cdot \sinh\left(\sqrt{\frac{n^2}{L_1} + \frac{m^2\pi^2}{L_2}} \cdot z\right)$$

(on homework).

Ex 8.8 (the disk) the Laplacian in polar coordinates is

$$\Delta u(r,\theta) = \left(\frac{\partial}{\partial r}\right)^2 + \frac{1}{r}\left(\frac{\partial}{\partial r}\right) + \frac{1}{r^2}\left(\frac{\partial}{\partial \theta}\right)^2$$

Solve

$$\begin{cases} \Delta u = 0 \\ u|_{r=1} = \sum_{n=1}^{\infty} \cos(n\theta) \cdot a_n. \end{cases}$$

$$u(r,\theta) = R(r)\Theta(\theta).$$

$$\Delta u = \frac{R''(r)}{\Theta(\theta)} + \frac{1}{r}R'(r)\Theta(\theta) + \frac{1}{r^2}\Theta''(\theta)R(r). \quad \Rightarrow \frac{r^2R'' + rR'}{R} = -\frac{\Theta''(\theta)}{\Theta(\theta)}$$

Solving the Θ equation

$$\Theta''(\theta) = -\lambda\Theta(\theta)$$

$$\Theta(\theta) = C_n\cos(n\theta) + D_n\sin(n\theta) \qquad \lambda = n^2$$

$$n = 0, \pm 1, \pm 2, \ldots$$

Solving the R equation

$$\left[r^2\frac{d^2}{dr^2} + r\frac{d}{dr} + -n^2\right]R(r) = 0.$$

$r = e^t$

$$\frac{d}{dt} = \frac{dr}{dt}\frac{d}{dr} = e^t\frac{d}{dr} = r\frac{d}{dr}$$

$$\left[\frac{d^2}{dt^2} + \frac{d}{dt} + -n^2\right]R(r) = 0.$$

$$e^{\alpha t} \longrightarrow \alpha(\alpha-1) + \alpha - n^2 = 0$$

$$\alpha^2 - n^2 = 0$$

$$\alpha = \pm n.$$

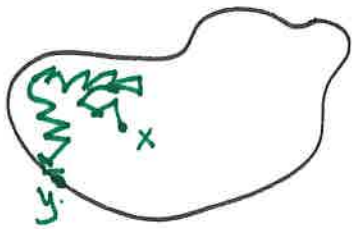
$$R(r) = A_n r^n + B_n r^{-n}.$$

If we require solutions to be bounded across 0,

$$u(r,\theta) = C_n r^n\cos(n\theta) + D_n r^n\sin(n\theta).$$

$$= (C_n - iD_n)z^n.$$

8.iii) Einstein's Brownian Motion



Brownian motion is a discontinuous path $X_0, X_1, X_2 \ldots$ where $X_{n+1} - X_n \sim N(0,\sigma)$.

- Sample brownian paths at X_0, w/ $\sigma \to \infty$.
- Call y the exit point.

is  drawn from a gaussian

$$\mathbb{E}(f(y)) = u(x) \quad \text{where} \quad \begin{cases} \Delta u = 0 \\ u|_{\partial R} = f. \end{cases}$$

Einstein's PhD showed Brownian motion of particles $\Longleftrightarrow$ heat/diffusion eq and allowed estimates of size of atoms <u>before</u> the discovery of the Nucleus by Rutherford 5-6 years later.

Lecture 9 Fourier Series I: Orthogonality + Coefficients

Recall that we solved the heat + wave equations for initial conditions

$$u(0,x) = \sum_{n=0}^{\infty} B_n \sin\left(\frac{n\pi}{L}x\right)$$

on $[0,L]$ (resp. cos for Neumann BC).

Question 9.1: What functions can be expressed as infinite sums of sin, cos?

The incredible answer (due to Fourier) is all of them!

← he's French, the last R is silent.

9.i) Periodic Functions

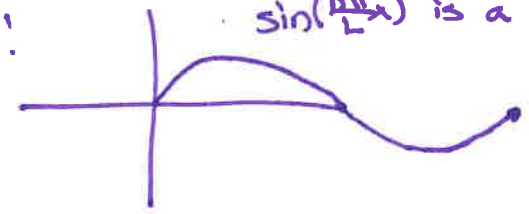
Def 9.2: a function $f: \mathbb{R} \to \mathbb{R}$ (or $\mathbb{C}$) is called L-periodic if

$$f(x+L) = f(x)$$

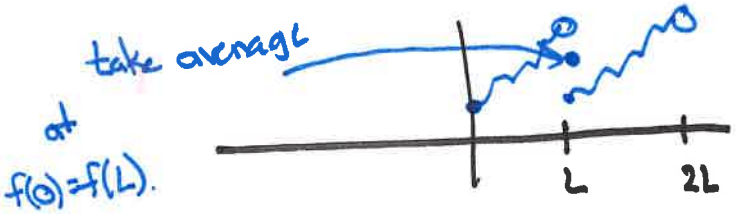
for all x.

Ex 9.3: $\sin\left(\frac{2n\pi}{L}x\right) = \sin\left(\frac{2n\pi}{L}x + 2n\pi\right) = \sin\left(\frac{2n\pi}{L}(x+L)\right)$.

↑ note the 2!



Ex 9.4: Functions need not be continuous



Prop 9.5: Periodic functions form a vector space. If f, g L-periodic,

i) $f(x+a)$ is periodic $\forall a$

ii) $f(mx)$ is periodic $m = 1, 2, \ldots$

iii) $\alpha f + \beta g$ is L-periodic

Ex 9.12 : Compute the Fourier series of

$$f(x) = \begin{cases} 1 & 0 \le x \le \pi \\ 0 & \pi \le x \le 2\pi \end{cases}$$



$$a_0 = \frac{1}{2\pi}\int_0^{2\pi} f(x)\,dx = \frac{\pi}{2\pi} = \frac{1}{2}$$

$$a_n = \frac{2}{2\pi}\int_0^{2\pi} f(x)\cos(nx)\,dx = \frac{1}{\pi}\int_0^{\pi} \cos(nx)\,dx = \frac{1}{\pi}\frac{\sin(nx)}{n}\Big|_{x=0}^{\pi} = 0.$$

$$b_n = \frac{2}{2\pi}\int_0^{2\pi} f(x)\sin(nx)\,dx = \frac{1}{\pi}\int_0^{\pi} \sin(nx)\,dx = \frac{1}{\pi}\left[\frac{-\cos(nx)}{n}\right]_0^{\pi}$$

$$= \frac{1}{\pi}\left[\frac{-\cos(n\pi)+1}{n}\right] = \begin{cases} \frac{2}{n\pi} & n \text{ odd} \\ 0 & \text{even.} \end{cases}$$

$$f(x) = \frac{1}{2} + \sum_{n=1,3,5\ldots} \frac{2}{n\pi}\sin(nx).$$

Trick 9.13 : put f on $[-\frac{1}{2}, \frac{1}{2}]$ by translation, use even/odd to see some are 0.

Ex 9.14 : Let $f(x) = \begin{cases} 0 & 0 \le x \le \pi \\ x-\pi & \pi \le x \le 2\pi \end{cases}$

$$a_0 = \frac{1}{2\pi}\int_0^{2\pi} f(x)\,dx = \frac{1}{2\pi}\int_{\pi}^{2\pi} (x+\pi)\,dx = \frac{1}{2\pi}\int_0^{\pi} u\,du = \frac{u^2}{4\pi}\Big|_0^{\pi} = \frac{\pi^2}{4\pi} = \frac{\pi}{4}$$

$u = x+\pi$

$$a_n = \frac{1}{\pi}\int_0^{2\pi} f(x)\cos(nx) = \frac{1}{\pi}\int_{\pi}^{2\pi} (x-\pi)\cos(nx)\,dx = \frac{1}{\pi}\int_0^{\pi} u\cos(n(u+\pi))\,dx.$$

b_n

$$= \frac{1}{\pi}\int_0^{\pi} u\sin(n(u+\pi))\,dx.$$

$= \sin(nu)$ even
$-\sin(nu)$ odd.

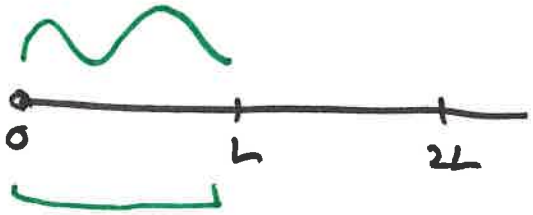
$t = nu$

$$= \frac{1}{n^2}\int t\sin t\,dt = \frac{1}{n^2}\left[-t\cos t + \int \cos t\,dt\right]_0^{n\pi} = \frac{(-1)^n}{\pi n^2}\left(-n\pi\cos(n\pi)\right)$$

$$= \frac{(-1)^n}{n}\left[-(-1)^n\right] = -\frac{1}{n}.$$

$$\Rightarrow \quad f(x) = \frac{\pi}{4} + \sum_{\text{odd}} \frac{2}{n^2\pi}\cos(nx) - \sum_{n=0}^{\infty} \frac{1}{n}\sin(nx).$$

Ex 9.6 : Every function $f:[0,L] \to \mathbb{R}$ has an L-periodic extension



9.ii) Fourier Series

Consider $\mathbb{R}^n$. It has basis $e_1 = \begin{pmatrix}1\\0\\\vdots\\0\end{pmatrix}$ $e_2 = \begin{pmatrix}0\\1\\0\\\vdots\\0\end{pmatrix}$ ... $e_n = \begin{pmatrix}0\\\vdots\\0\\1\end{pmatrix}$

and a dot product $\langle v,w\rangle = v\cdot w = v_1w_1 + \dots + v_nw_n$.

Key observation 9.7 : Any vector $v = \begin{pmatrix}v_1\\\vdots\\v_n\end{pmatrix}$ can be written

$$v = \sum_{i=1}^{n} v_i e_i = v_1\begin{pmatrix}1\\0\\\vdots\\0\end{pmatrix} + \dots + v_n\begin{pmatrix}0\\\vdots\\0\\1\end{pmatrix}$$

$$= \langle v,e_1\rangle\begin{pmatrix}1\\0\\\vdots\\0\end{pmatrix} + \dots + \langle v,e_n\rangle\begin{pmatrix}0\\\vdots\\0\\1\end{pmatrix}$$

$$= \sum_{i=1}^{n} \langle v,e_i\rangle e_i .$$

$\Rightarrow$ Writing a vector in the basis, coefficients are the dot products.

Def 9.8 : a basis $\{e_i\}$ of a vector space V w/ dot product $\langle -,-\rangle : V\times V \to \mathbb{R}$ is called orthonormal if

1) $\langle e_i, e_i\rangle = 1 \quad \forall i$
2) $\langle e_i, e_j\rangle = 0 \quad i \neq j$
3) e_i are linearly ind. and span.

Def 9.9 : Given two functions $f,g:[0,L]\to\mathbb{R}$, define the L^2-inner product by

$$\langle f,g\rangle_{L^2} = \frac{1}{L}\int_0^L f(x)g(x)\,dx .$$

Observe i) $\langle \alpha f_1 + \beta f_2, g\rangle_{L^2} = \alpha\langle f_1,g\rangle_{L^2} + \beta\langle f_2,g\rangle_{L^2}$

ii) same for g

iii) $\langle f,g\rangle = \langle g,f\rangle$.

Thm 9.10: the functions

$$\frac{1}{2L},\ \frac{1}{L}\sin\left(\frac{2\pi}{L}x\right),\ \frac{1}{L}\cos\left(\frac{2\pi}{L}x\right),\ \frac{1}{L}\sin\left(\frac{4\pi}{L}x\right),\ \ldots,\ \frac{1}{L}\sin\left(\frac{2\pi}{L}nx\right),\ \frac{1}{L}\cos\left(\frac{2\pi}{L}nx\right)$$

are an orthonormal basis of L-periodic functions.

Proof: Take $L = 2\pi$, general similar

- $\langle \cos(nx), \cos(nx)\rangle_{L^2} = \frac{1}{2\pi}\int_0^{2\pi} \cos(nx)\cos(nx)\,dx$
$$= \frac{1}{2\pi}\int_0^{2\pi} \frac{1}{2} + \frac{1}{2}\cos(2nx)\,dx$$
$$= \frac{\pi}{2\pi} + \frac{1}{2\pi}\left[\frac{1}{4n}\sin(2nx)\right]_0^{2\pi} = \frac{1}{2}$$

- $\langle \sin(nx), \sin(nx)\rangle = \frac{1}{2}$ same

- $\langle \cos(nx), \cos(mx)\rangle = \frac{1}{2\pi}\int_0^{2\pi} \cos(nx)\sin(mx)\,dx$
$$= \frac{1}{2\pi}\int_0^{2\pi} \frac{1}{2}\sin((n+m)x) - \frac{1}{2}\sin((n-m)x)\,dx$$
$$= 0.$$

others same.

- linearly independent + span in a few lectures.

Thm 9.11 (Fourier Series) If $f:[0,L] \to \mathbb{R}$ or is L-periodic

$$f(x) = a_0 + \sum_{n=1}^{\infty} a_n \cos\left(\frac{2n\pi}{L}x\right) + \sum_{n=1}^{\infty} b_n \sin\left(\frac{2n\pi}{L}x\right)$$

where

- $a_0 = \langle f, 1\rangle = \frac{1}{L}\int_0^L f(x)\,dx$
- $a_n = \langle f, \cos\left(\frac{2n\pi}{L}x\right)\rangle = \frac{2}{L}\int_0^L f(x)\cos\left(\frac{2n\pi}{L}x\right)dx$
- $b_n = \langle f, \sin\left(\frac{2n\pi}{L}x\right)\rangle = \frac{2}{L}\int_0^L f(x)\sin\left(\frac{2n\pi}{L}x\right)dx.$

Lecture 10 Fourier Series II: Solving PDEs.

Recall $f:[0,L] \to \mathbb{R}$ is given in Fourier series by

$$f(x) = a_0 + \sum_{n=1}^{\infty} a_n \cos\left(\frac{2n\pi}{L}x\right) + \sum_{n=1}^{\infty} b_n \sin\left(\frac{2n\pi}{L}x\right)$$

Def 10.1 $\{a_n\}$ and $\{b_n\}$ are called the Fourier coefficients of f.

10.i) More on odd/even functions

Recall f is odd if $f(-x) = -f(x)$
even if $f(-x) = f(x)$.

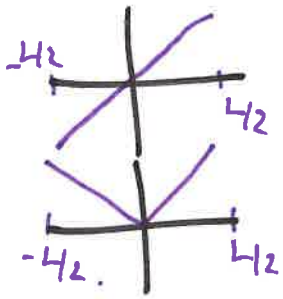
Since sin, cos are odd, even resp.

Prop 10.2: If $f:[0,L] \to \mathbb{R}$ and its periodic extension $f:[-\frac{L}{2}, \frac{L}{2}] \to \mathbb{R}$

- is odd then $a_n = 0 \ \forall n$ $\quad b_n = \frac{2}{L} \cdot 2 \int_0^{L/2} f(x) \sin(nx)\,dx$
- even then $b_n = 0 \ \forall n$ $\quad a_n = \frac{2}{L} \cdot 2 \int_0^{L/2} f(x) \cos(nx)\,dx$.

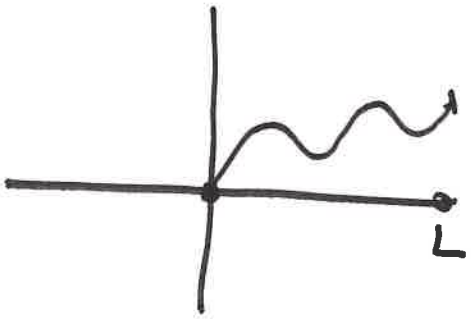
Ex 10.3: $f(x) = x$ has $a_n = 0$

$f(x) = |x|$ has $b_n = 0$

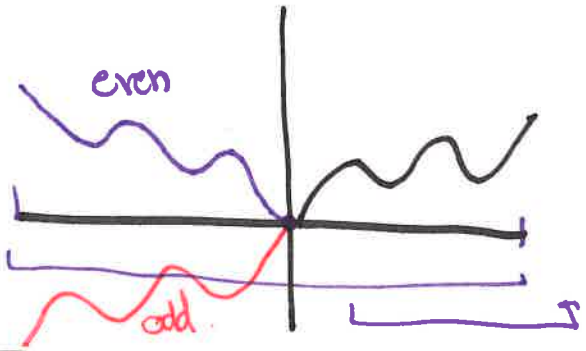


Extending functions $[0,L] \to \mathbb{R}$.

(I) L-periodic ext.



(II) 2L-periodic even extension.



(III) 2L-periodic odd extension.

Choose the extension

Ⓘ is L-periodic (neither even nor odd) $\quad a_0 + \sum_{n=1}^{\infty} a_n \cos\left(\frac{2n\pi}{L}x\right) + \sum_{n=1}^{\infty} b_n \sin\left(\frac{2n\pi}{L}x\right)$

Ⓘ is $2L$-periodic (odd) $\quad \sum_{n=1}^{\infty} B_n \sin\left(\frac{2n\pi}{2L}x\right)$

Ⓘ is $2L$-periodic (even) $\quad \sum_{n=1}^{\infty} A_n \cos\left(\frac{2n\pi}{2L}x\right)$

extension on $[-L, L]$.

Observation 10.4
Ⓘ automatically satisfies Dir. BC
Ⓘ automatically satisfies Neumann

10.ii) Fourier Series for the heat eq + wave eq.

We knew how to solve for $u(0,x) = \sum_n B_n \sin(\quad)$.

now write in Fourier series.

Thm 10.5: The sol. of the IVP

$$\begin{cases} \partial_t u = \Delta u \\ u(0,x) = f(x) \end{cases}$$

w/ Dirichlet BC is

$$u(t,x) = \sum_{n=1}^{\infty} B_n \sin\left(\frac{2n\pi}{2L}x\right) e^{-\frac{n^2\pi^2}{L^2}+}$$

where $\quad B_n = \frac{2}{L}\int_0^L f(x) \sin\left(\frac{2n\pi}{2L}x\right) dx.$

(same for Neumann, w/ A_n and cos).

Thm 10.6 (wave eq) the sol of the IVP

$$\begin{cases} \partial_t^2 u = c^2 \Delta u \\ u(0,x) = f(x) \\ \partial_t u(0,x) = g(x) \end{cases}$$

w/ Dir. BC is

$$u(t,x) = \sum_{n=1}^{\infty} \left[A_n \cos\left(\frac{2n\pi}{2L} \cdot tc\right) + \frac{B_n}{\left(\frac{2n\pi c}{2L}\right)} \sin\left(\frac{2n\pi}{2L} ct\right) \right] \sin\left(\frac{2n\pi}{2L} x\right).$$

where $A_n = \frac{2}{L}\int_0^L f(x)\sin\left(\frac{2n\pi}{2L}x\right)dx$

$B_n = \frac{2}{L}\int_0^L g(x)\sin\left(\frac{2n\pi}{2L}x\right)dx.$

For neumann replace the sin(x) w/ cos (not the t's).

10.iii) Full examples

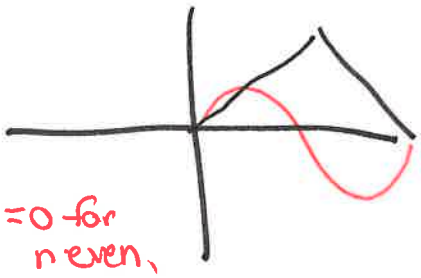
Ex 10.7 (Plucked String)



Pluck a string on $[0,L]$ so that

$$u(0,x) = -\frac{a}{L/2}|x - L/2| + a.$$

$$A_0 = \frac{1}{L}\int_0^L f(x)dx = \frac{1}{L}\cdot\frac{L}{2}\cdot a = \frac{a}{2}$$



$$A_n = \frac{2}{L}\int_0^L f(x)\sin\left(\frac{2n\pi}{2L}x\right)dx$$

$$= \frac{4}{L}\int_0^{L/2} -\frac{a}{L/2}x\sin\left(\frac{n\pi}{L}x\right)dx$$

$$= -\frac{4a}{L^2}\cdot 2\int_0^{L/2} x\sin\left(\frac{n\pi}{L}x\right)dx \qquad y = x\frac{\pi}{L}$$

$$= -\frac{8a}{L\pi}\int_0^{\pi/2} x\sin(nx)dx = -\frac{8a}{L\pi}\left[\frac{(-1)^{\frac{n-1}{2}}}{n^2}\right] \quad n \text{ odd}.$$

$$u(t,x) = \sum_{n=1,3,5}^{\infty} -\frac{8a}{L\pi}\left[\frac{(-1)^{\frac{n-1}{2}}}{n^2}\right]\cos\left(\frac{2n\pi}{2L}\cdot tc\right)\sin\left(\frac{2n\pi}{2L}x\right).$$

Ex 10.8: Assume $u(0,x) \equiv 1$ is constant. Solve the heat eq. w/ Dir. + Neumann BC at $x=0,1$.

Dir:

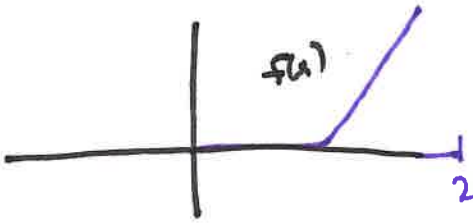
$$B_n = 2\int_0^1 \sin(n\pi x)\,dx = \begin{cases} 0 & n \text{ even} \\ \frac{4}{n\pi} & n \text{ odd.} \end{cases}$$

$$u(t,x) = \sum_{n=1,3,\ldots} \frac{4}{n\pi} \sin(n\pi x)\, e^{-n^2\pi^2 t}.$$

Neumann: It's already in equilibrium.

$$u(t,x) \equiv 1.$$

Intriguing Application 10.9: We saw for



$$f(x) = \frac{\pi}{4} + \sum_{n \text{ odd}} \frac{2}{n^2\pi}\cos(nx) - \sum_{n=0}^{\infty} \frac{1}{n}\sin(nx).$$

Thm 10.10: $\frac{\pi^2}{8} = 1 + \frac{1}{9} + \frac{1}{25} + \frac{1}{49} + \ldots$

Proof: We have in this case the series $\to f(x)$ $\forall x$ in interior.

$$\frac{\pi}{2} = f(0) = \frac{\pi}{4} + \sum_{n \text{ odd}} \frac{2}{n^2\pi} \Rightarrow \frac{\pi}{4} = \frac{2}{\pi}\left[1 + \frac{1}{9} + \frac{1}{25} + \ldots\right].$$

avg of endpoints.

Lecture 11 Fourier Series III + Function Spaces I.

We have done Fourier series in 1d.

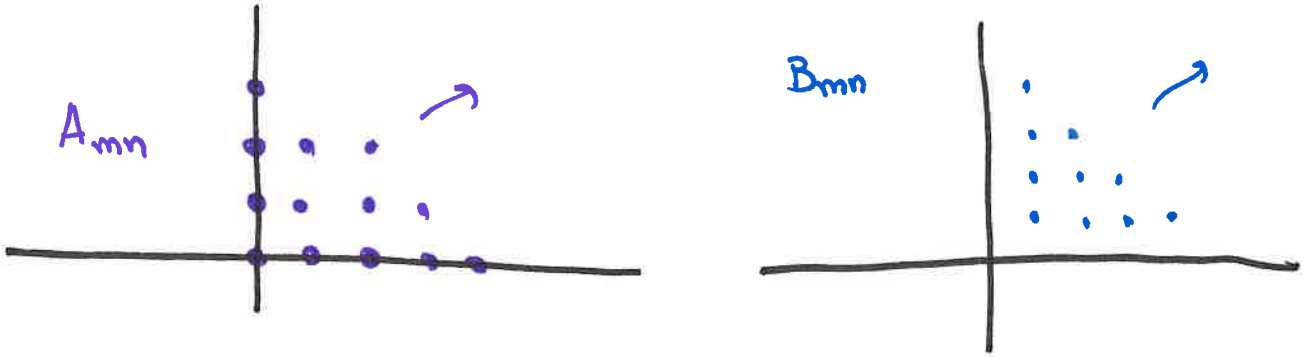
Thm 11.1: the functions

$$\cos\left(\frac{2n\pi x}{2L_1}\right)\cos\left(\frac{2m\pi y}{2L_2}\right)\cos\left(\frac{2p\pi z}{2L_3}\right) \qquad n,m,p = 0,1,2,\ldots$$

$$\sin\left(\frac{n\pi x}{2L_1}\right)\sin\left(\frac{2m\pi y}{2L_2}\right)\sin\left(\frac{2p\pi z}{2L_3}\right)$$

are an orthonormal basis of functions w/ Neumann, Dir boundary conditions on $[0,L_1]\times[0,L_2]\times[0,L_3]$. respectively. Same for 2d w/o the z terms.

Diagram 11.2: Picture the Fourier coefts as lattices



Thm 11.2: heat eq. the solution to the IVP

$$\begin{cases} \partial_t u = \Delta u \\ u(0,x,y,z) = f(x,y,z) \end{cases}$$

w/ Dir. BC on $\Omega = [0,L_1]\times[0,L_2]\times[0,L_3]$ is

$$u(t,x,y,z) = \sum_{m,n,p=1}^{\infty} B_{mnp}\sin\left(\frac{2n\pi}{2L_1}x\right)\sin\left(\frac{2m\pi}{2L_2}y\right)\sin\left(\frac{2p\pi}{2L_3}z\right) \cdot e^{-\left[\frac{n^2}{L_1^2}+\frac{m^2}{L_2^2}+\frac{p^2}{L_3^2}\right]\pi^2 t}$$

$$B_{mnp} = \frac{8}{\mathrm{Vol}(\Omega)} \int_\Omega f(x,y,z) \sin\left(\frac{2m\pi}{2L_1}x\right) \dots \sin\left(\frac{2p\pi}{2L_3}\right) dV.$$

Same for Neumann w/ cosines, and for wave equation (w/ Both A_{mnp} B_{mnp}).

Function Spaces I

11.ii) Hilbert spaces + Banach Spaces

Recall that $C^0(\Omega;\mathbb{R})$, $C^1(\Omega;\mathbb{R})$... are complete vector spaces w/ a norm. A norm obeys on V

$$\|\cdot\| : V \longrightarrow \mathbb{R}^{\geq 0}$$

i) $\|x\| = 0$ iff $x = 0$.

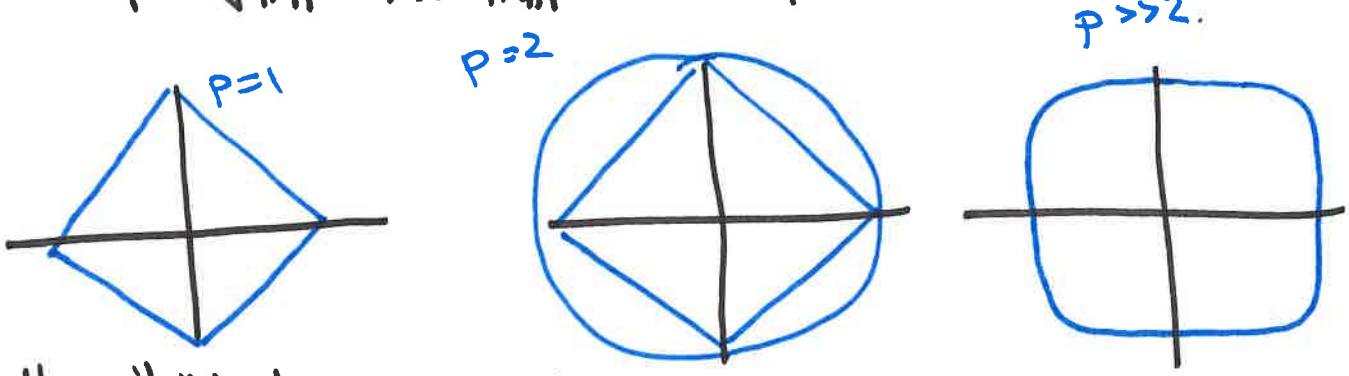
ii) $\|\alpha x\| = c\|x\|$

iii) $\|x+y\| \leq \|x\| + \|y\|$.

Ex 11.4 : Norms on $\mathbb{R}^n$.

$\|x\|_2 = \sqrt{x_1^2 + \dots + x_n^2}$ is the Euclidean or 2-norm.

$\|x\|_p = \sqrt[p]{|x_1|^p + \dots + |x_n|^p}$ is the p-norm.



the unit "circle" in different norms.

Def 11.5 : a Banach space is a complete, normed vector space.

Ex 11.6 : $C^0, C^1, \dots C^k$ are Banach spaces ($C^\infty(\Omega;\mathbb{R})$ is not).

An <u>inner product</u> is a bilinear function

$$\langle\ ,\ \rangle : V \times V \longrightarrow \mathbb{R}.$$

If V is a complex vector space, then $\longrightarrow \mathbb{C}$, and $\langle v, \alpha w\rangle = \bar{\alpha}\langle v, w\rangle$ conjugate linear in second variable.

That obeys

1) $\langle v, v\rangle \geq 0$ is a norm (the "induced" norm)

2) bilinear.

<u>Lemma 11.7</u> (Cauchy-Schwartz) $\forall v, w$ in V,

$$|\langle v, w\rangle| \leq \|v\| \cdot \|w\|.$$

in the norm from 1).

<u>Proof</u> : $\forall t$

$$0 \leq \|v + t\langle v, w\rangle w\|^2 = \langle v + t\langle v, w\rangle w,\ v + t\langle v, w\rangle w\rangle$$
$$= \|v\|^2 + t^2 |\langle v, w\rangle|^2 \|w\|^2 + 2t|\langle v, w\rangle|^2$$

minimize right side wrt $t_0 = \dfrac{-2|\langle v, w\rangle|^2}{|\langle v, w\rangle|^2 \|w\|^2} = \dfrac{-1}{\|w\|^2}$.

so

$$0 \leq \|v\|^2 + \frac{|\langle v, w\rangle|^2}{\|w\|^2} - \frac{2}{\|w\|^2}|\langle v, w\rangle|^2$$

$$0 \leq \|v\|^2 \|w\|^2 - |\langle v, w\rangle|^2.$$

□.

<u>Def 11.8</u> : A Hilbert space $(H, \langle\ ,\ \rangle)$ is a vector space w/ an inner product which is a Banach space (i.e complete in the induced norm).

Ex 11.9 : the L^2-inner product on $[0,L]$ was

$$\langle f,g \rangle_{L^2} = \frac{1}{L}\int_0^L f(x)g(x)\,dx$$

and on $\Omega \subset \mathbb{R}^n$

$$\langle f,g \rangle_{L^2} = \int_\Omega f(x_1 \dots x_n)\, g(x_1 \dots x_n)\, dV.$$

the induced **L^2-norm** is

$$\|f\|_{L^2} = \int_\Omega |f|^2\, dV.$$

Non-example 11.10 : there are two norms on continuous functions

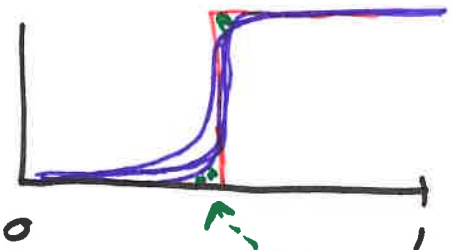
$$\|f\|_{C^0} = \max_{\Omega} |f(x)|. \quad x \in \Omega.$$

$$\|f\|_{L^2} = \int_\Omega |f|^2\, dV. = \langle f,f \rangle.$$

we saw $(C^0, \|\cdot\|_{C^0})$ is a Banach space

$(C^0, \|\ \|_{L^2})$ is NOT a Banach space (so $(C^0, \langle\ \rangle_{L^2})$ not Hilbert).

Let $f_n(x) =$



$\|f_n(x) - f_m(x)\|_{C^0} \sim \frac{1}{2}$ for some $m \gg n$.

$\|f_n(x) - f_m(x)\|_{L^2} \to 0$ $\left(= \text{Area}\right)$

but the limit is not continuous.

$\Rightarrow f_n$ doesn't converge in C^0-norm

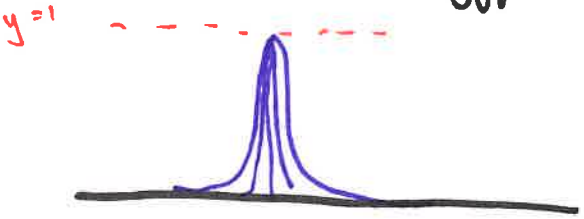
$\Rightarrow f_n$ does converge in L^2-norm, but limit not in C^0.

⇒ We need to find a bigger space

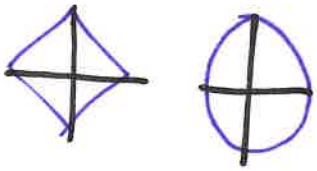
$$C^\infty(\Omega;\mathbb{R}) \hookrightarrow L^2(\Omega;\mathbb{R})$$

including more functions than just continuous for in to be complete.

Ex 11.11 : there are functions w/ $\|f_n\|_{L^\infty} = 1$ but $\|f_n\|_{L^2} \to 0$ (equiv $\|f_n\|_{L^\infty} \to \infty$. $\|f_n\|_{L^2} = 1$).



contrast this w/ p-norms on $\mathbb{R}^n$ where always with a factor of 2.

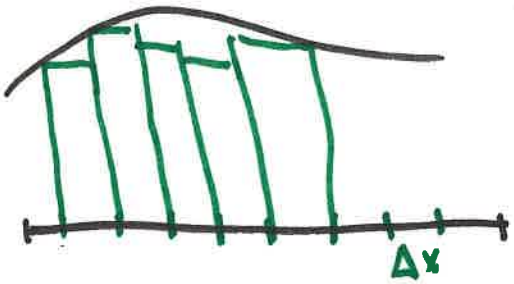


Lecture 12 | Function Spaces II : Hilbert space.

Recall we wish to find a complete space $L^2(\Omega;\mathbb{R})$ wrt the L^2 norm.

12.i) Lebesgue Integration

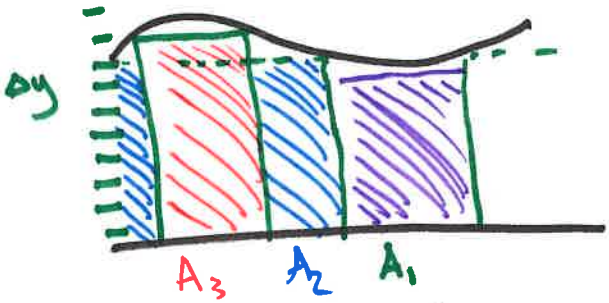
History 12.1 : Lebesgue developed the modern theory of integration in early 1900s.



Riemann Integral

$$\int_a^b f(x)\,dx = \lim_{N\to\infty} \sum_{i=1}^{N} \Delta x \cdot f(x_i)$$

$$\Delta x = \frac{b-a}{N}.$$



Lebesgue Integral

$$\int_a^b f(x)\,dx = \lim_{N\to\infty} \sum_{i=1}^{N} (i\Delta y) \cdot m(A_n)$$

↑ a measure of the size of the set.

Ex 12.2 : Consider $f(x) = \begin{cases} 0 & x \notin \mathbb{Q} \\ 1 & x \in \mathbb{Q} \end{cases}$.



$I = [a,b]$

not Riemann Integrable, but

$$\int_0^b f(x)\,dx = 1 \cdot m(\mathbb{Q}) + 0 \cdot m(I \setminus \mathbb{Q}).$$
$$= 0 \qquad (\mathbb{Q} \text{ has no area}).$$

Thm 12.3 :

$$L^2(\Omega;\mathbb{R}) = \left\{ f : \Omega \to \mathbb{R} \;\middle|\; \int_\Omega |f|^2 dm < \infty \text{ . is integrable in the sense of Lebesgue} \right\}$$

is complete, thus a Hilbert space.

Rem 12.4: We don't actually really need to know what elements of L^2 look like, just like we can't really "write down" most irrational numbers.

12.ii) Density

Recall that smooth functions on a closed, bounded set obtain a maximum.

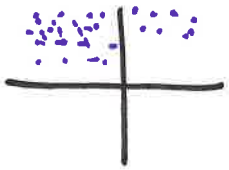
Define

$$C^\infty_{cpct}(\Omega;\mathbb{R}) = \{f : E \to \mathbb{R} \mid f \text{ is } \infty\text{-differentiable and } f = 0 \text{ except on a closed, bounded subset } E \subseteq \Omega \subseteq \mathbb{R}^n\}.$$

We know what elements of C^∞_{cpct} look like.

Def 12.5: A subset $\mathcal{D} \subseteq (H, \langle \cdot, \cdot \rangle)$ of a Hilbert space is called <u>dense</u> if $\forall x \in H \ \exists v_n \in \mathcal{D}$ such that $\lim_{n\to\infty} v_n = x$. ← ie $\lim_{n\to\infty} \|x_i - v_n\| = 0$.

Ex 12.6: $\mathbb{Q}^n \subseteq \mathbb{R}^n$ is a dense subset.



Thm 12.7: $C^\infty_{cpct}(\Omega;\mathbb{R}) \subseteq L^2(\Omega;\mathbb{R})$ is a dense subset. ie

1) $\forall f \in L^2 \quad \exists \psi_k \in C^\infty_{cpct}$ w/ $\lim_{k\to\infty} \|f - \psi_k\|_{L^2} = 0$

2) $\forall \varepsilon > 0 \quad \exists \psi \in C^\infty_{cpct}$ s.t. $\|f - \psi\|_{L^2} < \varepsilon$.

Prop 12.8: Suppose $V \subseteq H$ is a vector subspace that is closed (in the sense that it contains its limit points) then V is itself closed (thus a Hilbert space) or Banach.

Proof: Suppose that $\{v_n\}$ is Cauchy in V. We need to show it has a limit in V. B/c H is complete, $\exists x \in H$ so

$$v_n \longrightarrow x$$

in H. Then since $v_n \in V$, $x \in V$ by closedness.

Ex 12.9: Consider $\{f : \Omega \to \mathbb{R} \mid f|_{\Omega'} \equiv 0 \text{ on } \Omega' \subseteq \Omega\} \subseteq C^0(\Omega : \mathbb{R})$.

This is closed. If $f_n|_{\Omega'} \equiv 0$, and $f_n \to f$ in C^0 then

$$\max_{x \in \Omega} |f_n(x) - f(x)| \longrightarrow 0.$$

Therefore

$$f(x_0) = \lim_{n \to \infty} f_n(x_0) = 0 \quad \forall x_0 \in \Omega'.$$

12.iii) Bases: Recall e_n are a basis of $\mathbb{R}^n$

v_n eg. eigenvectors are another

For Hilbert spaces like $L^2(\Omega : \mathbb{R})$ there is no single "preferred" basis.

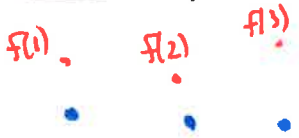
Def 12.10: A basis of a Hilbert space is a set $\{v_i\}_{i \in I}$ such that

i) $\{v_i\}_{i \in I}$ span a dense subspace

ii) linearly independent i.e. $\sum_{j=1}^{N} \alpha_j v_j = 0 \Rightarrow$ all α_j are 0.

Def 12.11 : · If I is a finite set, $|I|$ is called the dimension of H.
· If I is countable, H is called separable.

Ex 12.12 (spaces of sequences)



$= \{\mathbb{R}\text{-valued sequences on 3 points}\} = (\mathbb{R}^3, \|\ \|_2)$



$= \left\{\mathbb{R}\text{-valued sequences such that } \sum_{n=1}^{\infty} |a_n|^2 < \infty\right\} = \left(\ell^2(\mathbb{N};\mathbb{R}), \sum_{n=1}^{\infty} |a_n|^2\right)$

q

$\langle a_n, b_n \rangle = \sum_{n=1}^{\infty} a_n b_n$.

Rem 12.13 : Non-separable Hilbert spaces are gross and have many absurd/unintuitive counterexamples. We will never consider them.

Def 12.14 : A basis is called $\{v_i\}_{i=1}^{\infty}$
i) orthogonal if $\langle v_i, v_j \rangle = 0 \quad i \neq j$
ii) normal if $|v_i|^2 = 1 \quad \forall i$
orthonormal if both.

12.iv) L^p Spaces

Def 12.15 : the L^p spaces for $1 \leq p < \infty$.

$$L^p(\Omega;\mathbb{R}) = \left\{ f : \Omega \to \mathbb{R} \;\middle|\; \int_{\Omega} |f|^p dV < \infty \right\}$$

$\|f\|_{L^p} = \uparrow$ ← Lebesgue integral.

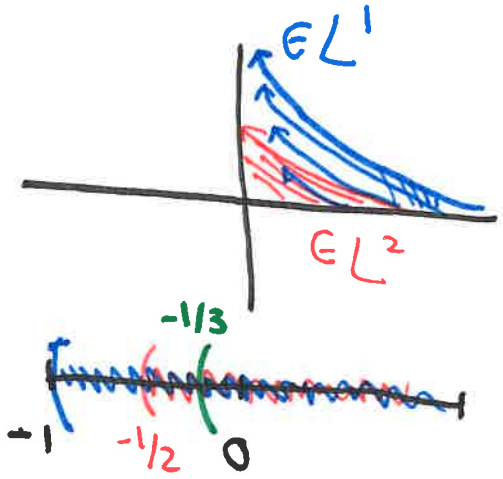
is a Banach space. It is not a Hilbert space unless $p=2$. All are complete.

$\langle f, g \rangle_{L^p} = \int |f|^{p/2} |g|^{p/2} dV$ is not bilinear so not an inner product.

Ex 12.16: Consider x^{α} on $[0,1]$.

$$\int_0^1 |x|^{-\alpha}\,dx = -\frac{1}{\alpha+1}x^{\alpha+1}\Big|_0^1 < \infty \quad \text{if } \alpha > -1.$$

- $x^{-\alpha} \in L^1([0,1],\mathbb{R})$ if $\alpha < 1$
- $x^{-\beta} \in L^2([0,1],\mathbb{R})$ if $2\beta > -1$, $\beta > -1/2$.
- $x^{-\delta} \in L^p([0,1],\mathbb{R})$ if $p\delta > -1$, $\delta > -\frac{1}{p}$.



In particular, there are functions in L^1 that are not in $L^2, L^3, \ldots$
in L^2 not in $L^3, L^4, \ldots$
in L^p not in $L^{p+\delta}$ $\delta > 0$.

Lecture 13 Function Spaces III: bases.

Recall that v_i $i=1,\dots n,\dots\infty$ is a basis of a Hilbert space $(H, \langle\ \rangle)$ if

i) v_i span and are linearly independent

ii) orthonormal if $\langle v_i, v_j\rangle = \delta_{ij}$

13.i) Some facts about convergence

Suppose that $F_N = \sum_{n=1}^{N<\infty} f_n \in H$. Then this means, if

$$\{F_N\}_{N=1}^{\infty} \in H \qquad := \sum_{j=1}^{\infty} f_j$$

is Cauchy, then by completeness, $\exists F$ such that $F_N \to F$, in which case

$$\| \sum_{j=1}^{N} f_n - F \| = \| \sum_{j=N+1}^{\infty} f_j \| \xrightarrow{N\to\infty} 0$$

Fact 13.1 (Iterated triangle inequality)

$$\| \sum f_j \| \leq \sum \| f_j \|$$

even for infinite sums.

Proof: $\| \sum_{j}^{N} f_j \| \leq \sum^{N} \|f_j\|$ $\forall N$ finite, take $\lim N\to\infty$. □

Def 13.2: A vector $v \in H$ is represented in a basis $\{v_j\}$ if $\exists c_i$ such that

$$\sum_{j=1}^{\infty} c_j v_j = v.$$

in which case $\langle v, v_j\rangle = c_j$.

Prop 13.3: The representation of a vector in an orthonormal basis exists and is unique.

Proof: By definition of span/dense, it exists. Suppose

$$v = \sum_{j=1}^{\infty} c_j v_j$$

$$v = \sum_{j=1}^{\infty} d_j v_j.$$

then $\exists N_1, N_2 \leq N = \max(N_1, N_2)$ such that

$$\|v - \sum_{j=1}^{N} c_j v_j\| < \frac{\varepsilon}{2}$$

$$\|v - \sum_{j=1}^{N} d_j v_j\| < \frac{\varepsilon}{2}$$

so

$$\|\sum_{j=1}^{N} c_j v_j - \sum_{j=1}^{N} d_j v_j\| \leq \|\sum_{j=1}^{N} c_j v_j - v + v - \sum_{j=1}^{N} d_j v_j\|$$

$$\leq \|\sum_{j=1}^{N} c_j v_j - v\| + \|v - \sum_{j=1}^{N} d_j v_j\| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

and

$$① = \|\sum_{j=1}^{N} (c_j - d_j) v_j\| = \sqrt{\langle (c_1 - d_1)v_1 + \dots + (c_N - d_N)v_N, (c_1 - d_1). \dots \rangle}$$

$$= \sqrt{\sum_{j=1}^{N} |c_j - d_j|^2}$$

$$\geq |c_j - d_j| \quad \forall j.$$

$$\Rightarrow |c_j - d_j| < \varepsilon \quad \forall j, \varepsilon \quad \Rightarrow c_j = d_j. \qquad \square$$

Thm 13.4 (Bessel Inequality): Suppose $\{e_j\}$ is an orthonormal sequence (not necessarily spanning) in a separable Hilbert space $(H, \langle\ \rangle)$. Then $\forall v \in H$

$$\sum_{j=1}^{\infty} |\langle v, e_j \rangle|^2 \leq \|v\|.$$

And $=$ only if $\sum_{j=1}^{N} \langle v, e_j \rangle e_j \longrightarrow v$.

Proof: $\langle v - S_N v, v - S_N v \rangle = \|v\|^2 + \|S_N v\|^2 - 2\langle v, S_N v \rangle$.

$$= \|v\|^2 + \|S_N v\|^2 - 2\|S_N v\|^2$$

$S_N = \sum_{j=1}^{N} \langle v, e_j \rangle e_j$ since $\langle v, \sum_j \langle e_j, v \rangle e_j \rangle = \sum^{N} \langle e_j, v \rangle \langle v, e_j \rangle = \|S_N v\|^2$

so $0 \le \|v - S_N v\|^2 = \|v\|^2 - \|S_N v\|^2$

Take lim $\Leftrightarrow$

$$0 \le \lim_{N\to\infty} \left\| v - \sum_{j=1}^{\infty} \langle v, e_j\rangle e_j \right\|^2 = \|v\|^2 - \left\| \sum_{j=1}^{\infty} \langle v, e_j\rangle e_j \right\|^2$$

norm is cont so $\lim \|f_n\| = \| \lim f_n \|$

so



and if equal, difference is zero. □

Thm 13.5: If $\{e_j\}$ is an orthonormal set, its a basis if and only if 0 is the only thing orthogonal to all.

Proof: Suppose basis, by Prop 13.3 if $v \perp$ all then

$$v = \sum_{j=1}^{\infty} \langle v, e_j\rangle e_j$$

then

$$0 = \langle e_j, v\rangle = \sum_{j=1}^{\infty} \langle v, e_j\rangle \langle e_j, v\rangle$$

$$= \sum_{j=1}^{\infty} |\langle v, e_j\rangle|^2 \quad \text{so} \quad \langle v, e_j\rangle = 0 \ \forall j. \ v = 0.$$

$\Leftarrow$ Suppose 0 is the only thing orthogonal to all. Then

$$S_N(v) \to v.$$

Why? As before, Bessel shows $\sum_{j=1}^{\infty} |\langle v, e_j\rangle|^2 \le \|v\|^2$

so $\|S_N - S_M\|^2 = \left\| \sum_{N=j}^{M} \langle v, e_j\rangle v_j \right\|^2 \le \sum_{N}^{M} |\langle v, e_j\rangle|^2$

tri ineq + orthogonality; Cauchy in $\mathbb{R}$.

so $\exists v$ st $S_N \to \tilde{v}$.

Then $\langle v-\tilde{v}, e_j\rangle = \langle v, e_j\rangle - \left\langle \sum_{k=1}^{\infty} \langle v, e_k\rangle e_k, e_j \right\rangle$

$= \langle v, e_j\rangle - \sum_k \langle v, e_k\rangle \underbrace{\langle e_k, e_j\rangle}_{=\delta_{kj}}$

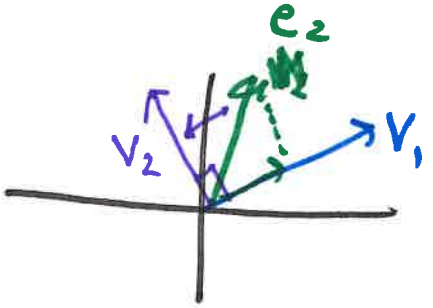
$= \langle v, e_j\rangle - \langle v, e_j\rangle = 0.$

so $v - \tilde{v} = 0.$

□

13.ii) Gram-Schmidt

Algorithm 13.6 : Any linearly independent set of vectors $\{e_j\}$ can be amended to an orthonormal basis.



Proof :

$v_1 = e_1 / \|e_1\|$

$v_2 = e_2 - \text{proj}_{v_1}(e_2)$

$= \left(e_2 - \langle e_2, v_1\rangle v_1\right) / \text{norm}$

$v_3 = e_3 - \langle e_3, v_2\rangle v_2 - \langle e_3, v_1\rangle v_1 \Big/ \text{norm}.$

$\vdots$

Ex 13.7 (Legendre Polynomials)

$1, x, x^2, x^3, \ldots$ are linearly ind. in $L^2([-1,1]:\mathbb{R})$.

$P_0 = 1$

$P_1 = \frac{3x}{2}$ $\qquad \int_{-1}^{1} x \cdot 1\, dx = 0, \quad \int_{-1}^{1} x^2 dx = \frac{2}{3}$

$P_2 = \frac{5}{2}(1 - 3x^2)$

$P_3 = \frac{7}{2}(3x - 5x^3)$

$\vdots$

$P_9 = \frac{19}{256}(315x - 4620x^3 + 18018x^5 - 25740x^7 + 12155x^9)$

$$P_n = \frac{2n+1}{2} \frac{1}{2^n n!} \left(\frac{d}{dx}\right)^n (x^2-1)^n$$

Lecture 14 Function Spaces IV : Fourier Series

To determine if something is a basis, we must check span + linear ind.

Lemma 14.1: If a collection $\{e_j\}$ in a Hilbert space $(H, \langle\ \rangle)$ are orthogonal, then they are automatically linearly independent.

Proof: If linearly dependent, there is a relation

$$e_1 = \sum_{j=2}^{n} c_j e_j$$

(up to re-ordering indices). Then

$$0 < \|e_1\|^2 = \langle e_1, e_1 \rangle = \langle e_1, \sum_{j=2}^{n} c_j e_j \rangle = \sum_{j=2}^{n} c_j \langle e_1, e_j \rangle = 0$$

a contradiction. ∎

Our Fourier Series can now be phased as

Thm 14.2: On $\Omega = [0, L_1] \times \dots \times [0, L_k] \subseteq \mathbb{R}^k$ the collection of functions

$$\sin\left(\frac{\pi n_1}{L_1} x_1\right) \sin\left(\frac{\pi n_2}{L_2} x\right) \dots \sin\left(\frac{\pi n_k}{L_k} x_k\right) \qquad n_1, n_2, \dots n_k \geq 1.$$

or

$$\cos\left(\frac{\pi n_1}{L_1} x_1\right) \dots \cos\left(\frac{\pi n_k}{L_k} x_k\right) \qquad n_1, \dots n_k \geq 0.$$

with the correct normalization form an orthonormal basis of $L^2(\Omega; \mathbb{R})$. Similarly for mixed (each variable is either of $\sin\left(\frac{2\pi n_1}{L_1} x_1\right)$ or $\cos\left(\frac{2\pi n_1}{L_1} x_1\right)$.

(proof shortly).

Prop 14.3: Suppose a vector $v \in (H, \langle\ \rangle)$ is written in a basis $\{e_j\}$ as

$$v = \sum_{j=1}^{\infty} c_j e_j.$$

Then

$$\|v\|^2 = \sum_{j=1}^{\infty} |c_j|^2.$$

Proof:

$$\|v\|^2 = \left\langle \sum_{j=1}^{n} c_j e_j \,,\, \sum_{k=1}^{\infty} c_k e_k \right\rangle$$

$$= \langle c_1 e_1 + c_2 e_2 + \dots \,,\, c_1 e_1 + c_2 e_2 + \dots \rangle$$

$= |c_1|^2$ $\quad$ $= 0$ $\quad$ $= 0$

$$= |c_1|^2 + |c_2|^2 + \dots \qquad \square$$

Thm 14.4: (Parseval's Theorem) If $f \in L^2([a,b]; \mathbb{R})$ w/ Sine/cos Fourier coefficients $\{A_n\}$ and $\{B_n\}$, then these are square summable, and

$$2|A_0|^2 + \sum_{n=1}^{\infty} |A_n|^2 = \frac{2}{\ell} \int_a^b |f(x)|^2 \, dx \qquad \ell = b-a$$

same for $\{B_n\}$ and $\{a_n\} + \{b_n\}$.

More generally for $f \in L^2([a_1,b_1] \times \dots \times [a_k,b_k])$

$$\sum_{n_1=0}^{\infty} \sum_{n_2=0}^{\infty} \dots \sum_{n_k=0}^{\infty} |B_n|^2 = \left(\frac{\ell_1 \dots \ell_k}{2^k} \right)^{-1} \int_{\Omega} |f(x_1, \dots x_k)|^2 \, dV$$

Rem 14.5: this can be written succinctly as

$$\|\{B_n\}\|_{\ell^2(\mathbb{N};\mathbb{R})} = \left(\frac{\ell}{2}\right)^{-1} \|f\|_{L^2([a,b])}$$

$$\|\{B_{n_1 \dots n_k}\}\|_{\ell^2(\mathbb{N}^k;\mathbb{R})} = \left(\frac{\ell_1 \dots \ell_k}{2^k}\right)^{-1} \|f\|_{L^2(\Omega)}$$

Ex 14.6: What is $\sum_{n=1}^{\infty} \frac{1}{n^4}$? Consider $g(x) = x^2$ on $[0,\pi]$.

Fourier cosine series is $\quad A_0 = \frac{\pi^2}{3}$ $\quad$ $A_n = \frac{(-1)^n \cdot 4}{n^2}$

$$g(x) = \frac{\pi^2}{3} + \sum_{n=1}^{\infty} \frac{(-1)^n \cdot 4}{n^2} \cos(nx).$$

$$\|f\|_{L^2} = \frac{2}{\pi} \int_0^{\pi} x^4 \, dx = \frac{2}{\pi} \frac{x^5}{5} \Big|_{x=0}^{\pi} = \frac{2\pi^4}{5}$$

$$2\frac{A_0^2}{2} + \sum A_n = 2\left(\frac{\pi^2}{3}\right) + 16 \sum_{n=1}^{\infty} \frac{1}{n^4} \Rightarrow$$

$$\frac{2\pi^4}{5} = \frac{2\pi^4}{9} + 16 \sum_{n=1}^{\infty} \frac{1}{n^4}$$

$$\boxed{\sum_{n=1}^{\infty} \frac{1}{n^4} = \frac{\pi^4}{90}}$$

14.ii) Constructing Bases

We will have two methods of confirming a set spans

1) Stone-Weierstauss theorem (today)
2) spectral theorem (later)

Def 14.7: An algebra of functions $\mathcal{A} \subseteq L^2(\Omega : \mathbb{R})$ is a vector subspace such that $\forall f(x), g(x) \in \mathcal{A}$ then

$$f(x)g(x) \in \mathcal{A}$$

as well.

Ex 14.8:

- polynomials form an algebra b/c $p(x)q(x)$ still a polynomial.
- trig functions $\cos(\frac{n\pi}{L}x)$, $\sin(\frac{n\pi}{L}x)$ form an algebra by identities

$$\cos(nx)\sin(mx) = \sin((m+n)x) \pm \sin((m-n)x).$$

Def 14.8: An algebra $\mathcal{A}$ is said to separate points if $\forall x, y \in \Omega$, $\exists f \in \mathcal{A}$ such that $f(x) \neq f(y)$.

Thm 14.9 (Stone-Weierstauss) If $\mathcal{A} \subseteq C^0(\Omega : \mathbb{R})$ (closed, bounded) is an algebra that

- contains constants (or $\nexists x$ st. $f(x) = 0 \ \forall f \in \mathcal{A}$)
- separates points.

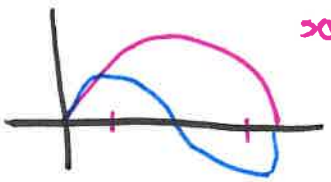
Then $\mathcal{A}$ is dense (in the C^0 norm).

Ex 14.10: polynomial $f(x) = x$ obviously separates points. $\Rightarrow$ polynomials

Proof (of thm 14.2) We wish to show $\sin(\frac{n\pi x}{L})$ span + are LI.

1.) LI by Lemma 14.1 since we checked orthogonality in Lecture 9.

2.) enough to show dense in C^0 b/c $C^0 \subseteq L^2$ dense ($X \subseteq Y$ dense, $Y \subseteq Z$ dense $\Rightarrow X \subseteq Z$ dense)



separates all but $\frac{\pi}{2} \pm x$.

$\sin(2x)$ separates those

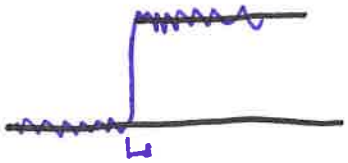
$\Rightarrow$ dense by Stone-Weierstrauss.

add const. in to show dense in $C^0([a,b])$ | end points same

14.iii) Convergence of Fourier series

Ex 14.11 : $f(x) =$  $= \frac{1}{2} - \sum_{n=1,3,5} \frac{2}{n\pi} \sin(nx)$

$$\sum_{n=0}^{\infty} a_n \cos(nx) + b_n \sin(xn) \xrightarrow{L^2} f(x)$$



but $f(x) \notin C^0$.

$\Rightarrow$ Fourier series converges in L^2, but not pointwise.

Question 14.12 : When does $\sum_{n=1}^{\infty} A_n \cos(n\pi x) \longrightarrow f(x)$

in C^0 (uniformly)
C^1 ? C^α?

This question is very hard and motivated many decades of analysis.
Modern formulation only crystalized in 1940s/50s.

Graphic 14.13 : Gibbs phenomena.

Lecture 15 Hyperbolic PDEs II : the drum

Let $\Omega \subseteq \mathbb{R}^2$ be the disk of radius r_0. Consider the IVP for the wave eq.

$$\begin{cases} \partial_t^2 u = \Delta u \\ u(0,r,\theta) = g \\ \partial_t u(0,r,\theta) = h \end{cases}$$

subject to vanishing Dirichlet boundary conditions at $r = r_0$.

Application 15.1 : this setting models a vibrating drum when the mallet strike chooses g, h.



Question 15.2 : How does changing the diameter of the drum affect pitch?
How does the mallet strike affect what overtones vibrate?

15.1) Separation of Variables (again)

Suppose $u(t,r,\theta) = T(t)\Phi(r,\theta)$.

$$T''(t)\Phi(r,\theta) = T(t)\Delta\Phi(r,\theta).$$

$$\frac{T''(t)}{T(t)} = \frac{\Delta\Phi(r,\theta)}{\Phi(r,\theta)} = -\lambda \qquad \lambda \in \mathbb{R}.$$

$$\Rightarrow \begin{cases} T''(t) = -\lambda T(t) \\ \Delta\Phi = -\lambda\Phi \end{cases} \Rightarrow \begin{array}{l} T(t) = C(\lambda)\cos(\sqrt{\lambda}t) + D(\lambda)\sin(\sqrt{\lambda}t) \\ \Phi(r,\theta) = \text{eigenfunction of } \Delta \text{ w/ eigenvalue } \lambda. \end{array}$$

Calculation 15.2 (separation of variables for $\Phi(r,\theta)$)

Suppose $\Phi(r,\theta) = R(r)\Theta(\theta)$. Recall $\Delta = \left(\frac{\partial}{\partial r}\right)^2 + \frac{1}{r}\frac{\partial}{\partial r} + \frac{1}{r^2}\frac{\partial}{\partial\theta^2}$

Plug in, $$\left[R''(r) + \frac{1}{r}R'(r)\right]\Theta(\theta) + \frac{1}{r^2}\Theta''(\theta)R(r) = -\lambda R(r)\Theta(\theta)$$

$$\frac{r^2R''(r) + rR'(r) + \lambda r^2 R(r)}{R(r)} = -\frac{\Theta''(\theta)}{\Theta(\theta)} = \alpha \qquad \text{and } \Theta(\theta) = \Theta(\theta + 2\pi).$$

So 1) $\Theta''(\theta) = -\alpha\Theta(\theta) \Rightarrow \alpha = n^2$

$\Theta(\theta) = A_n\cos(n\theta) + B_n\sin(n\theta)$.

2) $r^2R''(r) + rR'(r) + (\lambda r^2 * - n^2)R(r) = 0$. ← almost Bessel ODE.

Trick 15.3 (Rescaling the Bessel ODE)

Take $\rho = \sqrt{\lambda}\, r$.

$$\frac{d}{dr} = \frac{d\rho}{dr}\frac{d}{d\rho} = \sqrt{\lambda}\frac{d}{d\rho}$$

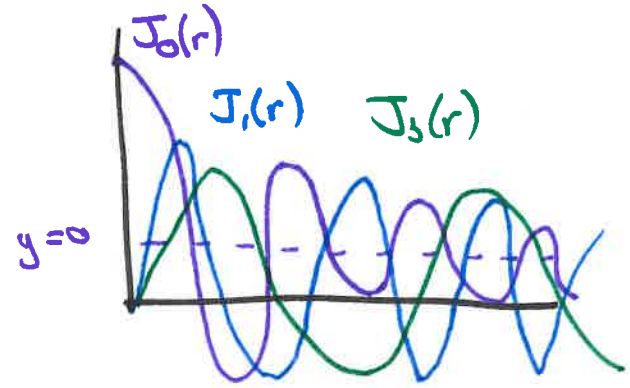
$$r\frac{d}{dr} = \sqrt{\lambda}\, r\frac{d}{d\rho} = \rho\frac{d}{d\rho}$$

$$\Rightarrow \rho^2R''(\rho) + rR(\rho) + (\rho^2 - n^2)R(\rho) = 0.$$

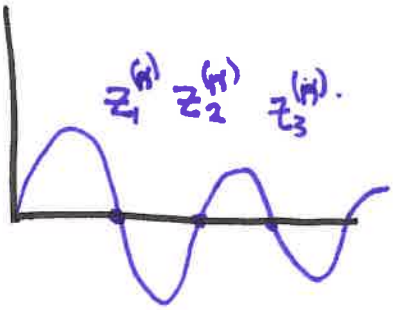
$$R(\rho) = a_{n\lambda}J_n(\rho) + b_{n\lambda}Y_n(\rho)$$

$$= a_{n\lambda}J_n(\sqrt{\lambda}\, r)$$

asymptote at $r=0$, throw it out.



Step 15.4: Solving for λ.



We imposed

$$0 = R(r_0) = a_{n\lambda}J_n(\sqrt{\lambda}\, r_0)$$

so $\sqrt{\lambda}\, r_0 = z_k^{(n)}$ for $k = 1, 2, 3, \ldots$ Set $\lambda_{nk} = \dfrac{(z_k^{(n)})^2}{r_0^2}$

We conclude

$$\Phi(r,\theta) = \left[A_{nk}\cos(n\theta) + B_{nk}\sin(n\theta)\right]J_n(\sqrt{\lambda_{nk}}\, r).$$

are the eigenfunctions.

Thm 15.5: the general solution of the wave equation on the disk are

$$u(t,r,\theta) = \sum_{n,k}^{\infty} \left[A_{nk}\cos(n\theta) + B_{nk}\sin(n\theta)\right] J_n(\sqrt{\lambda_{nk}}\, r)\cdot \cos(\sqrt{\lambda_{nk}}\, t)$$
$$+ \left[C_{nk}\cos(n\theta) + B_{nk}\sin(n\theta)\right] J_n(\sqrt{\lambda_{nk}}\, r)\cdot \sin(\sqrt{\lambda_{nk}}\, t).$$

where λ_{nk} is the kth zero of J_n.

Corollary 15.6: If $g(r,\theta)$, $h(r,\theta)$ are expressed as sums

$$g(r,\theta) = \sum_{n,k}^{\infty} \left[a_{nk}\cos(n\theta) + b_{nk}\sin(n\theta)\right] J_n(\sqrt{\lambda_{nk}}\, r)$$

then we can solve by matching coefficients.

15.ii) Fourier-Bessel or Hankel Series

Thm 15.7: For all n,k, the functions

$$\cos(n\theta)\, J_n(\sqrt{\lambda_{nk}}\, r) \quad \text{or } \sin$$

are pairwise orthogonal. i.e.

$$\int_0^{r_0}\int_0^{2\pi} \cos(n\theta)\, J_n(\sqrt{\lambda_{nk}}\, r)\cdot \cos(m\theta) J_m(\sqrt{\lambda_{mj}}\, r)\, r\, dr\, d\theta = \delta_{nm}\delta_{jk}.$$

Proof: We know $=0$ unless $n=m$ from Fourier. So show

$$\int_0^{r_0} J_n(\sqrt{\lambda_{nk}}\, r) J_m(\sqrt{\lambda_{mj}}\, r)\, r\, dr = 0.$$

Call these y_j, y_k. they obey

$$(ry_j')' + (\lambda_j r - \tfrac{n}{r})y_j = 0 \qquad \cdot y_k$$
$$-\quad (ry_k')' + (\lambda_k r - \tfrac{n}{r})y_k = 0 \qquad \cdot y_j$$

$$y_k(ry_j')' - y_j(ry_k')' + (\lambda_j - \lambda_k) r y_j y_k = 0.$$

$$\lambda_j - \lambda_k \int_0^{r_0} y_j y_k \, r\,dr = \int_0^{r_0} y_k (r y_j')' - y_j (r y_k')' \, dr$$

$$= \underbrace{\int_0^r r(y_k' y_j' - y_j' y_k')\,dr}_{=0} - \Big[r y_k y_j' - r y_j y_k' \Big]_{r=0}^{r=r_0}$$

$$= 0$$

□

Question 15.8 : Do they span? Hard with Stone-Weierstraup.

Lecture 16 Eigenvalues of the Laplacian

Recall that $\varphi : \Omega \to \mathbb{R}$ is an eigenfunction of the Laplacian if

$$\Delta\varphi = -\lambda\varphi.$$

16.i) Self-Adjointness

Recall that a finite matrix is self-adjoint if

$$A = A^+ \quad \leftarrow \text{Hermitian conjugate.}$$
$$= \overline{A}^T$$

In this case $\langle Av, w\rangle = (Av)^T\overline{w}$

$$= v^T A^T \overline{w}$$
$$= v^T(\overline{A^T w}) = \langle v, Aw\rangle.$$

Def 16.1: A 2nd order linear elliptic differential operator L is said to be (formally) self-adjoint if

$$\langle Lu, f\rangle_{L^2(\Omega;\mathbb{R})\text{ or }\mathbb{C}} = \langle u, Lf\rangle_{L^2(\Omega;\mathbb{R})}$$

Thm 16.2 (Green's Identity) If φ, ψ are functions on Ω then

$$\int_\Omega (\varphi\Delta\psi - \psi\Delta\varphi)\,dV = \int_{\partial\Omega}\left(\varphi\frac{\partial\psi}{\partial\nu} - \psi\frac{\partial\varphi}{\partial\nu}\right)d\sigma$$

Proof: Recall that via integration by parts

$$\int_\Omega \nabla\varphi\nabla\psi\,dV = \int_\Omega -\psi\Delta\varphi\,dV + \int_{\partial\Omega}\psi\frac{\partial\varphi}{\partial\nu}\,d\sigma.$$

$$-\int_\Omega \nabla\psi\nabla\varphi\,dV = -\int_\Omega \varphi\Delta\psi\,dV + \int_{\partial\Omega}\varphi\frac{\partial\psi}{\partial\nu}\,d\sigma$$

$$0 = \int_\Omega \varphi\Delta\psi - \psi\Delta\varphi\,dV + \int_{\partial\Omega}\left[\psi\frac{\partial\varphi}{\partial\nu} - \varphi\frac{\partial\psi}{\partial\nu}\right]d\sigma.$$

∎

Corollary 16.3: Subject to vanishing Dirichlet (or Neumann) boundary conditions, Δ is formally self-adjoint.

$$\langle \Delta\varphi, \psi \rangle_{L^2(\Omega)} = \langle \varphi, \Delta\psi \rangle_{L^2(\Omega)}.$$

Proof:

$$\langle \Delta\varphi, \psi \rangle_{L^2(\Omega)} = \int_\Omega \psi \Delta\varphi \, dV$$

$$= \int_\Omega \varphi \Delta\psi \, dV - \int_{\partial\Omega} \left(\varphi \frac{\partial \psi}{\partial \nu} - \psi \frac{\partial \varphi}{\partial \nu} \right) d\sigma.$$

($\varphi = 0$, $\frac{\partial \varphi}{\partial \nu} = 0$ by Dirichlet)

$$= \langle \varphi, \Delta\psi \rangle_{L^2(\Omega)}.$$

□

16.ii) The Spectrum:

Def 16.4: The Spectrum of Δ on Ω (more generally of any

$$L = \sum_{i,j} a_{ij} \partial_i \partial_j + \sum_k b_k \partial_k + c$$

on Ω is

$$\mathrm{Spec}(L) = \{ \lambda \in \mathbb{C} \mid \lambda \text{ is an eigenvalue of } L, \text{ i.e. } \exists \varphi \text{ s.t. } L\varphi = \lambda\varphi \}.$$

Prop 16.5: $\mathrm{Spec}(\Delta)$ lies on the negative real axis.

Proof: If $\Delta\varphi = -\lambda\varphi$ then ($\varphi : \Omega \to \mathbb{C}$ here) (self adjoint)

$$\lambda \langle \varphi, \varphi \rangle_{L^2} = \langle \Delta\varphi, \varphi \rangle_{L^2} = \langle \varphi, \Delta\varphi \rangle_{L^2} = \langle \varphi, \lambda\varphi \rangle = \bar{\lambda} \langle \varphi, \varphi \rangle$$

so $\lambda = \bar{\lambda}$ and λ is real. $-\lambda \leq 0$ by Thm 5.9.

□

Prop 16.6: If φ_1, φ_2 are eigenfunctions with distinct eigenvalues, then $\langle \varphi_1, \varphi_2 \rangle_{L^2} = 0$.

Proof:
$$\lambda_1 \langle \varphi, \psi \rangle_{L^2} = \langle \Delta\varphi, \psi \rangle_{L^2}$$
$$= \langle \varphi, \Delta\psi \rangle_{L^2}$$
$$= \langle \varphi, \lambda_2 \psi \rangle_{L^2} = \lambda_2 \langle \varphi, \psi \rangle_{L^2}$$
$$(\lambda_1 - \lambda_2)\langle \varphi, \psi \rangle_{L^2} = 0$$
$\neq 0$ so $= 0$.

□

Thm 16.7: (The Spectral Theorem for self-adjoint elliptic operators)

Suppose Ω is closed and bounded in $\mathbb{R}^d$, and
$$L = \sum_{ij} a_{ij}(x, \dots)\partial_i\partial_j + \sum_k b_k(x, \dots)\partial_k + c(x, \dots)$$
is a self-adjoint elliptic operator.

Then
1. $\mathrm{Spec}(L) \subseteq \mathbb{R}$ is discrete, countable, and bounded above
2. The eigenfunction $\varphi_1, \varphi_2, \dots$ of L form an orthonormal basis of $L^2(\Omega; \mathbb{R})$.

Proof: In a few weeks. Needs more functions spaces + operator theory.

Ex 16.8: $\Omega = [0, L]$ we saw $\left(\frac{\partial}{\partial x}\right)^2 \varphi = -\lambda\varphi$ so
$$-\lambda = \left(\frac{\pi}{L}\right)^2, \left(\frac{\pi \cdot 2}{L}\right)^2, \dots \left(\frac{\pi n}{L}\right)^2 \leq 0.$$
and the orthogonormal basis is Fourier Sine series.

Ex 16.9: $\Omega = [0, L_1] \times [0, L_2]$
$$\mathrm{Spec}(\Delta) = \{-\lambda\} = \left\{ \left(\frac{n\pi}{L_1}\right)^2 + \left(\frac{m\pi}{L_2}\right)^2 \mid n, m \geq 0 \right\}.$$
and the orthonormal basis is
$$\sin\left(\frac{n\pi}{L_1}x\right)\sin\left(\frac{m\pi}{L_2}y\right).$$

Ex 16.10 : We saw last time that for $\Omega = D_{r_0}$

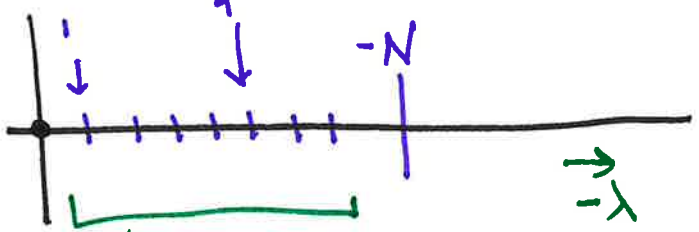
$$\left(\left(\frac{\partial}{\partial r}\right)^2 + \frac{1}{r}\frac{\partial}{\partial r} + \frac{1}{r^2}\frac{\partial^2}{\partial \theta^2}\right)\varphi = -\lambda\varphi$$

then

$$\mathrm{Spec}(\Delta|_{D_{r_0}}) = \left\{ -\left(\frac{z_k^{(n)}}{r_0}\right)^2 \;\middle|\; z_k^{(n)} \text{ is } k^{\text{th}} \text{ zero of } J_n \right\}.$$

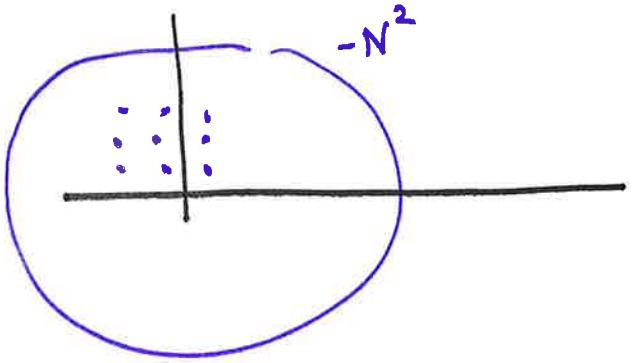
and the orthonormal basis was Fourier-Bessel Series.

16.iii) The Weyl Law



$\#\lambda \geq -N \sim \mathcal{O}(\sqrt{N})$.

Δ on $[0,L]$



$\#\lambda^2 \leq -N^2$

$\sim \mathcal{O}(N^2)$ (area)

so $\#\lambda \geq -N$

Let $N(\lambda) = \{ \#\lambda \mid |\lambda| \leq N \}$. for Δ.

Thm 16.11 (Weyl Law). For a closed, bounded $\Omega \subseteq \mathbb{R}^d$,

$$N(\lambda) \sim \lambda^{d/2} \quad \text{as } N \to \infty.$$

In fact

$$\lim_{N\to\infty} \frac{N(\lambda)}{\lambda^{d/2}} = \frac{1}{(2\pi)^d} \cdot \mathrm{Vol}(B_1(0)) \cdot \mathrm{Vol}(\Omega).$$

Rem 16.12 : $\Delta = \nabla^2 = \sum \partial_{x_i}^2$ ← analysts Laplacian

$\Delta = -\nabla^2 = -\sum \partial_{x_i}^2$ ← geometers Laplacian. ✓

I feel very strong that is the correct convention. Alas, book uses first.

Lecture 17 The Hydrogen Atom

17.i) A crash course on quantum mechanics

Idea 17.1: at microscopic scales, the motion of particles cannot be known deterministically, only probabilistically.

Def 17.2: the wavefunction of an electron $\Psi: \Omega \times \mathbb{R}^{\geq 0} \to \mathbb{C}$ is a complex function in the Hilbert space

$$\Psi \in L^2(\Omega : \mathbb{C}).$$

Interpretation 17.3 (Born, 1926) Since $\Psi \in L^2$ we can normalize so that

$$\int_{\Omega} |\Psi(t,x,y,z)|^2 \, dx\,dy\,dz = 1 \quad \forall t.$$

In this case, for a region $A \subseteq \Omega$ we interpret

$$\|\Psi(t,-)\|_{L^2}^2 = \int_A |\Psi(t,x,y,z)|^2 \, dV = \left\{ \begin{array}{l} \text{probability to find the electron} \\ \text{in } A \text{ at time } t \end{array} \right\}.$$

Def 17.4: The time-dependent Schrödinger equation shows how Ψ evolves.

$$i\hbar \frac{d\Psi}{dt} = H\Psi$$

(second-order elliptic)

where H is the Hamiltonian, the 2nd order linear elliptic PDE

$$H\Psi = \left[\underbrace{\frac{-\hbar^2}{2m}\Delta}_{\text{Kinetic}} + \underbrace{V(t,x,y,z)}_{\text{Potential}} \right] \Psi.$$

By the Spectral theorem, since H is self-adjoint, we deduce there is an orthonormal basis Ψ_n, $n = 1, 2, 3, \ldots$

$$H\Psi_n = E_n \Psi_n$$

where E_n are discrete, countable, bounded below.

Def 17.5: In this case,

Ψ_n are called the eigenstates

eigenvalue → E_n are called the energy levels

Thm 17.6 (general solution) The solution of the IVP

$$\begin{cases} i\hbar \frac{\partial \Psi}{\partial t} = H\Psi \\ \Psi(0,x,y,z) = \Psi_0 \end{cases}$$

is

$$\Psi = \sum_{n=1}^{\infty} C_n e^{-iE_n t/\hbar} \Psi_n. \qquad \text{where } \Psi_0 = \sum_{n=1}^{\infty} c_n \Psi_n.$$

$$\sum_{n=1}^{\infty} |c_n|^2 = 1.$$

where Ψ_n are the eigenfunction of H w/ Dir/Neumann BC.

Prop 17.7: the condition that $\|\Psi\|_{L^2} = 1$ is preserved by the Schrödinger equation.

$$i\hbar \frac{d}{dt}\|\Psi\|_{L^2}^2 = i\hbar \frac{d}{dt}\int_\Omega \Psi \cdot \overline{\Psi}\, dV = i\hbar \frac{d}{dt}\langle \Psi, \Psi \rangle$$

$$= i\hbar \int_\Omega \partial_t \Psi \overline{\Psi} + \Psi \partial_t \overline{\Psi}\, dV$$

$$= \left\langle i\hbar \frac{\partial \Psi}{\partial t}, \Psi \right\rangle + \left\langle \Psi, -i\hbar \frac{\partial \Psi}{\partial t} \right\rangle$$

$i\hbar \overline{\partial_t \Psi} = \overline{-i\hbar \partial_t \Psi}$

$$= \langle H\Psi, \Psi \rangle + \langle \Psi, -H\Psi \rangle$$

$$= 0 \quad \text{by self-adjointness of } H.$$

17.ii) The Hydrogen Atom, setup



Potential energy from electric field of proton is

$$V(x,y,z) = -\frac{1}{r}$$

$$= -\frac{\alpha^2}{r}$$

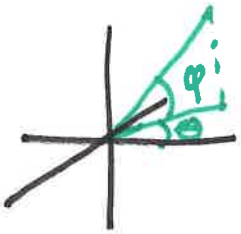
← general nucleus of any atom.

We therefore wish to find the eigenfunctions/values of

$$\left[-\frac{\hbar^2}{2m}\Delta - \frac{\alpha^2}{r}\right]\Psi = E\Psi.$$

Suppose $\Psi(r,\varphi,\theta)$ is in polar coordinates. Assume

$$\Psi(r,\varphi,\theta) = R(r)Y(\varphi,\theta).$$



- BC is that $\|\Psi\|_{L^2} < \infty$ on $\Omega = \mathbb{R}^3$, ie Dirichlet BC at $r = \infty$.

Lemma 17.8: In polar coordinates

$$\Delta = \left(\frac{\partial}{\partial r}\right)^2 + \frac{2}{r}\frac{\partial}{\partial r} + \frac{1}{r^2}\underbrace{\left(\frac{\partial^2}{\partial\varphi^2} + \frac{\cos(\varphi)}{\sin(\varphi)}\frac{\partial}{\partial\varphi} + \frac{1}{\sin(\varphi)^2}\frac{\partial^2}{\partial\theta^2}\right)}_{:=\Delta_{S^2}}.$$

Proof: change of variables.

Lemma 17.9: If Ψ is an eigenvalue then $R(r)$ and $Y(\varphi,\theta)$ individually solve

$$\Delta_{S^2}Y(\varphi,\theta) = -\mu Y(\varphi,\theta)$$

$$\frac{\hbar^2}{2m}\left[r^2R'' + 2rR' + \left(-Er^2 + \alpha^2 r - \mu\right)R\right] = 0.$$

Proof:

$$R''(r)Y(\varphi,\theta) + \frac{2}{r}R'(r)Y(\varphi,\theta) + \frac{1}{r^2}\left[\Delta_{S^2}Y(\varphi,\theta)\right]R(r) = -ER(r)Y(\varphi,\theta)$$
$$\frac{\alpha^2}{r}R(r)Y(\varphi,\theta)$$

$$\frac{r^2R''(r) + 2rR' + \alpha^2 rR + Er^2R}{R} = -\frac{\Delta_{S^2}Y(\varphi,\theta)}{Y(\varphi,\theta)} = \mu. \qquad \square$$

Rem 17.10: Both these equations are hard, the first one is asking if we can solve a 2d elliptic equation but

$$\Omega = \{\text{surface of Sphere}\}.$$

$\mu \geq 0$ * by similar positivity as Thm 5.9.

17.iii) Solving the angular equation

The third separation

$$\Delta_{S^2} Y(\varphi,\theta) = -\mu\, Y(\varphi,\theta) \qquad 0 \le \theta \le 2\pi \text{ periodic}, \quad 0 \le \varphi \le \pi.$$

Assume

$$Y(\varphi,\theta) = p(\varphi)q(\theta).$$

Lemma 17.11 : The above is satisfied if and only if p, q obey

1) $q''(\theta) + \nu q(\theta) = 0$

2) $\sin^2\varphi\, p''(\varphi) + \cos(\varphi)\sin(\varphi)\, p'(\varphi) + \left(\mu\sin^2(\varphi) - \nu\right)p(\varphi) = 0.$

Proof :

$$\left(\frac{\partial^2}{\partial\varphi^2} + \frac{\cos(\varphi)}{\sin(\varphi)}\frac{\partial}{\partial\varphi} + \frac{1}{\sin^2(\varphi)}\frac{\partial^2}{\partial\theta^2}\right)p(\varphi)q(\theta) = -\mu\, p(\varphi)q(\theta).$$

$$\frac{\sin^2\varphi \cdot p'' + \cos\varphi\sin\varphi\, p' + \mu\sin^2\varphi\, p}{p} = -\frac{q''(\theta)}{q(\theta)} = \nu.$$

□.

Lemma 17.12 : (Solving the θ) equation

$$q(\theta) = e^{im\theta} \qquad \nu = m^2 \qquad m = 0, 1, -1, 2, -2, \ldots$$

Proof : periodic boundary conditions, same as before. Recall

$$\cos\theta = \frac{e^{i\theta} + e^{-i\theta}}{2}$$

$$\sin\theta = \frac{e^{i\theta} - e^{-i\theta}}{2i}.$$

□

Lecture 18 The Hydrogen Atom Part II.

Recall we're finding the eigenfunctions / values of

$$\left[\frac{\hbar^2}{2m}\Delta + \frac{\alpha^2}{r}\right]\Psi = -E\Psi.$$

We found if

$$\Psi(r,\varphi,\theta) = R(r)Y(\varphi,\theta)$$

then

- $\Delta_{S^2} Y(\varphi,\theta) = -\mu Y(\varphi,\theta)$
- $r^2R''(r) + 2rR'(r) + [-Er^2 + \alpha^2 r - \mu]R(r) = 0.$

18.1) The angular eigenfunction continued

If $Y(\varphi,\theta) = p(\varphi)q(\theta)$ we found

- $q''(\theta) = -\nu q(\theta) \Rightarrow q(\theta) = e^{im\theta}$ $\quad \nu = m^2$, $m = -1, 0, 1, 2, \ldots$
- $p(\varphi)$ solves

$$\sin^2\varphi \cdot p''(\varphi) + \cos\varphi \sin\varphi\, p'(\varphi) + [\mu\sin^2\varphi - m^2]p(\varphi) = 0.$$

Substitution 18.1 : Let $t = \cos\varphi$ so $-1 \le t \le 1$

$$p(\varphi) = P(t).$$

then

$$(1-t^2)^2 P''(t) - 2t(1-t^2)P'(t) + [\mu(1-t^2) - m^2]P = 0.$$

called the Legendre ODE.

Proof : $t = \cos\varphi$, $\sin\varphi = \sqrt{1-t^2}$

$$\frac{dp}{d\varphi} = \frac{dP}{dt}\frac{dt}{d\varphi} = -\sin\varphi\frac{dP}{dt} = -\sqrt{1-t^2}\frac{dP}{dt}$$

$$\frac{d^2p}{d\varphi^2} = -\sin\varphi\frac{d}{dt}\left[-\sqrt{1-t^2}\frac{dP}{dt}\right] = (1-t^2)P''(t) - tP'(t).$$

and substituting gives the ODE for $P^m(t)$.

Thm 18.2 (Legendre): In the case that $m=0$, the only solutions obeying

$$|P(1)| < \infty \,, \quad |P(-1)| < \infty$$

are for $\mu = [redacted]$ an
$= \ell(\ell+1)$ for $\ell = 0, 1, 2, \ldots$

Proof: Assume that $P(t) = \sum_{k=0}^{\infty} c_k t^k$ ← by Stone-Weierstrauss this can converge in L^2!

Substituting and solving term by term shows that

$$c_{k+2} = - \frac{\mu - k(k+1)}{(k+1)(k+2)} c_k .$$

Therefore, unless $\mu = k(k+1)$ for some k, $c_{k+2} \to 1$ and $|P(1)| = \infty$.

If $\mu = (k+1)k$, then series truncates. The solutions are the Legendre polynomials

$$P_\ell(t) = \frac{(-1)^\ell}{2^\ell \ell!} \left(\frac{d}{dt}\right)^\ell (1-t^2)^\ell \qquad \ell = 0, 1, 2, \ldots$$

Thm 18.3 (Legendre $m>0$). For $m \neq 0$ the only solutions obeying $|P(1)| < \infty$ $|P(-1)| < \infty$ are

$$P_\ell^m(t) = (-1)^\ell \frac{(1-t^2)^{m/2}}{2^\ell \ell!} \frac{d^{\ell+m}}{dt^{\ell+m}} (1-t^2)^\ell$$

for $\ell = 0, 1, 2$

$$-\ell(\ell+1) \leq m \leq \ell(\ell+1)$$

Putting this back together.

Def 18.4 : the eigenfunctions $Y_\ell^m(\varphi,\theta)$ obeying

$$\Delta_{S^2} Y_\ell^m(\varphi,\theta) = -\ell(\ell+1) Y_\ell^m(\varphi,\theta)$$

are called the spherical harmonics.

$\mu = 0$	$\ell = 0$	m
$\mu = 2$	$\ell = 1$	$m = -1, m = 0, m = 1$
$\mu = 6$	$\ell = 2$	$m = -2, -1, 0, 1, 2.$

Thm 18.5 : The spherical harmonics $Y_\ell^m(\varphi,\theta)$ for

$\ell = 0, 1, 2,$

$m = -\ell(\ell+1), \ldots, \ell(\ell+1)$

form an orthonormal basis of $L^2(S^2; \mathbb{C})$.

Ex 18.6 :

$$Y_0^0 = \frac{1}{2}\sqrt{\frac{1}{\pi}}$$

$$Y_1^{-1} = \frac{1}{2}\sqrt{\frac{3}{2\pi}} \sin\varphi\, e^{-i\theta}$$

$$Y_1^0 = \frac{1}{2}\sqrt{\frac{3}{\pi}} \cos\varphi$$

$$Y_1^1 = \frac{1}{2}\sqrt{\frac{3}{2\pi}} \sin\varphi\, e^{i\theta}$$

$$\vdots$$

$$Y_8^3 = \frac{1}{64}\sqrt{\frac{19635}{2\pi}} \left[\sin^3\varphi\left(39\cos^5\varphi - 26\cos^3\varphi + 3\cos\varphi\right)\right] \cdot e^{3i\theta}.$$

Rem 18.7 : this same derivation can be done many increasingly fancy ways

- harmonic polynomials
- representations of $SO(3)$
- $\vdots$

18.ii) Solving the radial equation

$$r^2R'' + 2rR' + [-Er^2 + \alpha^2 r - \mu]R = 0.$$

Set $s = \sigma r$, $\sigma = \frac{2\sqrt{2mE}}{\hbar}$

Rescaling, $R(r) = p(\frac{s}{\sigma})$ where p solves

$$p'' + \frac{2}{s}p' - \left(\frac{1}{4} \cdot \frac{n}{s} + \frac{\mu}{s^2}\right)p = 0. \qquad n = \frac{2m\alpha^2}{\sigma\hbar^2}$$

Lemma 17.11: (a slightly tedious substitution)

If $p(s) = s^{\ell} e^{-s/2} q(s)$

solves, then $q(s)$ solves above

$$s\,q''(s) + (2+2\ell-s)q'(s) + (n-\ell-1)q(s) = 0$$

(associated) Laguerre equation.

where $\mu = \ell(\ell+1)$.

Proof: product rule + substitute.

Thm 17.12: The only solutions to the above equation obeying

$$|q(0)| < \infty \qquad \lim_{s\to\infty} q(s) = 0$$

occur if n is an integer. They're given by

$$L^{2\ell+1}_{n-\ell+1}(s) = \text{[redacted]}$$

where

$$L^j_k(s) = \frac{s^{-j}e^s}{k!}\left(\frac{d}{ds}\right)^k [s^{j+k}e^{-s}]$$

Conclusion 17.13 • E_n is discrete, with $n = \frac{2m\alpha^2}{2\sqrt{2mE}}\frac{\hbar}{\hbar^2}$ $n = 1,2,3,\ldots$

$$= -\frac{\alpha^2}{2a}\frac{1}{n^2} \qquad a = \frac{\hbar^2}{m\alpha^2} = .529\cdot 10^{-10}\,m \text{ Bohr radius.}$$

$$\psi_{nlm}(r,\varphi,\theta) = C_{nlm} \cdot \left(\frac{2r}{na}\right)^{\ell} e^{-r/na} L^{2\ell+1}_{n-\ell-1}\left(\frac{2r}{na}\right) \cdot Y^m_{\ell}(\varphi,\theta)$$

Lecture 20 Numerical PDEs I : Finite Differences.

As we have seen, many ODEs/PDEs are too hard to solve by hand, and others may not have analytic solutions.

Def 20.1 : A numerical method is an algorithm implemented on a computer which calculates an approximate representation of the solution of a PDE.

Questions 20.2 : 1) How accurate is the approximation? Is it biased?
2) How efficient is the algorithm?
3) How do 1)-2) scale as we increase e.g. dimension, resolution?

20.1) Discretizations of function Spaces:

Function spaces like $L^2(\Omega : \mathbb{C})$ are infinite-dimensional, but computers handle finite data.

Def 20.3 : A discretization method is a linear map

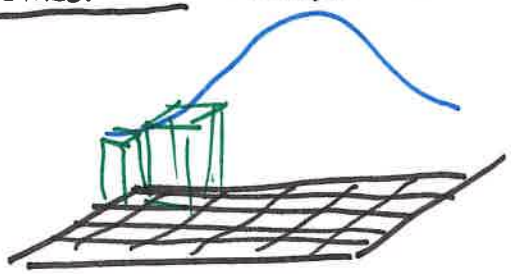
$$\mathcal{D} : \mathbb{R}^N \longrightarrow (H, \langle\ \rangle)$$

from a large finite dimensional vector space

$$\sum a_i e_i \longrightarrow f(x).$$

And a projection $\pi : H \longrightarrow \mathrm{Im}(\mathbb{R}^N)$.

Method 20.4 (Spatial Methods)



$$\mathrm{Im}(\mathbb{R}^N) = \left\{ \begin{array}{l} \text{Piecewise constant functions} \\ \text{on square grid} \end{array} \right\}.$$

$$\chi_i = \begin{cases} 1 & \text{on } Q_{ij} \\ 0 & \text{else.} \end{cases} \qquad \begin{array}{l} i = 1, \dots m \\ j = 1, \dots m \\ N = m^2 \end{array}$$

$$\mathcal{D}\left(\sum_{\substack{i=1 \\ j=1}}^{m} a_{ij} e_{ij} \right) = \sum_{\substack{i=1 \\ j=1}}^{m,m} a_{ij} \chi_{ij}.$$

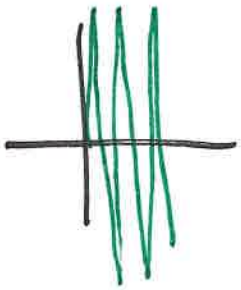
And $\pi(f) = A_{ij} e_{ij}$ $\qquad A_{ij} = \langle f, \chi_{ij} \rangle_{L^2}$

$$= \int_{a_{ij}} f \, dV.$$

Notation: $\pi(f) = \underline{f}$ is the discretized version.

Remark 20.5: The error of this method can be arbitrarily high

On $\Omega = [0,10] \times [0,10]$. Take $m = 10$, $f = 10^{10^{10}} \sin(100x)$.



$$\Rightarrow \int_{a_{ij}} f \, dV = 0 \quad \text{for all } ij$$

but obviously

$$\| f - \underline{f} \|_{L^2(\Omega)} = \text{very large}.$$

$\Rightarrow$ choosing appropriate resolution requires knowledge of lengthscales.

Rem 20.6: Many standard methods are spatial

- Finite difference (simplest, boxes)
- Finite volume (non-square, good energy/mass conservation)
- Finite element (graph structures ).

Method 20.7: (Spectral Methods)

$$\mathbb{R}^N \longrightarrow L^2(\Omega; \mathbb{C})$$

Let f_i be an orthonormal basis.

$$\mathcal{D}(\Sigma a_i e_i) \longrightarrow \underline{f} = \Sigma a_i f_i$$

For example, $m = 100$, $N = m^2$, $\Omega = [0,\pi] \times [0,\pi]$.

$$\Sigma a_{ij} e_{ij} \longrightarrow \sum_{ij} a_{ij} \cos(i \cdot x) \cos(j \cdot y) \qquad \text{(Fourier cosine series)}$$

Other bases also okay (spherical harmonics, Fourier-Bessel)

20.ii) Discretizations of Operators

Recall $$L = \sum_{ij} a_{ij}(x)\partial_i\partial_j + \cdots + c(x)$$

is a linear operator acting on functions and returning functions.

The approximation of the operator becomes a linear map

$$\underline{L} : \mathbb{R}^N \longrightarrow \mathbb{R}^N.$$

Def 20.8 : It doesn't have to be the case that $\mathcal{D}\underline{L} = L\mathcal{D}$

$$\begin{array}{ccc} L^2(\Omega;\mathbb{C}) & \xrightarrow{L} & L^2(\Omega;\mathbb{C}) \\ \pi \downarrow \uparrow \mathcal{D} & & \downarrow \pi \uparrow \mathcal{D} \\ \mathbb{R}^N & \xrightarrow{\underline{L}} & \mathbb{R}^N \end{array}$$

← going around different ways isn't same!
• mixing of Fourier modes.

Ex 20.9 : The Laplacian in the spectral basis.

$\Omega = [0,\pi]$, $\mathcal{D} : \mathbb{R}^N \to L^2(\Omega;\mathbb{R})$ is Fourier sine series.

$$\underline{\Delta} = \begin{pmatrix} \cdot & & \\ & \cdot & \\ & & \cdot \end{pmatrix} \qquad \underline{\Delta} = \pi\Delta.$$

π : {truncate Fourier series at N}.

$e_j = \sin(jx)$

$V_1 = \underline{\Delta} e_1$

$$V_1 = \underline{\Delta} e_1 = \pi\Delta(\sin x) = \pi(-1\sin x) = \begin{matrix} -1 e_1 \\ -4 e_2 \end{matrix}$$

$$= \begin{pmatrix} -1 & 0 & & & \\ 0 & -4 & & & \\ 0 & 0 & -9 & & \\ \vdots & \vdots & & -16 & \\ & & & & \ddots \end{pmatrix} \begin{pmatrix} u_1 \\ u_2 \\ \vdots \\ \vdots \end{pmatrix} = \begin{pmatrix} b_1 \\ \vdots \\ \vdots \\ b_N \end{pmatrix}$$

Ex 20.10 : Solve $\underline{\Delta}\underline{u} = \underline{g}$ where $\underline{g} = \sum_{j}^{N} b_j \sin(jx) = \pi g = \pi\left(\sum_{n=1}^{\infty} b_n \sin(nx)\right)$

$$\underline{u} = \sum_{n=1}^{N} \frac{b_j}{-n^2} \sin(nx)$$

Ex 20.10 (continued)

We know the true solution for $g = \sum_{n=1}^{\infty} b_n \sin(nx)$ is

$$u = \sum_{n=1}^{\infty} \frac{b_n}{-n^2} \sin(nx).$$

So the error is

$$\|u - \underline{u}\|_{L^2} = \left\| \sum_{n=1}^{\infty} \frac{b_n}{-n^2} \sin(nx) - \sum_{n=1}^{N} \frac{b_n}{-n^2} \sin(nx) \right\|_{L^2}$$

$$= \left\| \sum_{n=N+1}^{\infty} \frac{b_n}{-n^2} \sin(nx) \right\|_{L^2} \leq \frac{1}{(N+1)^2} \left\| \sum b_n \sin(nx) \right\|_{L^2}$$

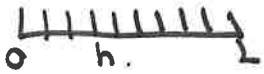
$$\leq \frac{\|g\|_{L^2}}{N^2}. \qquad \square$$

20.iii)

Finite Differences: Consider $f: [0, L] \to \mathbb{R}$.

$$f'(x)\Big|_{x=x_0} = \lim \frac{f(x_0+h) - f(x_0)}{h}$$

$$\approx \frac{f(x_0 + h) - f(x_0)}{h}$$



$$f'(x_n) = \frac{f(x_{n+1}) - f(x_{n-1})}{2h} \quad \text{or} \quad \frac{f(x_{n+1}) - f(x_n)}{h} \quad \text{or} \quad \frac{f(x_n) - f(x_{n-1})}{h}.$$

$$f''(x_n) = \frac{f'(x_{n+1}) - f'(x_n)}{h} = \frac{\left[\frac{f(x_{n+1}) - f(x_n)}{h}\right] - \left[\frac{f(x_n) - f(x_{n-1})}{h}\right]}{h}$$

$$= \frac{f(x_{n+1}) - 2f(x_n) + f(x_{n-1})}{h^2}$$

Now $e_n = y_n$. So

$$\underline{\Delta} e_n = [\underline{\Delta} y_n] = \frac{y_{n+1} - 2y_n + y_{n-1}}{2h}.$$

$$\underline{\Delta} = \begin{pmatrix} -2 & 1 & & \\ 1 & -2 & 1 & \\ & 1 & \ddots & \\ & & & \end{pmatrix}$$

Lecture 21 Numerical Methods II

Consider an ODE of the form

$$\frac{df}{dt} = F(f,t).$$

21.i) Numerical Methods for ODEs

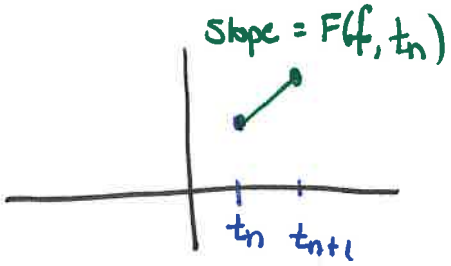
Consider $\Omega = [0,T]$, we saw $\mathbb{R}^N \longrightarrow L^2([0,T])$ finite spatial discretization.



$$f'(t) \approx \frac{f(t_{n+1}) - f(t_n)}{\tau}$$

by the ODE

$$\frac{f(t_{n+1}) - f(t_n)}{\tau} = F(f, t_n).$$



$\Rightarrow$ Two options, f_n or f_{n+1}.

Def 21.1: The Explicit Euler method yields an approximate solution $f = f(t_n)$ by

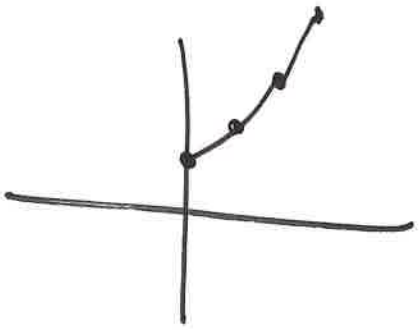
$$f(t_{n+1}) = f(t_n) + \tau \cdot F(f(t_n), t_n).$$

solved recursively.

Ex 21.2: $\dot{f}(t) = \alpha f \quad f(0) = 1. \quad \tau = 0.1.$

$= 3f.$

$f(0) = 1$	$F(f(0)) = 3$
$1 + .1 \cdot 3 = 1.3$	3.9
1.69	
2.197	
⋮	



Def 21.3 : the Implicit Euler method yields an approximate solution by

$$\frac{f(t_{n+1}) - f(t_n)}{\tau} = F(f(t_{n+1}))$$

$$f(t_{n+1}) - \tau \cdot F(f(t_{n+1})) = f(t_n).$$

i.e. $f(t_{n+1})$ solves the (nonlinear) equation $\quad \Phi \quad [1 - \tau \cdot F(\ )]\, f(t_{n+1}) - f(t_n) = 0.$

Ex 21.4 : For a vector-valued ODE, both methods work equally well.

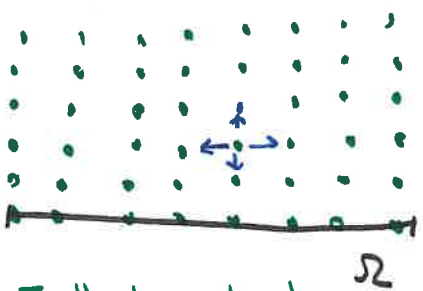
$$f'(t) = Af \qquad A = \begin{pmatrix} 3 & 1 & 0 \\ 0 & 2 & -1 \\ 0 & 1 & 1 \end{pmatrix}$$

$$f(0) = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} \qquad \tau = 0.01.$$

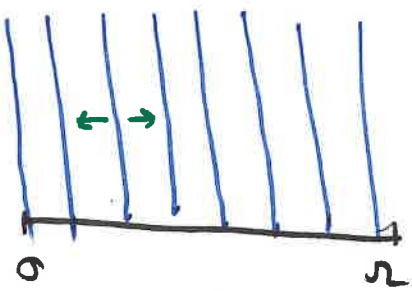
$$f(0.01) = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} + (0.01)\, A \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} \approx$$

21.ii) The Method of lines

Consider $(\partial_t - L)u = 0$ on $[0,\infty) \times \Omega$.



Full discretization



space only discretization.

$L^2(\Omega;\mathbb{C}) \xleftrightarrow{\Phi} \mathbb{R}^N$ discretization. $\qquad \underline{L} : \mathbb{R}^N \to \mathbb{R}^N.$

Def 21.5 : The method of lines is the reduction of a PDE to a (large) vector-valued ODE by applying spatial only discretization.

$\Rightarrow$ Any ODE solving can by applied to $\underline{u}_t = \underline{L}\,\underline{u}$ ← works for spatial and nonlinear!

Ex 21.6 : We saw that $\Delta = \begin{pmatrix} -1 & & \\ & -4 & \\ & & -9 \\ & & & \ddots \end{pmatrix}$ so the heat eq on $[0,\infty) \times [0,\pi]$

becomes $\underline{u} : \mathbb{R}^N \to \mathbb{R}$ solves

$$\frac{d}{dt}\underline{u} = \begin{pmatrix} -1 & & \\ & -4 & \\ & & -9 \\ & & & \ddots \end{pmatrix} \underline{u}$$

and euler's method reproduces discrete ODE approximations of $e^{-\lambda_n t}$ in each mode, then remember

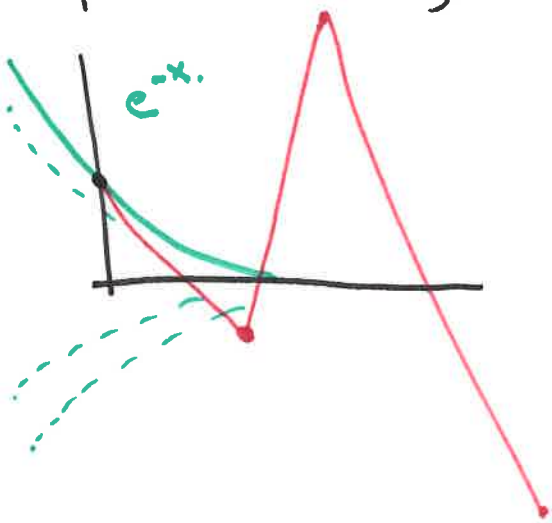
$$\underline{u} = \sum_{n=1}^{N} \underline{u}_n^{(t)} \sin(nx).$$

Similarly for finite differences

$$\frac{d}{dt}\underline{u} = \begin{pmatrix} -2 & 1 & & \\ 1 & -2 & 1 & \\ & 1 & \ddots & \ddots \end{pmatrix} \underline{u} \qquad \underline{u} = \sum_{n=1}^{N} \underline{u}_n \chi_n.$$

21.iii) Stiffness

Consider the simplest ODE w/ decay $f' = -f$.



Solving w/ an explicit method w/ large timestep leads to large instability

Thm 21.7 : The (numerical approx) solution $\underline{f}$ of the equation $f' = -\lambda f$ $\lambda > 0$ has numerical error

- (Explicit Euler) $\sim O(\tau)$ Provided $\tau \leq \frac{1}{2\lambda}$
- (Implicit Euler) $\sim O(\tau)$ no requirement!

Proof (of τ value) $\quad f' = -\lambda f.$

$$f_0 \to f_0 + \tau \cdot -\lambda f_0 = (1-\tau\lambda) f_0$$

$$f_1 \to f_1 + \tau \cdot -\lambda f_1 = (1-\tau\lambda)^2 f_0$$

So $f_n = (1-\tau\lambda)^n f_0$. diverges unless

$$|1-\tau\lambda| \leq 1$$

$$-1 \leq 1-\tau\lambda \leq 1$$

$$0 \leq \tau\lambda \leq 2$$

$$\tau \leq \frac{2}{\lambda}.$$

Def 21.8 : the ~~ratio~~ value $\tau = \frac{1}{2\lambda}$ is called the stability threshold.

Ex 21.9 : If $u' = \begin{pmatrix} \lambda_1 & & & \\ & \lambda_2 & & \\ & & \ddots & \\ & & & \lambda_n \end{pmatrix} u$ then stability threshold is is $\tau \leq \frac{1}{2\lambda_{max}}$.

⇒ If eigenvalues large $\tau \ll 1$ is needed. In nonlinear problems, λ_{max} can change along the way!

Def 21.10 : An ODE system or PDE via method of lines is called stiff if the stability threshold is small. The ratio (rel. to timescale)

$$\left(\frac{\lambda_{min}}{\lambda_{max}}\right)^{-1}$$

is called the stiffness ratio

Ex 21.11 : $\Delta = \begin{pmatrix} -2 & 1 & & \\ 1 & -2 & \ddots & \\ & \ddots & \ddots & \end{pmatrix}$ has $\lambda \sim \begin{pmatrix} -1 & & & \\ & -4 & & \\ & & -9 & \\ & & & \ddots \\ & & & -n^2 \end{pmatrix}$

so at resolution τ, h need $\tau \leq \frac{1}{2h^2}$ for stability

⇒ Heat eq very stiff.

	Speed	Stability	Use
Explicit	✓	×	Turbulent, fast evolving dynamics needed to resolve
Implicit	×	✓	only need equilibrium, take $\Delta t = \tau$ → large.

Lecture 22| Maximum principles I: Poisson formula.

Reading: Olver 4.3

22.i) Motivation

Recall the Poisson problem

$$\begin{cases} \Delta u = f \\ u|_{\partial\Omega} = h \end{cases} \qquad (I)$$

on a compact domain $\Omega \subseteq \mathbb{R}^d$.
(closed + bounded)

Recall we know explicit solutions for Ω = square, disk:

Thm (9.?): The general solution of

$$\begin{cases} \Delta u = 0 \\ u|_{\partial\Omega} = h \end{cases}$$

on $\Omega = \{x^2+y^2 \le 1\}$ is

$$u = \left[\sum_{n=1}^{\infty} A_n r^n \cos(n\theta) + B_n r^n \sin(n\theta)\right] + A_0.$$

and, since $u|_{r=1} = A_0 + \sum A_n \cos(n\theta) + B_n \sin(n\theta)$, A_n, B_n are the Fourier series of h.

Moving forward, we wish to know the following things about solutions on a general Ω.

Questions 22.1: Is the solution of (I) unique for given (f,h).

Question 22.2: For general domains, must there be a solution for every choice of $f, h \in L^2$ $\quad (f \in L^2(\Omega : \mathbb{R})$, $h \in L^2(\partial\Omega : \mathbb{R})$.

Question 22.3: What qualitative properties must the solution have?

22.ii) The Poisson Formula

Recall we can write $z = x+iy$ $\quad \mathbb{R}^2 = \mathbb{C}$.

$$\mathrm{Re}(z^n) = r^n \cos(n\theta)$$
$$\mathrm{Im}(z^n) = r^n \sin(n\theta)$$

Def 22.4 : A harmonic polynomial is a polynomial $p(x,y): \mathbb{R}^2 \to \mathbb{R}$ such that

$$\Delta p = 0.$$

Ex 22.5 :
$$\mathrm{Re}(z^n) = \mathrm{Re}\left((x+iy)^n\right)$$
$$= \mathrm{Re}\left((x+iy)(x+iy)(x+iy)^{n-2}\right)$$
$$= \mathrm{Re}\left((x^2 + 2ixy + -y^2)(x+iy)^{n-2}\right)$$
$$= \ldots$$

$$\mathrm{Re}(z^2) = x^2 - y^2$$

$\mathrm{Re}(z^n)$	$\mathrm{Im}(z^n)$
1	0
x	y
$x^2 - y^2$	$2xy$
$x^3 - 3xy^2$	$3x^2y - y^3$

Fact 22.6 : Once we show that our general solution

$$u = A_0 + \sum_{n=1}^{\infty} A_n \cos(n\theta) + \ldots$$

is the only one, we know the above list is a complete list (ie a basis, since we can take linear combinations)

Fact 22.7 : By binomial theorem,

$$\mathrm{Re}(z^n) = x^n - \binom{n}{2}x^{n-2}y^2 + \binom{n}{4}x^{n-4}y^4 + \ldots$$
$$\mathrm{Im}(z^n) = \binom{n}{1}x^{n-1}y - \binom{n}{3}x^{n-3}y^3 + \binom{n}{5}x^{n-5}y^5 + \ldots$$

Question 22.8 : Can we simplify the process

$$\begin{cases} \Delta u = 0 \\ u|_{\partial\Omega} = h \end{cases}$$

on disk? 1) compute Fourier series of h
2) use general solution.

To be only a single integral.

22.iii) Aside: functions defined via integrals

For an integral $\int_0^1 f(x)dx$ is a number, if $f_h(x) = 3e^{hx}$ then answer is a value in $\mathbb{R}$ that depends on h

$$\int_0^1 f(x,y)dx = \{\text{function of } y\}$$ ← dummy variables of integration

Def 22.9 : A function defined by an integral formula is one such that $f: \Omega_1 \to \mathbb{R}$

$K(x,y,\tilde{x},\tilde{y}) : \Omega_1 \times \Omega_2 \to \mathbb{R}$ (function of doubled variables).

$(x,y,\tilde{x},\tilde{y}) \in \Omega_1 \times \Omega_2$ and $g: \Omega_2 \to \mathbb{R}$

$$f(x,y) = \int_{\Omega_2} K(x,y,\tilde{x},\tilde{y}) \cdot g(\tilde{x},\tilde{y})\, d\tilde{x}\, d\tilde{y}$$

x,y: real variables. $\tilde{x},\tilde{y}$: dummy variables integrated out

more generally $f(x_1 \ldots x_d) = \int_{\Omega_2 \subseteq \mathbb{R}^d} K(x_1, x_2 \ldots x_d, \tilde{x}_1, \ldots \tilde{x}_d)\, g(\quad)\, d\tilde{x}_1 d\tilde{x} \ldots d\tilde{x}_d$.

22.iv) Back to Poisson Formula

We will express sol. to

$$\begin{cases} \Delta u = 0 \\ u|_{\partial\Omega} = h \end{cases}$$

as an integral formula.

Thm 22.10 (Poisson) The solution of

$$\begin{cases} \Delta u = 0 \\ u|_{\partial\Omega} = h \end{cases}$$

on the unit disk $\{x^2+y^2 \le 1\}$ is given by

$$u(r,\theta) = \frac{1}{2\pi}\int_0^{2\pi} \frac{1-r^2}{1+r^2-2r\cos(\theta-\varphi)}\, h(\varphi)\, d\varphi.$$

(φ: dummy variable)

Def 22.11 : the "$K(x,y)$" function is called the Poisson Kernel

Proof :

$$u(r,\theta) = \frac{A_0}{2} + \sum_{n=1}^{\infty} A_n r^n \cos(n\theta) + B_n r^n \sin(n\theta)$$

$$= \frac{1}{2\pi}\int_0^{2\pi} h(\varphi)\,d\varphi + \sum_{n=1}^{\infty}\left[\int_0^{2\pi} h(\varphi)\cos(n\varphi)\,d\varphi \cdot \frac{r^n\cos(n\theta)}{\pi} + \int_0^{2\pi} h(\varphi)\sin(n\varphi)\,d\varphi \cdot \frac{r^n\sin(n\theta)}{\pi}\right].$$

$$= \frac{1}{\pi}\int_0^{2\pi} h(\varphi)\left[\frac{1}{2} + \sum_{n=1}^{\infty} r^n\left[\cos(n\theta)\cos(n\varphi) - \sin(n\theta)\sin(n\varphi)\right]\right] d\varphi.$$

$$= \frac{1}{\pi}\int_0^{2\pi} h(\varphi)\left[\frac{1}{2} + \sum_{n=1}^{\infty} r^n\left[\cos(n(\varphi-\theta))\right]\right] d\varphi.$$

And $\sum_{n=1}^{\infty} r^n\cos(n\alpha) = \mathrm{Re}\left(\frac{1}{2} + \sum_{n=1}^{\infty} z^n\right) = \mathrm{Re}\left(\frac{1}{2} + \frac{z}{1-z}\right)$ (Geometric Series $\sum r^n = \frac{1}{1-r}$ for $|r|<1$.)

$$= \mathrm{Re}\left(\frac{1+z}{2(1-z)}\right)$$

$z = re^{i\alpha}$

$$= \mathrm{Re}\left(\frac{(1+z)(1-\bar{z})}{2|1-z|^2}\right) = \frac{\mathrm{Re}\left(1+z-\bar{z}-|z|^2\right)}{2|1-z|^2}$$



some geometry

$$= \frac{1-|z|^2}{2|1-z|^2} = \frac{1-r^2}{2(1+r^2-2r\cos\alpha)}.$$

take $\alpha = (\theta-\varphi)$.

∎

Lecture 23 Maximum Principle II

Recall we showed the solution of

$$\begin{cases} \Delta u = 0 \\ u|_{\partial\Omega} = h \end{cases}$$

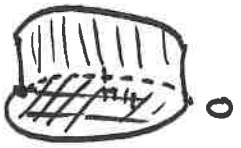
on $\Omega = B(0,1)$ is given by

$$u(r,\theta) = \frac{1}{2\pi}\int_0^{2\pi}\left[\frac{1-r^2}{1+r^2-2r\cos(\varphi-\theta)}\right]h(\varphi)\,d\varphi$$

or $\Omega = B(0,R)$

$$= \frac{1}{2\pi}\int_0^{2\pi}\left[\frac{R^2-r^2}{R^2+r^2-2rR\cos(\varphi-\theta)}\right]h(\varphi)\,d\varphi.$$

Ex 23.1 : $h(\theta) = \begin{cases} 1 & 0<\theta\le\pi \\ 0 & \text{else} \end{cases}$



Then $u(r,\theta) = \frac{1}{2\pi}\int_0^{\pi}\frac{1-r^2}{1+r^2-2r\cos(\varphi-\theta)}\,d\varphi$

$$= \begin{cases} 1 - \frac{1}{\pi}\tan^{-1}\left(\frac{1-r^2}{2r\sin\theta}\right) & 0<\theta\le\pi \\ \frac{1}{2} & \theta = 0,\pi \\ -\frac{1}{\pi}\tan^{-1}\left(\frac{1-r^2}{2r\sin\theta}\right) & \pi\le\theta<2\pi. \end{cases}$$

$\Rightarrow$ Answer by single harder integral (not ∞-many integrals depending on n).

23.i) The Mean-Value Theorem

Take $r=0$. The value at the center is

$$u(0,\theta) = \frac{1}{2\pi}\int_0^{2\pi}\left[\frac{R^2+0}{R^2+0-0}\right]h(\varphi)\,d\varphi = \frac{1}{2\pi}\int_0^{2\pi}h(\varphi)\,d\varphi.$$

Thm 23.2 : (Mean Value Property for Harmonic functions)

The $\Omega = B(x_0, R)$ and u is harmonic, ie $\Delta u = 0$ on Ω, then

$$u(x_0,y_0) = \frac{1}{2\pi R}\int_0^{2\pi} u(x_0+R\cos\theta, y_0+R\sin\theta)\, d\theta$$

$$= \frac{1}{2\pi R}\oint_C u\, ds. \qquad C = \partial R.$$

Proof: Consider the average

$$A(\rho) = \frac{1}{2\pi}\int_0^{2\pi} u(x_0+\rho\cos\theta, y_0+\rho\sin\theta)\, d\theta.$$

We claim A is constant, ie.

$$\frac{\partial}{\partial \rho}A(\rho) = \frac{1}{2\pi}\int_0^{2\pi} \cos\theta \frac{\partial u}{\partial x}\begin{pmatrix} x_0+\rho\cos\theta \\ y_0+\rho\sin\theta \end{pmatrix} + \sin\theta \frac{\partial u}{\partial y}\begin{pmatrix} x_0+ \\ y_0+\rho\sin\theta \end{pmatrix} d\theta$$

$$= \frac{1}{\pi R}\int_{\partial B(x_0;\rho)} \frac{\partial u}{\partial \nu}\cdot 1\, ds$$

$$= \frac{1}{2\pi R}\int_{B(x_0,\rho)} 1\cdot\Delta u - u\cdot\Delta 1\, dV = 0.$$

so $A(\rho)$ is const, and as $\rho \to 0$, u is assumed to be continuous, so

$$\to u(0).$$

Thm 23.3 (Mean Value Property for Balls)

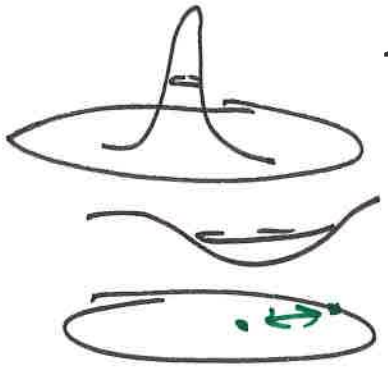
$$u(x_0,y_0) = \frac{1}{\pi R^2}\int_{B(x_0;R)} u\, dV$$

Proof :

$$\int_0^R u(x_0,y_0)\underbrace{dV}_{\rho\, d\rho} = \frac{1}{2\pi}\int_0^R\int_0^{2\pi} u(x_0+\rho\cos\theta, y_0+\rho\sin\theta)\, \rho\, d\theta\, d\rho.$$

$$\frac{R^2}{2} u(x_0,y_0) = \frac{1}{2\pi} \int_{B(x_0;R)} u \, dV$$

Interpretation 23.4: For a normal, arbitrary function can easily have $u \neq 0$, $u|_{\partial\Omega} = 0$.



For u harmonic there are strong constraints.

Center "talks to" the boundary. Remember harmonic means heat is balanced.

23.ii)

Maximum Principle:



Recall that at a crit point

$f'(x) = 0 \Rightarrow$ max, min or saddle

$f''(x) < 0$ max

$f''(x) > 0$ min.

for $f(x,y,z,..)$ must have $\nabla f = 0$

$\frac{\partial}{\partial x^2} f_{max} < 0 \quad \frac{\partial}{\partial y^2} f < 0 \ldots$

$\Rightarrow$ If (x,y,z) is a local max, cannot have

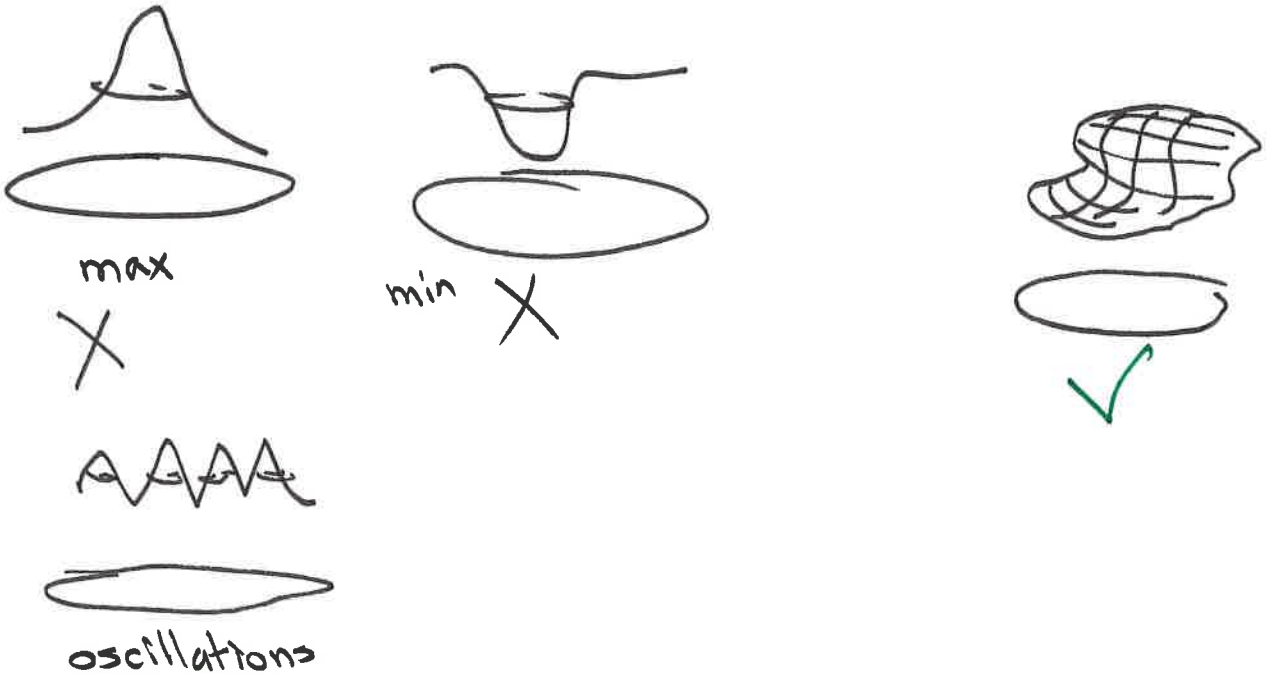
$$\Delta f = \sum \left(\frac{\partial}{\partial x_i}\right)^2 f < 0.$$

$\Rightarrow$ harmonic functions cannot have maximum, except on boundary.

Thm 23.5 (Maximum principle, harmonic functions).

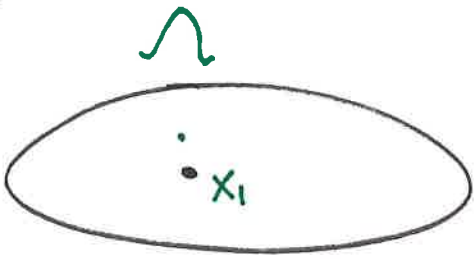
If a function $u: \Omega \to \mathbb{R}$ obeys $\Delta u = 0$, then the only time u achieves its max or min value is on $\partial\Omega$, unless $u = \text{const}$. (assume u continuous)

Ex 23.6 : the following qualitative behavior cannot happen for harmonic functions



Intuition 23.7 : Imagine the graph as a stretchy membrane held fixed at $u|_{\partial\Omega} = h$. Then the harmonic function snaps the membrane ~~tang~~ taut.

Proof of maximum principle: We prove by contradiction. Assume that $\overset{\text{local}}{\max}$ is obtained at x_1, y_1. Then for small radius



$$u(x,y) \leq u(x_1,y_1) \text{ on } B((x_1,y_1);\epsilon)$$

for small ϵ.

so

$$\frac{1}{2\pi}\int_{\partial B(\epsilon)} u(x,y) < u(x_1,y_1) \quad * \quad = \frac{1}{2\pi}\oint_{\partial B(\epsilon)} u\, ds$$

$u^{(x_1,y_1)}$ cannot be equal to the average of numbers all smaller than it! Unless all are constant.

Lecture 24 Uniqueness of Poisson problem

Recall we proved

Thm 23.5 (maximum principle)

If $u: \Omega \to \mathbb{R}$ obeys $\Delta u = 0$, then u only achieves an interior max/min if it is constant.

24.i) The shape of harmonic functions

The max principle places strong restrictions on harmonic functions.

Demonstration 24.1: Numerical examples of membrane problem.

On homework 4 you will prove the very famous

Thm 24.2 (Liouville) If $u: \mathbb{R}^2 \xrightarrow{\text{ord}} \mathbb{R}$ is harmonic and bounded, then u is constant.

Proof: Use mean value property on increasingly large disks

Ex 24.3: harmonic polynomials form an infinite-dimensional space of unbounded harmonic functions.

24.ii) Three theorems

Consider
$$\begin{cases} \Delta u = f \\ u|_{\partial\Omega} = g \end{cases}$$
on arbitrary $\Omega \subseteq \mathbb{R}^d$. We will show three strong theorems

i) uniqueness
ii) existence
iii) regularity

Thm 24.4: the solution u to the above is unique.

ie if $\begin{cases} \Delta u = f \\ u|_{\partial\Omega} = h \end{cases}$ and $\begin{cases} \Delta v = f \\ v|_{\partial\Omega} = h \end{cases}$

then $u = v$ on $\overline{\Omega}$.

Proof: Consider $u - v$. By linearity, it solves

$$\begin{cases} \Delta(u-v) = f - f = 0 \\ u-v|_{\partial\Omega} = h - h = 0. \end{cases}$$

$\Rightarrow$ u is harmonic and $=0$ on $\partial\Omega$. By max principle $(u-v)$ and $(v-u)$ both achieve maximum on boundary. Therefore

$$\begin{matrix} v-u \leq 0 \\ u-v \leq 0 \end{matrix} \Rightarrow \quad u - v \equiv 0.$$

□

Corollary 24.5: The solutions for Δ on disks/rectangles from Fourier series/ separation of variables and

$$u = \sum \frac{1}{\lambda_n} \langle f, \varphi_n \rangle \varphi_n$$

via eigenvalues are unique!

24.iii) **The existence**:

Thm 24.6: For $f \in L^2(\Omega : \mathbb{R})$ and $h \in H(\partial\Omega : \mathbb{R})$ (some Hilbert space $C^0 \subseteq H \subseteq C^2$ w/ technical conditions)

there is a solution to

$$\begin{cases} \Delta u = f \\ u|_{\partial\Omega} = h. \end{cases}$$

The proof has two lemmas.

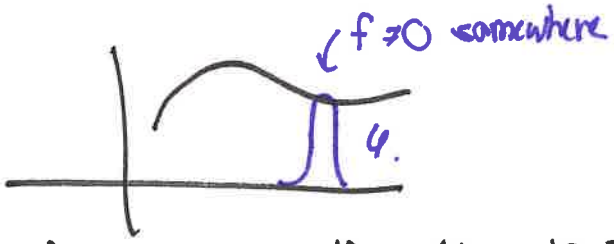
Lemma 24.7 (Fundamental Lemma of variations)

If for $f \in L^2(\Omega : \mathbb{R})$

$$\int_\Omega \varphi \cdot f \, dV = 0$$

for all $\varphi \in C^\infty_{cpct}(\Omega : \mathbb{R})$. Then $f = 0$.

Proof: Intuitively



let $\varphi_n \to f$ in L^2 (approximation thm 12.7)

Then $$\|f\|^2_{L^2} = \lim_{n\to\infty} \langle \varphi_n, f \rangle = \lim_{n\to\infty} \int_\Omega \varphi_n \cdot f \, dV = 0.$$

Lemma 24.8 (technical lemma). The subspace

$$\mathrm{Im}(\Delta) \subseteq L^2(\Omega : \mathbb{R})$$

is closed.

Proof: later, maybe.

Proof Step 24.9: It is enough to solve

$$\begin{cases} \Delta u = f \\ u|_{\partial\Omega} = 0 \end{cases}.$$

Take $\tilde{u}$ any function w/ $\tilde{u}|_{\partial\Omega} = h$. Then if u solves

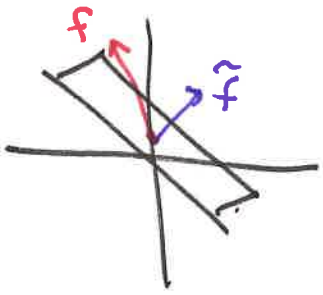
$$\begin{cases} \Delta u = \tilde{f} \\ u|_{\partial\Omega} = 0 \end{cases} \quad \text{for} \quad \tilde{f} = f - \Delta\tilde{u}$$

then $u + \tilde{u}$ solves $\begin{cases} \Delta(u+\tilde{u}) = f \\ u+\tilde{u}|_{\partial\Omega} = h. \end{cases}$

Proof of thm:

Idea 24.10: Use self adjointness to reduce to problem we've already solved.

Suppose there were an f so that $f \notin \mathrm{Im}(\Delta)$.



by adding $\Delta\tilde{u}$ we can assume that

$$f \perp \mathrm{Im}(\Delta)$$

$$\Rightarrow \quad \langle \Delta u, f \rangle_{L^2(\Omega)} = 0 \quad \text{for all } u.$$

$$\Rightarrow \quad \int_{\partial\Omega} \partial_n u \, f + \langle u, \Delta f \rangle_{L^2(\Omega)} = 0. \qquad \text{(self adjointness / Green's)}$$

$$\Rightarrow \quad \Delta f = 0 \qquad \text{(fundamental lemma)}$$

$$\text{and} \quad f|_{\partial\Omega} = 0.$$

$$\Rightarrow \quad f \equiv 0 \text{ by uniqueness.}$$

□

24.iv) Regularity Classes of functions

We have many different ways of measuring the differentiability or regularity of functions.

Hierarchy 24.11

In order of increasing regularity

$$L^2(\Omega:\mathbb{R}) \subsetneq C^0(\Omega:\mathbb{R}) \subsetneq C^1(\Omega:\mathbb{R}) \subsetneq \cdots \subsetneq C^\infty(\Omega:\mathbb{R})$$

(C^∞: only vector space.)

Ex 24.12 If u solves

$$\Delta u = f$$

w/ $f \in C^0$, we cannot expect u to be better than $u \in C^2$ (b/c $f = u_{xx}$ is a combination of derivatives that is continuous but not differentiable).

Question 24.13 (the regularity question)

Suppose that $f \in C^\infty(\Omega:\mathbb{R})$, and that u solves

$$\Delta u = f$$

on Ω. Can we conclude $u \in C^\infty(\text{Int}(\Omega):\mathbb{R})$?

Thm 24.14 (Elliptic Regularity). Yes.

Def 24.15: An analytic function is one such that the Taylor series $f:\Omega \to \mathbb{R}$ at a point $x \in \Omega$ converges on $[x-r_x, x+r_x]$ for some radius $r_x > 0$, for every x.

Fact 24.16

Not all smooth functions are analytic!

$$\begin{cases} e^{-\frac{1}{x}} & x > 0 \\ 0 & x \leq 0 \end{cases}$$



has all derivatives exist and are 0 at $x=0$, so Taylor says

$$f(x+h) = 0 + \frac{f'(0)}{2}h + \dots$$

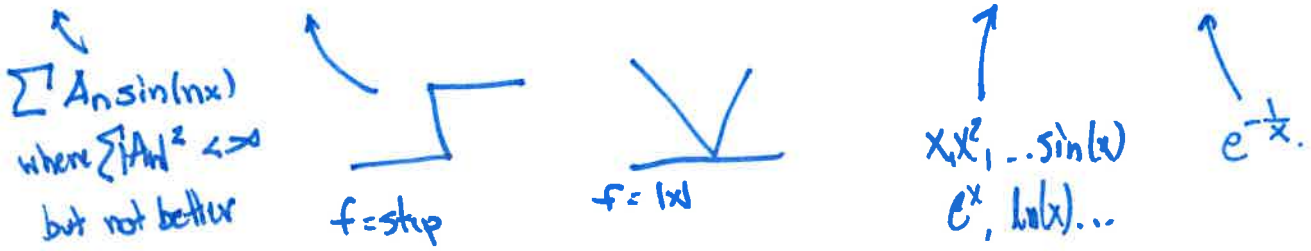
$$= 0 + 0 + \dots = 0 \quad \text{but} \quad f(x+h) \neq 0 \text{ for } h > 0$$

$\Rightarrow r_x$ radius of conv. is zero.

Summary 24.17

w = analytic

$$L^2(\Omega;\mathbb{R}) \supseteq C^0(\Omega;\mathbb{R}) \supseteq C^1(\Omega;\mathbb{R}) \supseteq \dots \supseteq C^{\infty}(\Omega;\mathbb{R}) \supseteq C^{w}(\Omega;\mathbb{R})$$



Thm 24.18: If an analytic function is $u \equiv 0$ on a neighborhood $\Omega' \subseteq \Omega$ w/ non-empty interior, then $u \equiv 0$ on Ω.



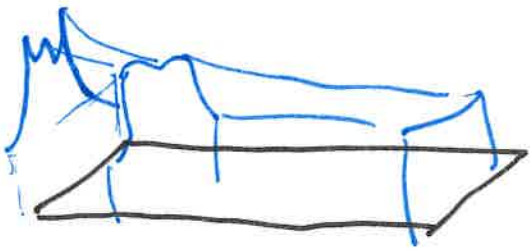
Thm 24.19 If u solves $\Delta u = 0$ on Ω, then u is analytic in $\text{Int}(\Omega)$.

Corollary 24.20: If $\Delta u = 0$ and $u \equiv 0$ on a subdomain w/ nonempty interior, then $u \equiv 0$ everywhere.

Lecture 24½ Maximum Principles for the heat equation

Suppose
$$\begin{cases} \frac{\partial u}{\partial t} - \Delta u = -F & \text{on } \Omega. \\ u|_{\partial\Omega} = h \\ u|_{t=0} = g \end{cases}$$
$-F$ ← heat sink

Then



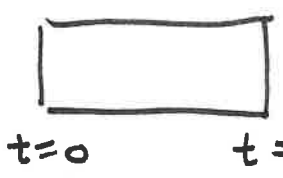
heat reaches equilibrium so intuitively, sharp peaks get smoothed out.

24½.1: Maximum principle for heat eq

Thm 24.5.1 (Heat max principle) Suppose $\Omega \subseteq \mathbb{R}^d$ is closed and bounded, $-F \leq 0$ is a heat sink on Ω. Then if u solves
$$\begin{cases} \frac{\partial u}{\partial t} - \Delta u = -F(x,t) \end{cases}$$
on $(x,t) \in \Omega \times [0,T]$, then the max of u occurs along $\partial\Omega$ or $t=0$.

Proof: We will prove in the case that $\Omega = [a,b]$.

(Case 1) $-F < 0$. Suppose (t_m, x_m) were the max. Then
$$\frac{\partial u}{\partial t} = 0$$
$$\frac{\partial u}{\partial x} = 0 \qquad \frac{\partial^2 u}{\partial x^2} \leq 0.$$
but $\frac{\partial u}{\partial t} - \frac{\partial^2 u}{\partial x^2} = 0 + -\frac{\partial^2 u}{\partial x^2} \geq 0$ so $\neq F$.



← must rule out here.

If (c,T) were max, must have $\frac{\partial u}{\partial t} \geq 0$ at $t=T$, and $\frac{\partial u}{\partial x^2}$ still max in x-direction.

$$\Rightarrow \frac{\partial u}{\partial t} - \frac{\partial^2 u}{\partial x^2} \geq 0 \quad \rightarrow\leftarrow \neq -F.$$

(Case II) $-F \leq 0$.

Replace $-F$ by $-\tilde{F} = -F - 2\varepsilon$ for $\varepsilon < 1$.

If u solves the original eq. then

$$v = u(t,x) + \varepsilon x^2$$

solves

$$\begin{cases} \frac{\partial v}{\partial t} - \Delta v = -\tilde{F}. \end{cases}$$

and $\tilde{F} < 0$. So v obtains max on boundaries. Call M max of u on $\partial\Omega \cup \{t=0\}$.

$$u(t,x) \leq v(t,x) \leq M + \varepsilon(a^2 + b^2) \quad \text{by def,}$$

take $\lim \varepsilon \to 0$.

Corollary 24½.2: If $F = 0$ ie normal heat eq, same holds for min.

Thm 24½.3: There is at most one solution to

$$\begin{cases} \frac{\partial u}{\partial t} - \Delta u = f \\ u|_{\partial\Omega} = h \\ u(0,x) = g \end{cases}$$

on $[0,T] \times \Omega$.

Proof: If u, v were two, $\tilde{u} = u - v$ would solve $\begin{cases} \frac{\partial \tilde{u}}{\partial t} - \Delta \tilde{u} = 0 \\ \tilde{u}|_{\partial\Omega} = 0 \\ \tilde{u}(0,x) = 0. \end{cases}$

so $\quad 0 \leq \tilde{u}$ min on bd

$\tilde{u} \leq 0$ max on bd ∎

⇒ If separation of variables gives a solution it is unique.

$u = \sum_n e^{-\lambda_n t} A_n \varphi_n$ solves IVP $u(0,x) = \sum_n A_n \varphi_n$ w/ vanishing Dir. BC.

Lecture 25 The delta function

The Dirac delta "function" is

$$\delta(x) = \begin{cases} \infty & x = 0 \\ 0 & x \neq 0 \end{cases}.$$

But what does this mean?

25.i) Dual vector spaces

Let V be a finite dimensional vector space e.g. $V = \text{span}\{e_1, e_2, e_3\}$.

We can form more vector spaces from V.

- $V \oplus V$ eg $\text{Span}\{e_1, e_2, e_3, e_1', e_2', e_3'\}$ dim 6
- $V \otimes V$ eg $\text{Span}\{e_1 e_2, e_1 e_3, e_1 e_1, \ldots\}$ dim 9
- $\text{Mat}(V,V) = \{$ linear maps $A : V \to V \}$ i.e. matrices.

Ex 25.1 : Given a vector space V

$$V^* = \{\xi : V \to \mathbb{R} \mid \xi \text{ is a continuous linear map}\}.$$

For example $\xi(x,y,z) = 3x - 4y + 7z \in \mathbb{R}$. Is a vector space.

$$\begin{aligned}(\xi + \eta)(\alpha v + \beta w) &= \xi(\alpha v + \beta w) + \eta(\alpha v + \beta w) \\ &= \alpha\xi(v) + \beta\xi(w) + \alpha\eta(v) + \beta\eta(w) \\ &= \alpha(\xi+\eta)(v) + \beta(\xi+\eta)(w).\end{aligned}$$

is also linear. So is $c\xi$. So vector space.

Ex 25.2 : When $V = \{\ \}$ column vectors, we can think of

$V^* = [\quad]$ as row vectors, so that $\xi \in V^*$

$\xi(v) = [\ \xi\ ] \cdot \begin{bmatrix} v \end{bmatrix}$ is matrix multiplication

$(1 \times n) \times (n \times 1)$

Def 25.3: V^* is called the dual (vector) space of V. (elements called ↓)

Ex 25.4: For each function $g \in L^2(\Omega:\mathbb{R})$, we get a continuous linear functional

$\xi_g \in (L^2)^*$ defined by $\xi_g[f] = \langle g, f \rangle_{L^2} = \int_\Omega gf \, dV$

Ex 25.5: Dual spaces reverse inclusions

$C^1(\Omega:\mathbb{R}) \subseteq C^0(\Omega:\mathbb{R}) \subseteq L^2(\Omega:\mathbb{R})$

$L^2(\Omega:\mathbb{R})^* \subseteq C^0(\Omega:\mathbb{R})^* \subseteq \dots \subseteq C^\infty(\Omega:\mathbb{R})^*$ (← not Banach!)

Ex 25.6: An element in $C^0(\Omega:\mathbb{R})^*$ is linear map F

$F : C^0(\Omega:\mathbb{R}) \to \mathbb{R}$

eating a function and giving a number. e.g.

$F[f(x)] = f(0)$ $\qquad F(f+g) = f(0)+g(0)$, $F(cf) = cf(0)$.

25.ii) Distributions

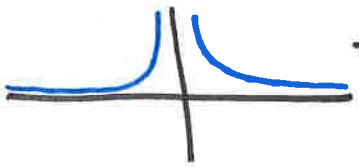
Def 25.7: An element of $C^\infty(\Omega:\mathbb{R})^*$ is called a distribution. (Schwartz)

Def 25.8: The δ-distribution is $\delta \in C^0(\Omega:\mathbb{R})^*$ as in Ex 25.6

$\delta[f(x)] = f(0)$.

(← continuous)

Rem 25.9: Notice L^2 functions don't have to have pointwise values,



$f(x) = \frac{1}{x^{1/3}} \in L^2(\Omega:\mathbb{R})$

but $f(0)$ isn't defined.

Notation 25.10: For function $g \in L^2$ we have $\xi_g \in (L^2)^*$. Pretending the definition carries over we usually write

$$\delta[f(x)] := \text{"} \int_{\Omega} \delta(x) f(x)\, dx \text{"} = f(0).$$

this doesn't technically make sense!

Rule 25.11: distributions can be formally manipulated like functions.

$$\delta_3[f] = \int \delta(x-3) f(x) dx = f(3)$$

$$13\delta(x-4) - 5\delta(x+1)[f] = \int [13\delta(x-4) - 5\delta(x+1)] f(x) dx$$

$$= 13 f(4) - 5 f(-1).$$

Ex 25.12: the derivative $\partial_x \delta(x)$ is defined by

$$\partial_x \delta[f] = -f'(0)$$

pretend $\partial_x \delta[f] = \int \partial_x \delta(x) \cdot f(x) dV$ fake int. by parts.

$$= -\int \delta(x) \cdot \partial_x f(x)\, dV$$

$$\partial_x \partial_y^2 \delta[f] = (-1)^3 \partial_x \partial_y^2 f(0).$$

25.iii) Fourier Series of Distributions

Recall that if $f \in L^2(\Omega : \mathbb{R})$ Ω = box, then

$$B_{nmk} = \frac{8}{\text{vol}\,\Omega} \iiint f(x,y,z) \cdot \sin\left(\frac{n\pi x}{L_2}\right) \sin\left(\frac{m\pi y}{L_2}\right) \sin\left(\frac{k\pi z}{L_3}\right) dx dy dz$$

$$= \frac{2}{L} \int f(x) \sin\left(\frac{n\pi x}{L}\right) dx \in \mathbb{R}$$

Key Observation 25.13: Fourier Coefficients make sense as long as $f(x), \in H^*$ (space*) $\sin\left(\frac{n\pi x}{L}\right) \in H$ ($= e^{\text{>}}$) are in dual spaces

$\Rightarrow$ Fourier series makes sense for distributions.

Observation 25.14: Let a_n, b_n be the Fourier coeff. of a distribution.

We know if $f \in L^2$ then $\sum |a_n|^2 + |b_n|^2 < \infty$ by Parseval.

$\Rightarrow$ if f is distribution, cannot converge



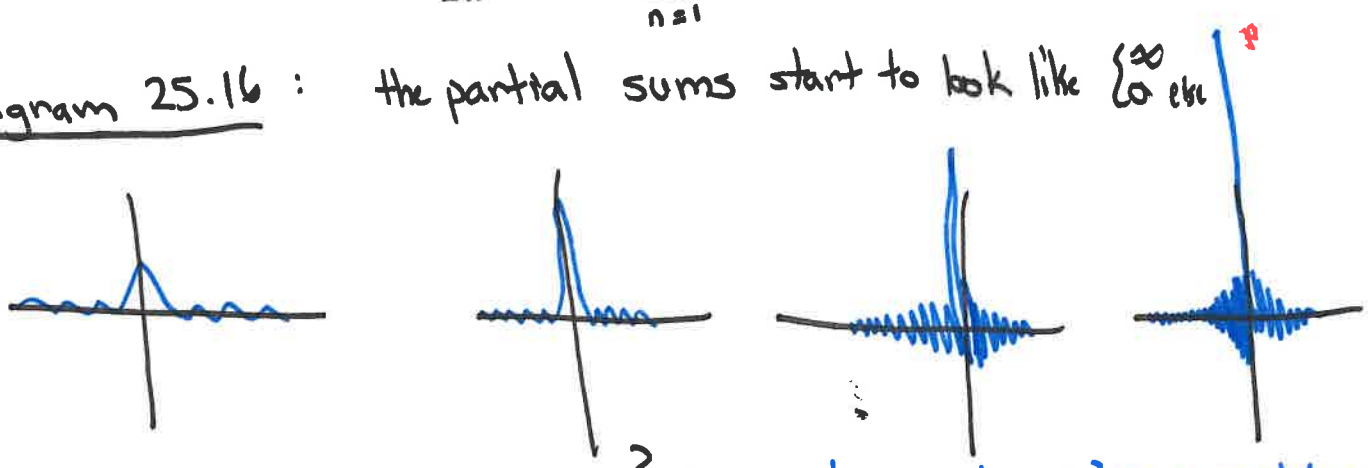
Lemma 25.15: The Fourier series of the δ-function is

$$a_k = \frac{1}{\pi}\int_{-\pi}^{\pi} \delta(x)\cos(kx)dx = \frac{1}{\pi}\cos(0) = \frac{1}{\pi} \qquad a_0 = \frac{1}{2\pi}\int_{-\pi}^{\pi} \delta \cdot 1 = \frac{1}{2\pi}$$

$$b_k = \frac{1}{\pi}\int_{-\pi}^{\pi} \delta(x)\sin(kx)dx = \frac{1}{\pi}\sin(0) = 0.$$

$$\Rightarrow \quad \delta(x) \text{ "=" } \frac{1}{2\pi} + \frac{1}{\pi}\sum_{n=1}^{\infty}\cos(nx)$$

Diagram 25.16: the partial sums start to look like $\sum_{0}^{\infty}$ etc



in what sense do they converge? ← we know not in L^2 by completeness.

$$S_N = \frac{1}{2\pi} + \frac{1}{\pi}\sum_{n=1}^{N}\cos(nx) = \frac{1}{2\pi}\,\frac{\sin\left((N+\frac{1}{2})x\right)}{\sin\left(\frac{1}{2}x\right)}$$

Def 25.17: a sequence of functions f_n converges weakly to a distribution f_x

if $$\lim_{n\to\infty}\int_{\mathbb{R}} f_n(x)g(x)dx = \int f(x)g(x)dx \qquad \forall g \in C^{\infty}([a,b];\mathbb{R})$$

We denote this $f_n \rightharpoonup f_x$

↖ half arrow.

Lecture 26 Green's Functions I: ODEs.

Recall $\delta(x-x_0)$ was the distribution obeying

$$\delta(x-x_0)[f(x)] = \text{"} \int_{\mathbb{R}} \delta(x-x_0) f(x)\,dx \text{"} = f(x_0).$$

And via integration by parts when $H(x) = \begin{cases} 0 & x \le 0 \\ 1 & x > 0 \end{cases}$,

$$\left[\frac{d}{dx} H(x)\right] = \delta(x) \qquad \int_{\mathbb{R}} \left(\frac{d}{dx}H\right) f = -\int_{\mathbb{R}} H \cdot \frac{\partial}{\partial x} f = \lim f(x) - (-f(0)).$$

26.i) The principle of superposition, revisited

Consider $\Delta u = f$ on $[0, L]$. We know that for each n, the solution of

$$\begin{cases} \Delta u = B_n \sin\left(n\frac{\pi x}{L}\right) \\ u|_{0,L} = 0 \end{cases}$$

is

$$u_n(x) = B_n \cdot -\left(\frac{L}{n\pi}\right)^2 \cdot \sin(n\pi x/L)$$

We can write this $u(x;n)$

And in general, the solution of $\Delta u = f$ w/ Dir. boundary conditions is

$$u(x) = \sum_{n=1}^{\infty} u(x;n) \cdot B_n \qquad f = B_n \cdot \sin\left(\frac{n\pi x}{L}\right) \quad B_n = \{f \text{ is Fourier sine basis}\}.$$

$$= \text{"} \int_{\mathbb{N}} u(x;n)\, B(n)\, dn \text{"}$$

we can facetiously write this sum as an "integral" over $n = 1, 2, 3, \ldots$

26.ii) The Fictitious Position basis

We know $\sin\left(\frac{\pi x}{L}\right), \sin\left(\frac{2\pi x}{L}\right), \ldots$ are a basis, and we calculate coefficients via

$$B_n = \left\langle \sin\left(\frac{n\pi x}{L}\right), f \right\rangle_{L^2} \cdot \frac{2}{\pi}$$

These are a "spectral" basis which pick out specific oscillatory frequencies.

Informal Definition 26.1: The position basis of $L^2(\Omega : \mathbb{R})$ is the collection of functions $\delta(x-x_0)$ for $x_0 \in \Omega$.



Calculation 26.2: For each x_0, the coefficient in the $\delta(x-x_0)$ direction is

$$\langle \delta(x-x_0), f\rangle_{L^2(\Omega)} = \int \delta(x-x_0) f(x)\,dx = f(x_0).$$

And

$$f(x) = \text{“}\sum_{x_0 \in \Omega} f(x_0)\,\delta(x-x_0)\text{”}$$

↓ upgrade to dummy variable.

$$= \int \delta(x-y) f(y)\,dy$$

Analogy 26.3:

	Basis element	dummy variable	expression
Sine	$\sin(\frac{n\pi x}{L})$	$n \in \mathbb{N}$	$\sum_n B_n \sin(\frac{n\pi x}{L})$
Position	$\delta(x-y)$	$y \in \Omega$.	$\int \delta(x-y) f(y)\,dy$

no need to calculate coefficient b/c it's the value itself.

Idea 26.4: Suppose we could solve

$$\begin{cases} \Delta u = \delta(x-y) \\ u|_{\partial\Omega} = 0 \end{cases}$$

for all y. Call the solution $u(x;y)$. Then the solution of

$$\begin{cases} \Delta u = f \\ u|_{\partial\Omega} = 0 \end{cases}$$

is just

↙ coeff in position basis.

$$u(x) = \text{“}\sum_y u(x;y) f(y)\text{”} = \int_\Omega u(x;y) f(y)\,dy.$$

Definition 26.5: the solution of

$$\begin{cases} \Delta u = \delta(x-y) \\ u|_{\partial\Omega} = 0 \end{cases}$$

is called the Green's function of Δ (with Dirichlet BCs) denoted

$$G(x;y) : \Omega \times \Omega \to \mathbb{R}.$$

Formula 26.6: The solution of $\begin{cases} \Delta u = f \\ u|_{\partial\Omega} = 0 \end{cases}$

therefore has the integral formula

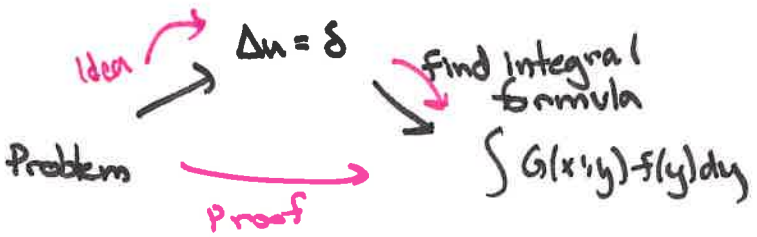
$$u(x) = \int_\Omega G(x;y) f(y)\, dy.$$

Strategy 26.7: If we can find $G(x;y)$ once and for all, then we can solve w/ Fourier Series (but its often need to find $G(x;y)$.

Comparison 26.8:

$\begin{cases} \Delta u = f \\ u|_{\partial\Omega} = 0 \end{cases}$ any Ω closed, bounded	$\begin{cases} \Delta u = 0 \\ u|_{\partial\Omega} = h(\theta) \end{cases}$ disk only
Integral Kernel $K(x,y) = G(x;y)$	$= K(r,\theta,\varphi)$ $= \frac{1-r^2}{1+r^2-2r\cos(\theta-\varphi)}$
$u = \int_\Omega G(x;y) f(y)\, dy$	$u = \int_0^{2\pi} \frac{1}{2\pi} K(r,\theta,\varphi) h(\varphi)\, d\varphi.$

Outline 26.9



We usually "Guess" the formula the verify it directly.

Informal Proof 26.10

$$\Delta u(x) = \Delta \int_\Omega G(x;y) f(y)\, dy$$

$$= \int_\Omega \Delta G(x;y) f(y)\, dy = \int_\Omega \delta(x-y) f(y)\, dy = f(x) \quad \square$$

$$u|_{\partial\Omega} = \int_\Omega G(x;y) f(y)\, dy \quad \text{for } x \in \partial\Omega \quad G(x;y) = 0 \quad \square$$

26.iii) Green's functions for ODEs

Consider $\begin{cases} -\left(\frac{d}{dx}\right)^2 u = f & \text{on } [0,1] \\ u|_{0,1} = 0 \end{cases}$

'jump discontinuity in derivative'

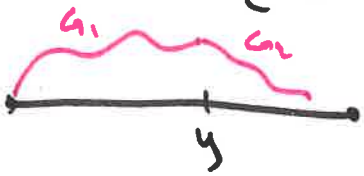


The Green's function obeys $-\left(\frac{d}{dx}\right)^2 u = \delta(x-y)$

$\left(\frac{d}{dx}H\right)(x,y) = -\int \delta(x-y) = H(x,y)$

Procedure 26.11: We know Green's function obeys

$$G(x;y) = \begin{cases} G_1(x;y) & x \le y \\ G_2(x;y) & x \ge y \end{cases}$$ where $G_1(x;y)$, $G_2(x;y)$ solve the ODE and



1) $G_1(0;y) = 0$, $G_2(1;y) = 0$ Dirichlet BC

2) $G_1(y;y) - G_2(y;y) = 0$ Continuity

3) $G_1'(y;y) - G_2'(y;y) = +1$

Prop 26.12: the Green's function of

$$\begin{cases} -\left(\frac{d}{dx}\right)^2 u = f \\ u|_{0,1} = 0 \end{cases}$$

is

$$G(x;y) = \begin{cases} (1-y)x & x \le y \\ y(1-x) & x > y. \end{cases}$$

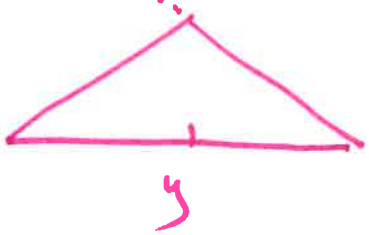
Proof: We know G_1, G_2 solve $\left(\frac{d}{dx}\right)^2 = 0$

$$G_1(x;y) = A(y)x + B(y)$$
$$G_2(x;y) = C(y)x + D(y).$$

⇒ Four unknowns, four constraints

1) $0 = G_1(0;y) = A(y)\cdot 0 + B(y)$ so $B(y) = 0$.

2) $0 = G_2(1;y) = C(y)\cdot 1 + D(y)$ $\quad D(y) = -C(y)$



3) $A(y)\cdot y = C(y)\cdot y + D(y)$ so $A(y) = C(y)$

4) $A(y) - C(y) = -1$

$A = C(y) - 1$

$$\Rightarrow \begin{cases} Ay = Cy - C \\ A - C = 1 \end{cases} \qquad C\left(\frac{y-1}{y}\right) = A$$

$$C\left(\frac{y-1}{y} - 1\right) = 1 \Rightarrow \begin{matrix} C(y-1-y) = y \\ C = -y \end{matrix} \qquad A + y = 1$$

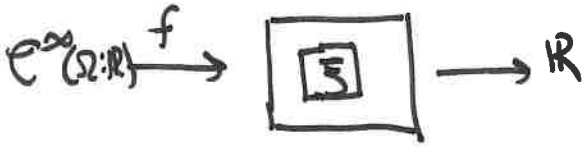
$$\begin{cases} G_1(x;y) = (1-y)x \\ G_2(x;y) = -yx + y = y(1-x) \end{cases}$$

Lecture 27 Green's Functions II : 2d

26.iii) Green's functions for ODEs.

27.i) More on weak convergence

Recall that $\xi \in C^\infty(\Omega:\mathbb{R})^*$ is called a distribution or generalized function.



Def 27.1 : A distribution ξ has weak derivative $\frac{d}{dx}\xi$ (or $\partial_x\partial_y..\xi$) if it obeys

$$\int_\Omega (\partial_x\partial_y..\xi)\,\varphi\,dV = (-1)^n \int_\Omega \xi\cdot\partial_x\partial_y...\varphi\,dV$$

(n partials)

for all $\varphi \in C^\infty_{cpt}(\Omega:\mathbb{R})$

Ex 27.2 :

$$\frac{d}{dx}H = \delta(x)$$

$$\int_{-L}^{L} \frac{d}{dx}H\cdot\varphi\,dx \overset{def}{:=} \int_{-L}^{L} -H\cdot\frac{\partial}{\partial x}\varphi\,dx \qquad C^\infty_{cpt}(\Omega:\mathbb{R})$$

($\varphi(-L)=0$)

$$= \int_0^L -\partial_x\varphi\,dx = -[\varphi(L)-\varphi(0)]$$

$$= \varphi(0) = \int_{-L}^{L}\delta(x-0)\varphi(x)dx$$

Def 27.3 : $u \in C^\infty(\Omega:\mathbb{R})^*$ is said to be a weak solution of a PDE

$$\begin{cases} Lu = f \\ u|_{\partial\Omega} = 0 \end{cases}$$

if $Lu=f$ holds in the weak sense, i.e.

$$\int_\Omega Lu \cdot \varphi \overset{\text{def}}{:=} \int \left[\sum_{i,j} a_{ij}(x) \partial_i \partial_j \ldots + c(x) u \right] \varphi \, dV$$

$$= \int u \cdot \left[\partial_j \partial_i \, a_{ij}(x) \varphi + \partial_k \left(b_k^{(x)} \cdot \varphi(x) \right) + c(x) \varphi \right] dV$$

is equal to

$$= \int f \cdot \varphi \, dV \qquad \forall \varphi \in C^\infty_{\text{pcpt}}.$$

Ex 27.4 The Green's function is a distribution $G(x;y) \in C^\infty(\Omega)^*$ that solves

$$Lu = \delta(x-y)$$

in the weak sense.

Any real solution

$$\begin{cases} \Delta u = f & \text{for } u, f \in C^k(\Omega; \mathbb{R}) \\ u|_{\partial\Omega} = 0 \end{cases}$$

$\varphi \in C^\infty_{\text{pcpt}}(\mathring{\Omega}; \mathbb{R})$

solves

literally true here

$$\int \Delta u \cdot \varphi \, dV = \int -\nabla u \cdot \nabla \varphi \, dV$$

$$= \int + u \cdot \Delta \varphi \, dV + \underbrace{\int \varphi \partial_\nu u - \partial_\nu \varphi \, u}_{=0.}$$

$$\int (\Delta u - f) \varphi \, dV = 0 \quad \forall \varphi \iff \Delta u - f = 0 \text{ by Fundamental Lemma of variations.}$$

Rem 27.4.5 : Recall $\xi_n \rightharpoonup \xi$ weakly if

$$\int_\Omega \xi_n \cdot \varphi \, dV \longrightarrow \int_\Omega \varphi \cdot \xi \, dV \quad \text{in } \mathbb{R} \quad \forall \varphi.$$

27.ii) Green's functions in 2d

Let's solve

$$\begin{cases} \Delta G(x,y,x',y') = \delta(x-x')\delta(y-y') \\ G \to 0 \text{ at } r \to \infty \end{cases}$$

on $\Omega = \mathbb{R}^2, \mathbb{R}^3$. Recall G depends on y, Δ, Ω, and the BC.

Def 27.5: The δ function at $\vec{x} = (x_1, \dots x_n)$ in $\mathbb{R}^n$ is

$$\delta(x-x_1)\delta(y-x_2)\delta(z-x_3)\cdots$$

Thm 27.6: The Green's function

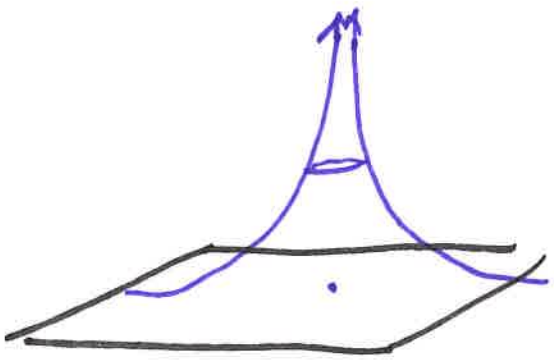
$$-\Delta G(\vec{x}, \vec{x}') = \delta(\vec{x}-\vec{x}')$$

is

$$G_2(\vec{x};\vec{x}') = -\frac{1}{4\pi}\log\|\vec{x}-\vec{x}'\|^2 \quad \text{on } \mathbb{R}^2$$

$$G_3(\vec{x};\vec{x}') = +\frac{1}{4\pi}\frac{1}{|\vec{x}-\vec{x}'|} \quad \text{on } \mathbb{R}^3$$

$$G_n(\vec{x};\vec{x}') = \frac{1}{C(n)}\frac{1}{|\vec{x}-\vec{x}'|^{n-2}} \quad \text{on } \mathbb{R}^n.$$



Remark 27.7: For Δ, heat, etc you often see the Green's function written

$$G(r) = \frac{1}{4\pi r}$$

meaning $G(\vec{x},\vec{x}') = \frac{1}{4\pi} G(|\vec{x}-\vec{x}'|)$

This only holds for a few nice cases w/ radial symmetry.

Rem 27.8: For reasonable elliptic PDEs (smooth coeff) Green's function is smooth for $\vec{x} \neq \vec{x}'$ w/ asymptote

$$G \sim \frac{1}{|\vec{x}-\vec{x}'|^{n-2}} \quad \text{as} \quad \vec{x} \to \vec{x}'$$

(only bounded in ODE case!).



Boundary Value.

Thm 27.9:

Suppose Ω is a closed and bounded domain.

Then

$$G_\Omega(\vec{x};\vec{y}) = G_{\mathbb{R}^n}(\vec{x};\vec{y}) + z(x;y) \quad \text{for Dir BC and } \Delta$$

where z solves

$$\begin{cases} \Delta_x z = 0 \\ z|_{\partial\Omega} = -G_{\mathbb{R}^n}(\vec{x};\vec{y})\big|_\Omega. \end{cases}$$

Same for Neumann, general L.

Proof: By linearity, it solves

$$\Delta\left(G_{\mathbb{R}^n}(\vec{x};\vec{y}) + z(x;y)\right) = \delta(\vec{x}-\vec{y}) + 0$$

$$G|_{\mathbb{R}^n} + z\Big|_{\partial\Omega} = G_{\mathbb{R}^n}\big|_{\partial\Omega} - G_{\mathbb{R}^n}\big|_{\partial\Omega} = 0.$$

weakly.

Corollary 27.10: If $\tilde{G}$ any Green's function for Δ (any $\tilde{\Omega}, \tilde{B}$) then on Ω,

$$u(\vec{x}) + \int_\Omega \tilde{G}(\vec{x},\vec{y}) \overset{f}{\Delta u}\, dV + \oint_{\partial\Omega} \tilde{G}(\vec{x},\vec{y}) \frac{\partial u}{\partial \nu}(\vec{y}) - \frac{\partial \tilde{G}(\vec{x},\vec{y})}{\partial \nu} u(\vec{y})\, dV$$

$= \int_\Omega u \cdot \Delta G\, dV = \int_\Omega u \cdot \delta(\vec{x}-\vec{y})\, dV$ by def.

Lecture 28 Computation with Green's functions

Recall that

$$G_{\mathbb{R}^2}(x;y) = -\frac{1}{2\pi}\log|x-y|$$

$$G_{\mathbb{R}^3}(x;y) = \frac{1}{4\pi}\frac{1}{|x-y|}$$

Then the solution of

$$\begin{cases} -\Delta u = f & f \in C^\infty_{\text{compact}}(\mathbb{R}^d;\mathbb{R}) \\ u|_{r\to\infty} = 0 \end{cases}$$

is given by

$$u(x) = \int_{\mathbb{R}^3} \frac{1}{4\pi}\frac{1}{|x-y|} f(y)\,dy.$$

From last time: Remark 27.7

28.i) Computational Examples

Seemingly unrelated question 28.1: Can you tell if an exoplanet is hollow or has a dense core from its gravitational influence on nearby objects?

Ex 28.1: Suppose $f(r,\theta,\varphi) = \begin{cases} C & \text{for } 0 \le r \le R \\ 0 & \text{else.} \end{cases}$

Find u st $-\Delta u = f$

By symmetry

$$u = u(r) = \frac{1}{4\pi}\iiint_0^{2\pi} \frac{1}{|(r,0,0)-(s,\varphi,\theta)|}\, s^2\sin\varphi\, d\theta\, d\varphi\, ds$$

$$= \frac{2\pi}{4\pi}\int_0^R\int_0^\pi \frac{1}{\sqrt{r^2+s^2-2rs\cos\varphi}}\, s^2\sin\varphi\, d\varphi\, ds.$$



Assume (by symmetry) that r lies along z axis's

By the law of cosines

$$(\text{dist})^2 = r^2 + s^2 - 2sr\cos\varphi.$$

set $z = \cos\varphi$

$dz = -\sin\varphi$

$\varphi = 0 \quad z = 1$

$$= -\frac{1}{2}\int_0^R \int_1^{-1} \frac{1}{\sqrt{r^2+s^2-2sr\cdot z}} s^2\,dz\,ds$$

$$= \frac{1}{2}\int_0^R \int_{-1}^{+1} \frac{1}{\sqrt{\alpha-\beta z}}\,dz \cdot s^2\,ds$$

$$= \frac{1}{2}\int_0^R \left[-\frac{2}{b}\sqrt{\alpha-\beta z}\right]_{-1}^{+1} s^2\,ds$$

$$= \frac{1}{2}\int_0^R \left[-\frac{2}{b}\sqrt{\alpha-\beta} + \frac{2}{b}\sqrt{\alpha+\beta}\right]$$

$$= \frac{1}{2}\int_0^R \frac{2}{2rs}\sqrt{r^2+s^2+2rs} - \frac{2}{2rs}\sqrt{r^2+s^2-2rs}\cdot s^2\,ds$$

$$= \frac{1}{2}\int_0^R \frac{1}{rs}\left[|r+s| - |r-s|\right] s^2\,ds$$

<u>Case 1</u> : $r \geq s$. $\quad |r+s| - |r-s| = r+s-(r-s) = 2s$

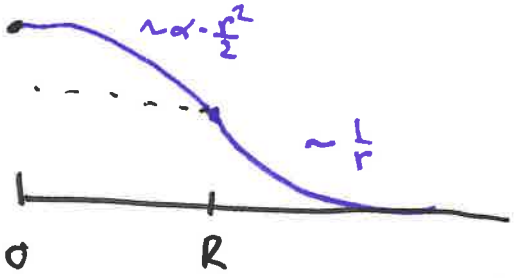
$$= \frac{1}{2}\int_0^R \frac{1}{rs}\cdot 2s\cdot s^2\,ds = \int_0^R \frac{s^2}{r}\,ds = \frac{R^3}{3}$$

<u>Case 2</u> : $s \leq r$

$$= |r+s| - |r-s| = r+s-(s-r) = 2r$$

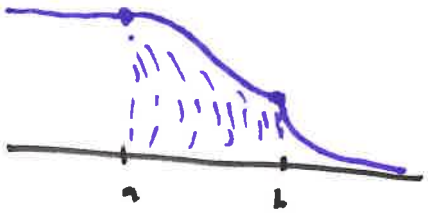
$$= \frac{1}{2}\left[\int_0^r \frac{1}{rs}\cdot 2s\cdot s^2 + \int_r^R \frac{1}{rs}\,2r\cdot s^2\right] = \left\{\frac{R^2}{2} - \frac{r^2}{6}\right.$$

$$u(r) = \begin{cases} \left(\frac{R^2}{2} - \frac{r^2}{6}\right)C & 0 \le r \le R \\ C\frac{R^3}{3r} & r \ge R \end{cases}$$



Ex 28.3 Suppose $f = \begin{cases} C & a \le r \le b \\ 0 & \text{else} \end{cases}$ hollow planet

$$u(r) = \begin{cases} \frac{1}{2}(b^2 - a^2) & 0 \le r \le a \\ \frac{1}{2}(b^2 - \frac{r^2}{2}) - \frac{a^3}{r^3} & a \le r \le b \\ \frac{b^3 - a^3}{3r} & r \ge b. \end{cases}$$

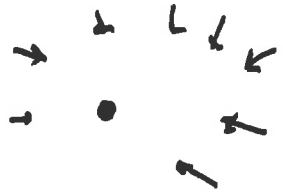


Answer 28.4 : Gravitational Potential V obeys

$$-\Delta V = 4\pi G \rho$$

ρ = mass density

$$F_{gravity} = \nabla V$$



solid



$$= \frac{4\pi}{3} R^3 \frac{1}{r} = \frac{m}{r}$$ outside

hollow



$$\frac{4\pi}{3}(b^3 - a^3)\frac{1}{r} = \frac{m}{r}$$ same.

$\Rightarrow$ No!

28.ii) The method of Images

Recall for $\Omega \subsetneq \mathbb{R}^3$ closed and sometimes bounded

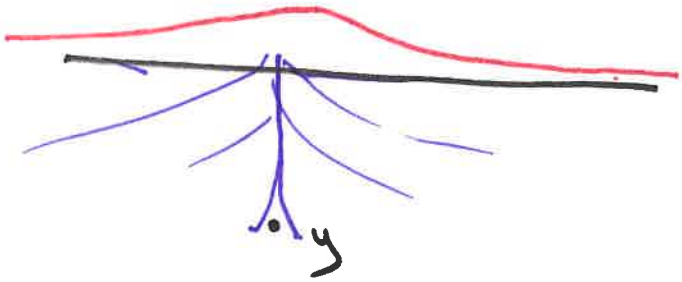
$$G_\Omega = G_{\mathbb{R}^3} + z(x;y) \quad \text{where} \quad \begin{cases} \Delta z = 0 \\ z|_{\partial\Omega} = -G_{\mathbb{R}^3}. \end{cases}$$

Ex 28.5: Solve $H = \{\vec{x} \in \mathbb{R}^3 \mid \vec{x} = (x,y,z),\ z \geq 0\}$.

Find

$$\begin{cases} \Delta G_H = \delta \\ G_H|_{z=0} = 0 \end{cases}$$





To cancel, we say by symmetry

$$G_H(x,y) = G_{\mathbb{R}^3}(x;y) - G_{\mathbb{R}^3}(x;-y).$$

$\Rightarrow$

~~Principle~~

$\Rightarrow$ Solution of $\begin{cases} -\Delta u = c \\ u|_H = 0 \end{cases}$ is the restriction of $\begin{cases} -\Delta u = c(y) - c(-y) \\ \text{H.C.} \end{cases}$ on $\mathbb{R}^3$



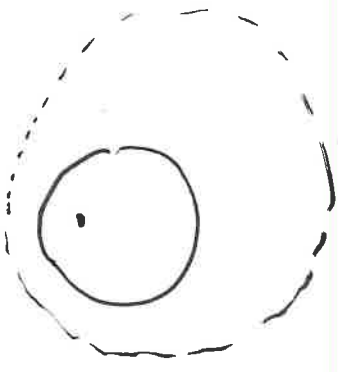
Principle 28.6 (Method of Images)



$u|_{\partial\Omega} = 0$ =



or



Lecture 29 The Poisson and Heat Kernels

Recall that if u obeys

$$\begin{cases} -\Delta u = 0 \\ u|_{\partial B} = h \end{cases}$$

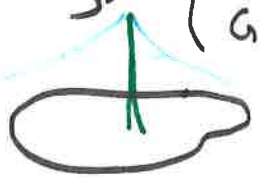
then u is given by

$$u(r,\theta) = \int_{\partial B} P(r,\theta;\varphi)\, h(\varphi)\, d\varphi$$

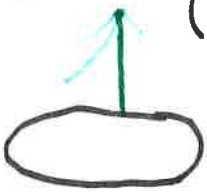
for $B = B(0,1)$ in $\mathbb{R}^2$.

29.i) Approximate Identities

Intuitively, $\begin{cases} \Delta G(x;y) = \delta(x-y) \\ G|_{\partial\Omega} = 0 \end{cases}$ ← n-dimensional delta function



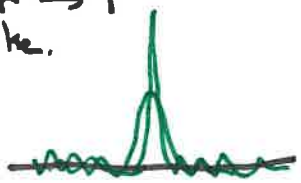
while $\begin{cases} \Delta P = 0 \\ P|_{\partial\Omega} = \delta(\varphi-\theta) \end{cases}$ ← $n-1$ dim delta function on the boundary



So the Poisson Kernel solves

$$\begin{cases} -\Delta P(r,\theta;\varphi) = 0 \\ P|_{\partial\Omega}(\theta;\varphi) = \delta(\theta-\varphi) \end{cases}$$

As $r \to 1$ → spike.

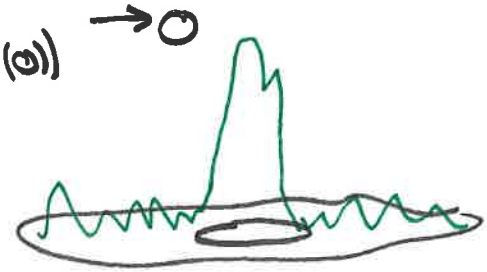


Def 29.1: A sequence of approximate identities is a sequence of (Kernel) functions $K_n : \mathbb{R}^d \to \mathbb{R}$ such that

1) $\int_{\mathbb{R}^d} K_n \, dV = 1$ (but $\int_{\mathbb{R}^d} |K_n| \, dV \le M \quad \forall n$)

2) For all $\delta > 0$

$$\| K_n \|_{C^0(\mathbb{R}^d - B_\delta(0))} \to 0$$



Lemma 29.2: If K_n are a sequence of approximate identities, then $K_n \to \delta$ in the weak sense.

Proof: We must show

$$\int_{\mathbb{R}^d} K_n(y)\varphi(y)dy \to \int_{\mathbb{R}^d} \delta(y)\varphi(y)dy = \varphi(0). \quad \forall \varphi \in C^\infty_{cpct}$$

$$\left| \int_{\mathbb{R}^d} K_n(y)\varphi(y)dy - \varphi(0) \right| = \left| \int_{\mathbb{R}^d} K_n(y)\varphi(y)dy - \varphi(0)\int_{\mathbb{R}^d} K_n(y)dy \right| \quad \text{by 1)}$$

$$= \left| \int_{\mathbb{R}^d} K_n(y)\left[\varphi(y)-\varphi(0)\right]dy \right|$$

$$= \left| \int_{|y|<\delta} K_n(y)\left[\varphi(y)-\varphi(0)\right]dy \right| \leq |\varphi(\delta)-\varphi(0)| \cdot M$$

(small by continuity)

$$+ \left| \int_{|y|>\delta} K_n(y)\left[\varphi(y)-\varphi(0)\right]dy \right|$$

$$\leq 2\max|\varphi(y)| \cdot \delta \|K_n\|_{\infty(\mathbb{R}^d - B_\delta)}$$

$\to 0$ by 2.

More generally,
- $K_n(x-y) \to \delta(x-y)$
- on general $\Omega \subseteq \mathbb{R}^d$ same.
- In particular $K_n : [0, 2\pi] \to \mathbb{R}$ can be approx identities

so $K_n(\theta) \to \delta(\theta)$.

Thm 29.3: For $r_n = 1 - \frac{1}{n}$ then

$$P_n = P(r_n; \alpha) = \frac{1}{2\pi} \frac{1 - r_n^2}{1 + r_n^2 - 2r_n\cos(\alpha)}$$

is a sequence of approximate identities.

Proof: Basically a lot of trig identities, for 1)

For 2) For $\theta \geq \varepsilon$ then $\cos(\theta) \leq 1-\varepsilon$.

$$\frac{1-r^2}{1+r^2-2r(\cos\theta)} \to 0$$

$$\to \geq 2\varepsilon.$$

denominator remains bounded away from 0.

Thm 29.5: The function

$$u(r,\theta) = \frac{1}{2\pi}\int_0^{2\pi} P(r,\theta;\varphi)\, h(\varphi)\, d\varphi$$

solves

$$\begin{cases} -\Delta u = 0 \\ u|_{\partial\Omega} = h \end{cases}$$

Remark 29.6: this ties up the loose end in the derivation about $\sum |z|^n$ not converging at the boundary.

Proof: At $r=1$

$$u|_{\partial\Omega} = \lim_{r\to 1} u(r,\theta) = \lim \int_0^{2\pi} P(r,\theta;\varphi)\, h(\varphi)\, d\varphi$$

$\Delta u = 0$ so smooth up to boundary

$$= \lim_{n\to\infty} \int_0^{2\pi} P(1-\tfrac{1}{n},\theta;\varphi)\, h(\varphi)\, d\varphi$$

approx identity.

$$= \int_0^{2\pi} \delta(\theta-\varphi)\, h(\varphi)\, d\varphi = h(\theta).$$



29. ii) The Heat Kernel.

Place an infinite pinprick of heat at $z \in \mathbb{R}^d$ at $t=0$. The heat evolution solves

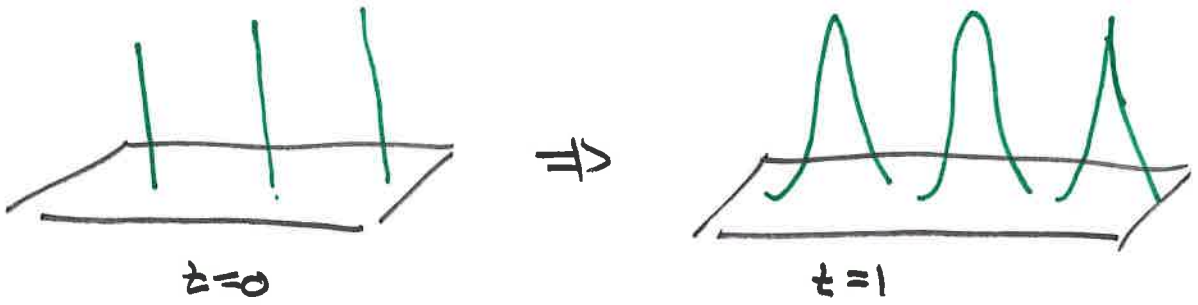
$$\begin{cases} (\partial_t - \Delta)u = 0 \\ u|_{r\to\infty} = 0 \\ u(0,x) = \delta(x-z) \end{cases}$$

More formally,

$$\lim_{t\to 0} u(t,x) \longrightarrow \delta(x-z)$$

in the weak sense.

The heat evolution, via superposition may be thought of as a collection of pinpricks



all evolving independently.

Def/Prop 29.4 : The Heat Kernel is the solution $K_*(t,x:y)$ that obeys

$$\begin{cases} (\partial_t - \Delta)K = 0 \\ K|_{r\to\infty} = 0 \end{cases}$$

on $(0,\infty)\times\mathbb{R}^d$. And for $t = \frac{1}{n}$

$K(\frac{1}{n},x:y)$ are a sequence of approximate identities $\to \delta(x-y)$.

Proof : Justify "the" not "a". If $K, \tilde{K}$ both obeyed, we show $K=\tilde{K}$ below.

Prop 29.5 : The solution of

$$\begin{cases} \partial_t u - \Delta u = 0 \\ u|_{r\to\infty} = 0 \\ u(0,x) = g \end{cases}$$

is -

$$u(t,x) = \int_{\mathbb{R}^d} K(t,x:y)\, g(y)\, dy.$$

Proof :

$$(\partial_t - \Delta)u(t,x) = \int_{\mathbb{R}^d} (\partial_t - \Delta)K(t,x:y)\, g(y)\, dy$$

$$= \int_{\mathbb{R}^d} 0 \cdot g(y)\, dy \qquad \forall t > 0.$$

$$u(0,x) = \lim_{t\to 0} \int_{\mathbb{R}^d} K(t,x:y)g(y)dy \to \int \delta(x-y)g(y)dy = g(x).$$

If both $K, \tilde{K}$ are kernels

$$\int_{\mathbb{R}^d} \tilde{K}(t,x;y)g(y)dy = u(t,x) = \int_{\mathbb{R}^d} K(t,x;y)g(y)\,dy$$

(uniqueness by max principle)

$$\Rightarrow \int_{\mathbb{R}^d} \underbrace{[\tilde{K}(t,x;y) - K(t,x;y)]}_{=0 \text{ by Fundamental Lemma of Variations.}} g(y)dy = 0 \quad \forall g$$

Lecture 30 Fourier Transforms I

Recall the solution of the heat equation is given by

$$\begin{cases} (\partial_t - \Delta)u = 0 \\ u(0,x) = g \\ u \to 0 \text{ at } \infty \end{cases}$$

$$u(t,x) = \int_{\mathbb{R}^d} K(t,x;y)\, g(y)\, dy$$

Thm 30.1: The Heat Kernel is given by

$$K(t,x;y) = \frac{1}{(4\pi t)^{d/2}} e^{-|x-y|^2/4t}.$$

Proof: We showed $K(t,x;y)$ solves the heat equation on $(0,\infty)\times\mathbb{R}^d$.

We still must show, $\lim_{t\to 0} K(t,x;y) = \delta(x-y)$ in the weak sense.

For $y=0$

$$\int_{\mathbb{R}^d} \frac{1}{(4\pi t)^{d/2}} e^{-x^2/4t}\, dx = 1$$

by integrating the Gaussian

$$\int e^{-a^2/b} = \sqrt{\pi b}$$

$$\int_{\mathbb{R}} e^{-x^2/4t} = \sqrt{\pi 4 \cdot t}$$

$$\int_{\mathbb{R}^d} e^{-x_1^2/4t} e^{-x_2^2/4t} \cdots dx_1 dx_2$$

And outside radius δ

$$\leq \max |K(t,x_0)|$$

$$= e^{-\delta^2/4t} \to 0 \quad \text{as } t \to 0.$$

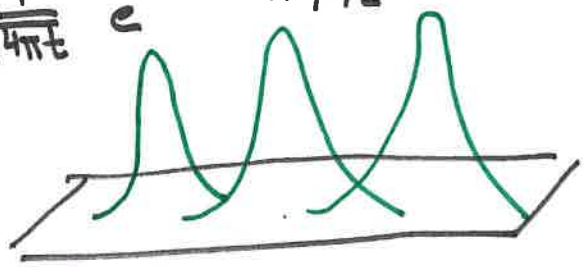
□

Ex 30.1: Consider $H = \sum c_i \delta(x - z_i)$ Solve $\begin{cases} (\partial_t - \Delta)u = 0 \\ u(0,x) = H \end{cases}$

$$u(t,x) = \int \frac{1}{(4\pi t)^{1/2}} e^{-|x-y|^2/4t} \sum_i c_i \delta(y - z_i)\, dy$$

$$= \sum_i c_i \int \frac{1}{\sqrt{4\pi t}} e^{-|x-y|^2/4t} \delta(y - z_i)\, dy = \sum_i c_i \left(\frac{1}{4\pi t}\right)^{1/2} e^{\cdots}$$

$$= \sum_i c_i \frac{1}{\sqrt{4\pi t}} e^{-|x-z_i|^2/4t}$$



Ex 30.2 : Solve $\begin{cases} (\partial_t - \Delta)u = 0 \\ u(0,x) = C_0. \end{cases}$

$$u(t,x) = \int \left(\frac{1}{4\pi t}\right)^{d/2} e^{-|x-y|^2/4t} \cdot C_0 \, dy$$

$z = y - x$

$$= C_0 \int_{\mathbb{R}^d} \left(\frac{1}{4\pi t}\right)^{d/2} e^{-z^2/4t} \, dz = C_0.$$

but we know that, already in equil.

Ex 30.3 : $H(x) = \begin{cases} 1 & x \geq 0 \\ 0 & \text{else} \end{cases}$ solve $u(t,x) \Leftrightarrow \begin{cases} (\partial_t - \Delta)u \\ u(0,x) = H \end{cases}$

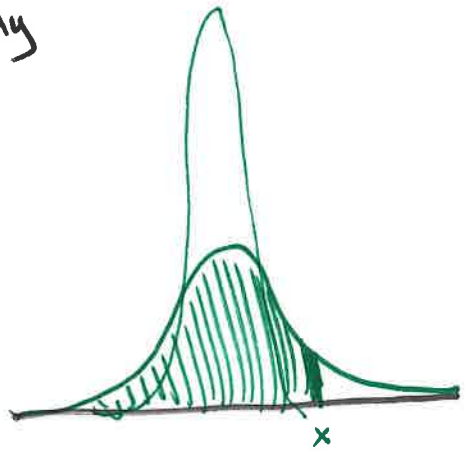
$$u(t,x) = \int_{-\infty}^{\infty} \frac{1}{(4\pi t)^{1/2}} e^{-|x-y|^2/4t} \cdot H(y) \, dy$$

$$= \frac{1}{\sqrt{4\pi t}} \int_0^{\infty} e^{-|x-y|^2/4t} \, dy \qquad z = y - x, \quad dz = dy$$

$$= \frac{1}{\sqrt{4\pi t}} \int_{-x}^{\infty} e^{-z^2/4t} \, dz$$

$$= \frac{1}{\sqrt{4\pi t}} \int_{-\infty}^{x} e^{-z^2/4t} \, dz$$

$$= \frac{1}{\sqrt{4\pi t}} \operatorname{Erf}_{4t}(x)$$





What about with heat source

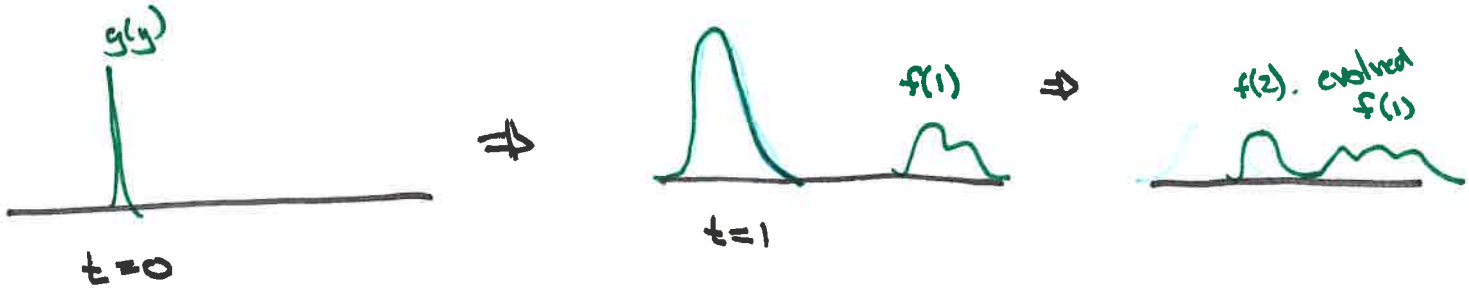
Thm 30.4: The solution of the inhomogeneous problem is

$$\begin{cases} (\partial_t - \Delta) u = f(t,x) \\ u(0,x) = g \\ u|_{\infty} = 0 \end{cases}$$

is

$$u(t,x) = \int_{\mathbb{R}^d} \frac{1}{\sqrt{4\pi t}} e^{-|x-y|^2/4t} g(y)\,dy + \int_0^t \int_{\mathbb{R}^d} \frac{1}{\sqrt{4\pi \tau}} e^{-|x-y|^2/4\tau} f(s,y)\,dy\,ds \qquad \tau = t - s.$$

Idea 30.5



$g(0)$ → evolve forward

$f(dt)$ → evolve heat source as IVP starting from $\tau = dt$, $2dt$ ...

use superposition on all.

30.ii) Computing the Green's function + Inverse Operators

Question 30.6: Given a more general PDE

$$\begin{cases} (\partial_t + L)u = 0 \\ u(0,x) = g \\ u|_{\partial\Omega} = 0 \end{cases}$$

L 2nd order elliptic (self-adjoint), eg $-\Delta + \Delta$, how do we find the Green's function on $\mathbb{R}^d$?

Intuition 30.7: When solving

$$Ax = b$$

on $\mathbb{R}^m$, if you know A^{-1}, you win. Think about the Green's function analogously. For elliptic

$$A^{-1} = \left[A^{-1}e_1, A^{-1}e_2 \dots\right]$$

$$Lu = f \iff u = L^{-1}f.$$

$$L^{-1} = \left[G(\cdot, y_1), G(\cdot, y_2) \dots\right].$$

30.8 Computing u With Fourier Series: We know for a compact domain, for any f,

$$\begin{cases} -\Delta u = f \\ u|_{\partial\Omega} = 0 \end{cases}$$ is given by

$$f \xrightarrow{\text{Fourier}} \sum B_n \sin(nx)$$

$$\downarrow$$

$$(-\Delta)^{-1} f = \sum^{*} \frac{B_n}{n^2} \sin(nx)$$

$$\Rightarrow L^{-1} = (\text{Fourier})^{-1} \circ (\text{divide by eigenvalues}) \circ (\text{Fourier}).$$

$$G(\cdot, y) = L^{-1}(\delta(y)).$$

$\Rightarrow$ How do we compute this on $\mathbb{R}^n$? Fourier transforms

Lecture 31 Complex Fourier Series and Transforms

Recall $e^{inx} = \cos(nx) + i\sin(nx)$.

For $\mathbb{C}$-valued functions complex Fourier Series is more convenient.

31.i Hermitian Vector Space

Consider a Hermitian vector space is a complex vector space $V = \mathbb{C}^n$ with a

Def 31.1 Hermitian Inner product

$$\langle v, w \rangle_{\mathbb{C}} = \alpha_1\overline{\beta}_1 + \cdots + \alpha_n\overline{\beta}_n \qquad v = \sum \alpha_i v_i \qquad \alpha_i \in \mathbb{C}.$$
$$w = \sum \beta_j v_j$$

Properties 31.2 : The Hermitian inner product obeys and a norm

i) $\langle v, v \rangle = \alpha_1\overline{\alpha}_1 + \cdots + \alpha_n\overline{\alpha}_n = \|v\|^2 \geq 0$ is real

ii) $\langle v, w \rangle_{\mathbb{C}} = \overline{\langle w, v \rangle}_{\mathbb{C}}$

iii) $\langle v_1 + v_2, w \rangle_{\mathbb{C}} = \langle v_1, w \rangle_{\mathbb{C}} + \langle v_2, w \rangle_{\mathbb{C}}$

iv) $\langle \alpha v, w \rangle = \alpha \langle v, w \rangle$ and $\langle v, \alpha w \rangle = \overline{\alpha} \langle v, w \rangle$

v) $v = \sum_{n=1}^{N} \langle v, e_j \rangle e_j$ basis e_j.

Ex 31.3 : The space $L^2(\Omega : \mathbb{C})$ of complex-valued function is the Hilbert space w/ inner product

$$\langle f, g \rangle_{L^2} = \int_{\Omega} f(x)\overline{g(x)}\, dV.$$

31.ii) Complex Fourier Series

Prop 31.4 : The functions $e^{inx \cdot w}$ for $n = -2, -1, 0, 1, 2, \ldots$

form an orthonormal basis of the L-periodic functions on $\mathbb{R}$ for $w = \frac{2\pi}{L}$.

- Span by Stone-Weierstrass.

$$\langle e^{inx}, e^{imx} \rangle = \int_0^L e^{inwx} e^{-imwx}\, dx = \int_0^L \cos((n-m)x) + i\sin((n-m)wx)\, dx$$
$$= 0 \text{ unless } n = m$$

Thm 31.5 : (Complex Fourier Series) An L-periodic function $f : \mathbb{R} \to \mathbb{C}$ can be expressed

$$f(x) = \sum_{n=-\infty}^{\infty} c_n e^{inwx}$$

when $$c_n = \langle f, e^{inwx} \rangle_{L^2([0,L],\mathbb{C})} = \frac{1}{L}\int_0^L f(x) e^{-inwx}\, dx$$

$$= \frac{1}{L}\int_0^L [h_1(x) + ih_2(x)][\cos(nwx) - i\sin(nwx)]\, dx.$$

And $$\lim_{N\to\infty} \sum_{n=-N}^{N} c_n e^{inwx} = f(x) \text{ in } L^2 \text{ i.e. } \left\| \sum_{n=-N}^{N} c_n e^{inwx} - f(x) \right\|_{L^2} \to 0.$$

Thm 31.6 : (Parseval's Thm)

$$\sum_{n=-\infty}^{\infty} |c_n|^2 = \|f\|^2_{L^2(\mathbb{R};\mathbb{C})} \qquad \text{now } \int |e^{inwx}|^2 dx = 1$$

w/o normalization.

Ex 31.7 (Equivalence) Suppose that $f(x) : \mathbb{R} \longrightarrow \mathbb{R} \subseteq \mathbb{C}$ is secretly real-valued.

Write $c_n = \alpha_n + i\beta_n$.

Then $$\boxed{\begin{aligned} a_n &= 2\alpha_n \\ b_n &= -2\beta_n \end{aligned}}$$ is the relation w/ the real fourier coefficients.

31.iii) Complex Fourier Series as a transform

Recall $$f : [0,L] \longrightarrow \mathbb{C}$$
$$x \longmapsto f(x)$$

eats a number and returns a number. Similarly an operator

$$L : \{\text{function space } A\} \longrightarrow \{\text{function space } B\}$$

eats a function + returns function.

Def 31.8 : A transform is an invertible operator that gives the relationship between two distinct but equivalent presentations of the data of a function.

Observation 31.9: A sequence is just a function $\mathbb{Z} \longrightarrow \mathbb{C}$.

Recall

$$l^2(\mathbb{Z};\mathbb{C}) = \left\{ \{c_n\} \in \mathbb{C} \;\middle|\; \sum |c_n|^2 < \infty \right\}$$

is the space of square integrable sequences.

Ex 31.10: Fourier Series is a transform

$$\mathcal{F}: L^2([0,L];\mathbb{C}) \longrightarrow l^2(\mathbb{Z};\mathbb{C})$$

$$f(x) \longmapsto \hat{f}(n): \mathbb{Z} \to \mathbb{C}$$

$\hat{f}(n) = c_n$ complex Fourier Coefficient.

And the inverse of Fourier Series

$$\mathcal{F}^{-1}: l^2(\mathbb{Z};\mathbb{C}) \longrightarrow L^2([0,L];\mathbb{C})$$

$$\hat{g}(n) \longmapsto g(x) = \sum_{n=-\infty}^{\infty} \hat{g}(n) e^{inwx}.$$

Lemma 31.11:

$$\mathcal{F} \circ \mathcal{F}^{-1}(\hat{g}) = \hat{g}$$

$$\mathcal{F}^{-1} \circ \mathcal{F}(f) = f.$$

Proof:

$$\mathcal{F} \circ \mathcal{F}^{-1}(\hat{g}) = \mathcal{F}\left(\sum_{n=-\infty}^{\infty} \hat{g}(n) e^{inwx} \right)$$

$$= \widehat{\mathcal{F}(\hat{g})}(n) = \left\langle \sum_{k=-\infty}^{\infty} \hat{g}(k) e^{ikwx},\ e^{ikwx} \right\rangle$$

$$= \hat{g}(n) \quad \text{by orthogonality.}$$

$$\mathcal{F}^{-1} \circ \mathcal{F}(f) = \mathcal{F}^{-1}(\{\hat{f}(n)\})$$

$$= \sum_{n=-\infty}^{\infty} \hat{f}(n) e^{inwx} \overset{\checkmark}{=} f(x)$$

note. $\hat{f}(n)$ is the data of the coefficients, not a function of x!

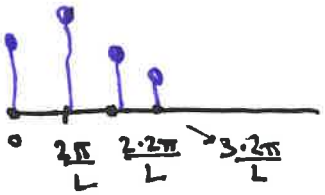
Rem: Parseval says $\mathcal{F}: L^2(\Omega;\mathbb{C}) \longrightarrow l^2(\mathbb{Z};\mathbb{C})$ is an **isometry**, invertible linear map and norm-preserving / inner product.

31.iv) Fourier Transform

Fourier Series applies for periodic functions. What about non-periodic?



function on $[0,L)$



Fourier coeff $\hat{f}$ on $\frac{2\pi}{L}\cdot\mathbb{Z}$.

Idea 31.12: Take limit as $L\to\infty$. Then the spacing $\frac{2\pi}{L}\to 0$.

$\Rightarrow$ Coefficients become also a function $\hat{f}:\mathbb{R}\to\mathbb{C}$.



Def 31.13: The Fourier transform of a function $f:\mathbb{R}\to\mathbb{C}$ is

$$\hat{f}(k) = \mathcal{F}(f) = \int_{-\infty}^{\infty} f(x)\, e^{-ikx}\, dx$$

and the inverse transform

$$g(x) = \mathcal{F}^{-1}(\hat{g}(k)) = \frac{1}{2\pi}\int_{-\infty}^{\infty} \hat{g}(k)\, e^{-ikx}\, dk.$$

Thm 31.14 (Fourier-Parseval)

$$\mathcal{F}: L^2(\mathbb{R}_x:\mathbb{C}) \longrightarrow L^2(\mathbb{R}_k:\mathbb{C})$$
$$f \longmapsto \hat{f}$$

is an invertible operation w/ inverse $\mathcal{F}^{-1}$. And the relationship

$$\int_{-\infty}^{\infty} |f(x)|^2\, dx = \frac{1}{2\pi}\int_{-\infty}^{\infty} |\hat{f}(k)|^2\, dk$$

holds.

Lecture 32 The Fourier Transform II.

Recall

$$\hat{f}(k) = \mathcal{F}[f(x)] = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x) e^{-ikx}\, dx.$$

$$g(x) = \mathcal{F}^{-1}[\hat{g}(k)] = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \hat{g}(k) e^{ikx}\, dx.$$

32.i) Examples of the Fourier Transform

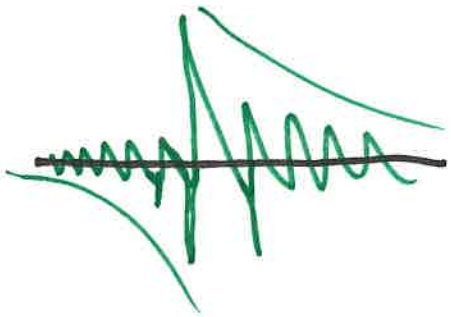
Ex 32.1: $f(x) = \begin{cases} 1 & |x| \le a \\ 0 & \text{else} \end{cases}$



$$\hat{f}(k) = \int_{-\infty}^{\infty} f(x) e^{-ikx}\, dx = \int_{-a}^{a} e^{-ikx}\, dx = \frac{1}{-ik}\left[e^{-ikx}\right]_{x=-a}^{a} = \frac{e^{ika} - e^{-ika}}{ik}$$

$$= 2\frac{\sin(ka)}{k} \cdot \frac{1}{\sqrt{2\pi}} \quad k \neq 0.$$

$$\hat{f}(0) = \int_{-a}^{a} f(x) e^{0}\, dx = \frac{2a}{\sqrt{2\pi}}.$$



Ex 32.2: $f(x) = \begin{cases} 1 - \frac{|x|}{a} & |x| \le a \\ 0 & \text{else} \end{cases}$



$$\hat{f}(k) = \int_{-\infty}^{\infty} f(x) e^{-ikx}\, dx = \frac{1}{\sqrt{2\pi}} \int_{-a}^{a} \left(1 - \frac{|x|}{a}\right) e^{-ikx}\, dx$$

$$= \int_{-a}^{0} \left(1 + \frac{x}{a}\right) e^{-ikx}\, dx + \int_{0}^{a} \left(1 - \frac{x}{a}\right) e^{-ikx}\, dx$$

$$= \int_{-a}^{a} e^{-ikx}\, dx + \frac{1}{a}\int_{-a}^{0} x e^{-ikx}\, dx - \frac{1}{a}\int_{0}^{a} x e^{-ikx}\, dx$$

$$= \frac{e^{-ikx}}{-ik}\Big|_{x=-a}^{a} + \frac{1}{a}\left[\frac{x e^{-ikx}}{-ik} + \frac{e^{-ikx}}{k^2}\right] - \frac{1}{a}\left[\frac{x e^{-ikx}}{-ik} + \frac{e^{-ikx}}{k^2}\right]$$

$$= \frac{e^{ika} - e^{-ika}}{ik} + \frac{1}{a}\left[\frac{1}{k^2} - \frac{a e^{-ika}}{ik} - \frac{e^{-ika}}{k^2}\right] - \frac{1}{a}\left[\frac{a e^{-ika}}{ik} + \frac{e^{-ika}}{k^2} - \frac{1}{k^2}\right]$$

$$= \frac{2 - (e^{ika} + e^{-ika})}{ak^2} = \frac{2(1-\cos(ka))}{ak^2}$$

$$\overset{!}{=} \frac{a}{\sqrt{2\pi}} \left[\frac{\sin(ka/2)}{(ka/2)}\right]^2 \quad \text{and}$$

$$\hat{f}(0) = \frac{a}{\sqrt{2\pi}}.$$

Ex 32.3: By inverse transform

$$\frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \left(\frac{\sin(\pi k)}{\pi k}\right)^2 e^{2\pi i kx/a}\, dk = \begin{cases} 1 - |x|/a & |x| \le a \\ 0 & \text{else.} \end{cases}$$

This integral is probably impossible to calculate w/ normal calculus.

Ex 32.4 (Gaussian) Recall $\int_{-\infty}^{\infty} e^{-x^2/2\sigma^2}\, dx = \sigma\sqrt{2\pi}$

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-x^2/2\sigma^2}.$$

$$\hat{f}(k) = \int_{-\infty}^{\infty} e^{-x^2/2\sigma} \cdot e^{-ikx}\, dx \cdot \frac{1}{\sqrt{2\pi}} \cdot \sigma\sqrt{2\pi}$$

$$= \frac{1}{2\pi\sigma} \int_{-\infty}^{\infty} e^{-\frac{1}{2\sigma^2}\left[x^2 + i\,2\sigma^2 kx\right]}\, dx$$

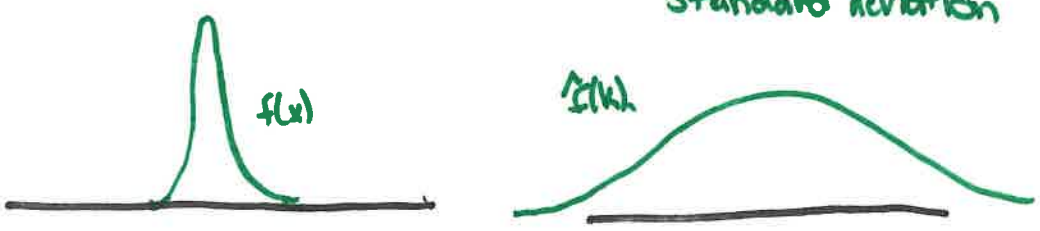
complete the square: $\left(x + \frac{2\sigma^2 ik}{2}\right)^2 = x^2 + ikx + \frac{i^2 k^2 \sigma^4}{4}$

$$= \frac{1}{2\pi\sigma} \int_{-\infty}^{\infty} e^{-\frac{1}{2\sigma^2}\left[\left(x + \frac{ik}{2}\cdot 2\sigma^2\right)^2 + \frac{k^2}{4}(2\sigma)^2\right]}\, dx.$$

$$= \frac{e^{-k^2(2\sigma^2)^2/(2\sigma^2 \cdot 4)}}{2\pi\sigma} \underbrace{\int_{-\infty}^{\infty} e^{-\frac{1}{2\sigma^2}\left[x + \frac{ik}{2}\right]^2}\, dx}_{= \sigma\sqrt{2\pi}}$$

$$= \frac{1}{\sqrt{2\pi}} e^{-k^2 \cdot \sigma^2/2}$$

standard deviation $\sigma \longleftrightarrow \frac{1}{\sigma}$



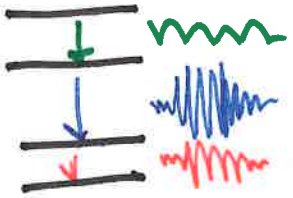
Ex 32.5 : The Fourier Transform also makes sense for distributions.

$$\mathcal{F}[\delta(x)] = \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty}\delta(x)e^{-ikx}dx = \frac{1}{\sqrt{2\pi}} \quad \text{or} \quad \widehat{\delta(x-\beta)} = \frac{e^{-ik\beta}}{\sqrt{2\pi}}$$

$$\mathcal{F}^{-1}[\delta(k-\alpha)] = \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty}\delta(k-\alpha)e^{ikx}dk = \frac{1}{2\pi}e^{ik\alpha}.$$

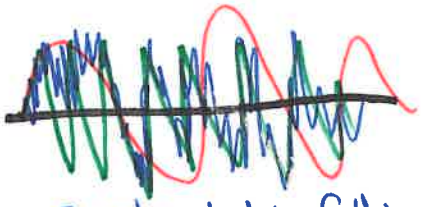
32.ii) Interpretation of the Fourier transform

Ex 32.6 : Recall that atomic transition produce light of fixed frequency



Emission light of three frequencies

$$E(x) = A_1\sin(w_{red}x) + A_2\sin(w_{blue}x) + A_3\sin(w_3x)$$



complex electric field in space

Fourier transform of the electric field is the frequency spectrum. Electric field describes the physics (wave eq) but the frequency spectrum is more digestable for signal processing. Fourier transform is the dictionary for converting.



32.iii) Properties of Fourier Transform

Proposition 32.7 : The Fourier transform obeys

i) $\widehat{f+g} = \hat{f} + \hat{g}$

ii) $\widehat{\alpha f} = \alpha\hat{f}$

] linearity

iii) $\widehat{f(x+a)} = e^{ika}\hat{f}(k)$

iv) $\widehat{e^{ibx}f(x)} = \hat{f}(k-b)$

v) $\widehat{f(cx)} = \frac{1}{c}\hat{f}\left(\frac{k}{c}\right)$

$$\int_{-\infty}^{\infty}f(x+a)e^{-ikx}dx = \int_{-\infty}^{\infty}f(y)e^{-ik(y-a)}dy$$

$$y = x+a \quad = e^{ika}\int_{-\infty}^{\infty}f(y)e^{-iky}dy$$

$$= e^{ika}\hat{f}(k)$$

Ex 32.8: Compute F.T. of $g(x) = \begin{cases} 2 & 2 \le x \le 11 \\ 0 & \text{else.} \end{cases}$

For $f(x) = \begin{cases} 1 & |x| \le 1 \\ 0 & \text{else,} \end{cases}$ $g(x) = 2f\left(\frac{x-7}{4}\right)$

so $\hat{g}(x) = 2\widehat{f\left(\frac{x-7}{4}\right)} = 2\left(\frac{1}{4}\right)^{-1}\widehat{f(x-7)}[4k]$

$= 8e^{-i7k}\hat{f}(4k)$

$= 8e^{-7ik}\frac{\sin(4k)}{k}$.

Key Idea 32.9: The Fourier transform turns differentiation into multiplication.

Thm 32.10: (Fourier Differentiation)

- $\widehat{\frac{d}{dx}f} = ik\hat{f}$
- $\frac{d}{dk}\hat{f} = \widehat{-ixf}$

Proof:

$$\mathcal{F}\left(\frac{d}{dx}f\right) = \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty}\frac{d}{dx}f(x)\,e^{-ikx}\,dx$$

$$= \frac{1}{\sqrt{2\pi}}\cdot \text{(*)}\int_{-\infty}^{\infty} f(x)\cdot -\frac{d}{dx}e^{-ikx}\,dx \qquad \text{(*)} \leftarrow \lim_{x\to\infty} f(x) = 0.$$

$$= \frac{1}{\sqrt{2\pi}}\cdot ik\int_{-\infty}^{\infty} f(x)e^{-ikx}\,dx = ik\hat{f}(k).$$

other way similar.

Main Idea 32.11

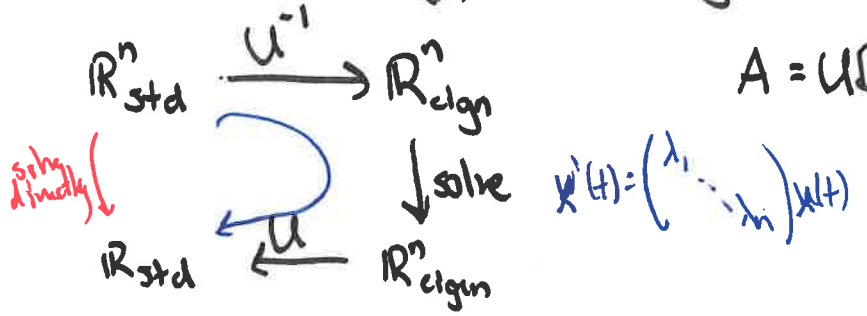
elliptic / ODE $\overset{FT}{\rightsquigarrow}$ algebraic eq

parabolic / hyperbolic PDE $\rightsquigarrow$ ODE.

Lecture 33 Fourier Transform III

33.i) Review of Decoupling ODEs.

Recall to solve $x'(t) = Ax$, we find eigenvectors/values.



Steps are 1) move to optimal basis 2) solve there 3) move back.

33.ii) Fourier Transform for ODEs

Ex 33.1 (FT reproduces characteristic polynomial)

Solve $\left(\frac{d}{dx}\right)^2 f + \alpha^2 f = 0$

using Fourier transforms.

$$\left(\frac{d}{dx}\right)^2 f + \alpha^2 f = 0$$

$$\mathcal{F}(f'') + \alpha^2 \mathcal{F}(f) = 0.$$

$$(ik)^2 \hat{f}(k) + \alpha^2 \hat{f}(k) = 0 \quad \Rightarrow (-k^2 + \alpha^2)\hat{f}(k) = 0 \quad \forall k.$$

either $\hat{f}(k) = 0$ or $k = \pm\alpha$.

$\hat{f}(k) =$



$$\Rightarrow f(x) = C_1 e^{i\alpha x} + C_2 e^{-i\alpha x}.$$

Ex 33.2: Solve $\left[\left(\frac{d}{dx}\right)^n f + \ldots + a_2\left(\frac{d}{dx}\right)^2 f + a_1 \frac{d}{dx} f + a_0 f\right] = 0.$

Take $\mathcal{F}$ of both sides

$$\left[(ik)^n + a_{n-1}(ik)^{n-1} + \ldots + a_0\right] \hat{f}(k) = 0$$

characteristic polynomial.

Ex 33.3: Solve

$$a\left(\frac{d}{dx}\right)^2 f + b\left(\frac{d}{dx}f\right) + c = e^{ix\gamma}$$

$$\left[a(ik)^2 + b(ik) + c\right]\hat{f} = \hat{g} = \delta(k-\gamma)/\sqrt{2\pi}$$

$$\hat{f} = \frac{1}{\sqrt{2\pi}} \frac{\delta(k-\gamma)}{a(i\gamma)^2 + b(i\gamma) + c}$$

$$f = \frac{1}{2\pi} \frac{e^{ix\gamma}}{a(i\gamma)^2 + b(i\gamma) + c}.$$

For general g,

$$\hat{f} = \frac{\hat{g}(k)}{a(ik)^2 + b(ik) + c}$$

general solution w/ an integral formula.

so

$$f(x) = \mathcal{F}^{-1}(\hat{f}) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \frac{\hat{g}(k)}{a(ik)^2 + b(ik) + c}\, dk.$$

33.iii) Fourier transform in higher dimension

$$\mathcal{F}: L^2(\mathbb{R}^d_x : \mathbb{C}) \longrightarrow L^2(\mathbb{R}^d_k : \mathbb{C})$$

$$\partial_{x_n} \longleftrightarrow ik_n \hat{f}$$

$$\mathcal{F}(f(x_1, \ldots, x_d)) = \left(\frac{1}{\sqrt{2\pi}}\right)^d \int_{-\infty}^{\infty} \cdots \int_{-\infty}^{\infty} f(x_1 \ldots x_d)\, e^{-i\vec{k}\cdot\vec{x}}\, dx_1 \ldots dx_d.$$

$[k_1 \ldots k_d]$

same for inverse.

Ex 33.4:

$$\widehat{-\Delta u} = \mathcal{F}\left(-\partial_x^2 u - \partial_y^2 u\right)$$

$$= -(ik_x)^2 \hat{u} - (ik_y)^2 \hat{u}$$

$$= \left((k_x)^2 + (k_y)^2\right)\hat{u} = |k|^2 \hat{u}.$$

More generally

$$\widehat{Lu} = \underbrace{\left[\sum a_{\alpha\beta} (ik_\alpha)(ik_\beta) + b_\delta (ik_\delta) + c\right]}_{P_L(k)} \hat{u}$$

for const coeff.

Thm 33.5 : For $f \in L^2(\mathbb{R}^d : \mathbb{C})$, the solution of

$$Lu = f$$

exists if $\sum a_{\alpha\beta}(ik_\alpha)(ik_\beta) + b_\delta(ik_\delta) + c \neq 0$ for k real.

And is given by

$$u = \mathcal{F}^{-1}\left[\frac{\hat{f}(k)}{P_L(k)} \right].$$

$$= \frac{1}{\sqrt{2\pi}^d} \int_{\mathbb{R}^d} \frac{1}{P_L(k)} \hat{f}(k) e^{ikx} \, dk_1 \ldots dk_d.$$

Proof :

$$\left\{ \begin{matrix} Lu = f \\ \text{PDE in } x \end{matrix} \right\} \xrightarrow{\mathcal{F}} \left\{ P_L(k)\hat{u} = \hat{f} \right\}$$

$$\downarrow \text{ easy to solve } \hat{u} = \frac{\hat{f}(k)}{P_L(k)}$$

$$\left\{ \begin{matrix} \text{solution} \\ u = \int \ldots dk. \end{matrix} \right\} \xleftarrow{\mathcal{F}^{-1}} \left\{ \hat{u} : \frac{\hat{f}}{P_L} \right\}$$

D.

33.iv) Convolutions

What is $\mathcal{F}(f(x)g(x))$ or $\mathcal{F}^{-1}\left(\frac{1}{P(k)} \cdot h(k)\right)$?

Def 33.6 : For two functions $f, g : \mathbb{R} \to \mathbb{C}$ define the convolution as

$$(f * g)(x) = \int_{-\infty}^{\infty} f(x-y) g(y) \, dy$$

in general

$$(f * g)(\vec{x}) = \int_{\mathbb{R}^d} f(\vec{x} - \vec{y}) \, g(\vec{y}) \, d\vec{y}.$$

Thm 33.7 (Fourier Product)

$$\widehat{f(x)g(x)} = (\hat{f} * \hat{g})(k)$$

$$\widehat{(f*g)(x)} = \hat{f}(k)\hat{g}(k).$$

Proof:

$$\hat{f} * \hat{g}(k) = \int_{-\infty}^{\infty} \hat{f}(k-q)\hat{g}(q)\,dq$$

$$= \int_{-\infty}^{\infty} \hat{g}(q) \left[\int_{-\infty}^{\infty} f(x) e^{-i(k-q)x}\,dx \right] dq$$

$$= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} \hat{g}(q) e^{iqx}\,dq \cdot f(x) e^{-ikx}\,dx$$

$$= \int_{-\infty}^{\infty} g(x) f(x) e^{-ikx}\,dx = \widehat{f(x)g(x)}.$$

Thm 33.8: The solution of $Lu = f$ on $\mathbb{R}^d$ (provided $P_L(k) \neq 0$) is

$$u = \mathcal{F}^{-1}\left(\frac{1}{P_L(k)} \hat{f}(k) \right) = \frac{1}{\sqrt{2\pi}^d} \int_{\mathbb{R}^d} \widehat{\frac{1}{P_L}}(x-y) f(y)\,dy$$

Def 33.9 / Corollary: If $f = \delta(x - y)$ then $\hat{f} = \text{const}\, e^{ik\cdot y}$

$$\Rightarrow u = \frac{1}{\sqrt{2\pi}^d} \int_{\mathbb{R}^d} \frac{1}{P_L(k)} e^{-ik(x-y)}\,dy$$

so

$$G(x:y) = G(x-y) = \mathcal{F}^{-1}\left(\frac{1}{P_L(k)} \right)(x-y).$$

is the Green's function.

Lecture 34 Fourier Transform IV: Green's functions

Recall that

$$\widehat{f(x)g(x)} = (\hat{f} * \hat{g})(k)$$

$$\widehat{(f*g)(x)} = \hat{f}(k)\cdot\hat{g}(k)$$

and after Fourier Transform

$$\widehat{Lu} = P_L(k)\,\hat{u}.$$

From last time

Prop/Def 34.1: The Green's function on $\mathbb{R}^d$ is given by

$$G(x;y) = \mathcal{F}^{-1}\left(\frac{1}{P_L(k)}\right)(x-y).$$

Ex 34.2: On homework, you will show that on $\mathbb{R}^3$.

$$\widehat{-\Delta u} = |k|^2\hat{u} \quad \text{so} \quad P_L(k) = |k|^2$$

and $$\mathcal{F}^{-1}\left(\frac{1}{|k|^2}\right) = \frac{1}{4\pi|x|}$$

Rem 34.3: For const. coeff. linear PDEs, one always has

$$G(x;y) = G(x-y) = \mathcal{F}^{-1}\left(\frac{1}{P_L(k)}\right)(x-y)$$

Rem 34.4: The assumption that $P_L(k) \neq 0$ can be eliminated, but takes more analysis at asymtotes. The equation

$$-\Delta + 1 \qquad (P_L = |k|^2 + 1 \neq 0$$

is much nicer than

$$-\Delta - 1 \qquad (P_L = |k|^2 \overline{+} 1).$$



(w/ BC)

Thm 34.5 (Fredholm): If L is a second order linear elliptic PDE on Ω, then there is a finite-dimensional spaces $V \subseteq L^2(\Omega;\mathbb{C})$ $K \subseteq C^\infty(\Omega;\mathbb{C})$ such that the following holds

1) There exist solutions to

$$Lu = f$$

if and only if $\langle f, v\rangle_{L^2} = 0 \quad \forall v \in V.$

2) All solutions, given some u_0 w/ $Lu_0 = f$ are of the form

$$u = u_0 + \sum_{j=1}^{N} c_j \varphi_j \qquad \varphi_j \text{ a basis of } K = \ker(L).$$

$\dim(K) - \dim(V)$ is called the Fredholm Index (of L).

34.ii) Green's function for the Heat + Wave equations.

For $$(\partial_t - \Delta)u = f$$

We Fourier Transform only in x:

Prop 34.6: If u solves $(\partial_t - \Delta)u = 0$, then

$$\partial_t \hat{u}(k) + |k|^2 \hat{u}(k) = 0$$

for all k, so

$$\hat{u}(k) = C(k)\, e^{-|k|^2 t}.$$

Proof:

$$\widehat{\partial_t u} - \widehat{\Delta u} = \int \partial_t u \cdot e^{-ikx}\, dx + |k|^2 \hat{u}$$

$$= \partial_t \hat{u} + |k|^2 \hat{u}. \qquad \square$$

Corollary 34.7 (which we already know) The Green's function of the Heat equation is

$$K(t, x; y) = e^{-|x-y|^2/4t}$$

and the solution of $\begin{cases} (\partial_t - \Delta)u = 0 \\ u|_{t=0} = g \end{cases}$ is

$$u(t,x) = \frac{1}{\sqrt{4\pi t}} \int_{\mathbb{R}^1} e^{-|x-y|^2/4t} g(y)\,dy$$

Proof: $\partial_t \hat{u} - |k|^2 \hat{u} = 0$ at $\hat{u}(0,k) = \widehat{u(0,x)} = \hat{g}(k)$.

so $\hat{u}(t,k) = \hat{g}(k)\, e^{-|k|^2 t}$.

Therefore

$$u(t,x) = g(x) * \mathcal{F}^{-1}\left(e^{-|k|^2 t}\right).$$

Recall $\mathcal{F}\left(e^{-x^2/2\sigma^2}\right) = \frac{1}{\sqrt{2\pi}\sigma} e^{-|k|^2\sigma^2/2}$

$\frac{\sigma^2}{2} = t$ $\quad \frac{1}{2\sigma^2} = \frac{1}{4t}$

$$= g(x) * \frac{1}{\sqrt{4\pi t}} e^{-x^2/4t}$$

$$= \int_{\mathbb{R}^d} g(y) \frac{1}{\sqrt{4\pi t}} e^{-|x-y|^2/4t}\,dy .$$

and if $g(y) * \delta(y)$ Green's function is recovered.

34.iii) The Wave Equation

$$\begin{cases} (\partial_t^2 - \Delta)u = 0 \\ u(0,x) = 0 \\ \partial_t u(0,x) = h \end{cases}$$

Fourier Transforming gives

$$\begin{cases} \partial_t^2 \hat{u} = -|k|^2 \hat{u} \\ \partial_t \hat{u}(0,k) = \hat{h}(k). \end{cases}$$

complex coeff.

$$= A(k)\cos(|k|t) + B(k)\sin(|k|t)$$

$$\hat{u}(k) = A_k e^{i|k|t} + B_k e^{-i|k|t} \quad \Rightarrow \quad A(k) + B(k) = 0.$$

$$\partial_t \hat{u}(k) = i|k|\left[A_k - B_k\right] \qquad i|k|\,2A(k) = \hat{h}(k).$$

$$\Rightarrow \quad \hat{u}(k) = 2i\sin(|k|t)\cdot \hat{A}(k)$$

$$= \sin(|k|t)\cdot \frac{\hat{h}(k)}{2i|k|}\cdot 2i$$

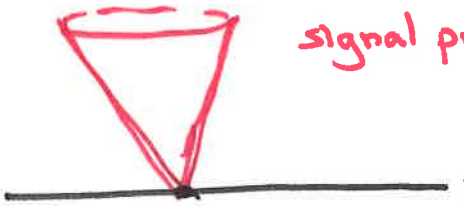
$$= \frac{\sin(|k|t)}{|k|}\,\hat{h}(k).$$

$$\Rightarrow \quad u(t,x) = \int_{\mathbb{R}^d} \frac{\sin(|k|t)}{|k|}\,\hat{h}(k)\,dk.$$

$$= \frac{1}{4\pi|x|c^2}\,\delta\left(t-\frac{|x|}{c}\right) * h(x)$$

Thm 34.8 : The above formula holds, more generally

$$u(t,x) = \frac{1}{4\pi c^2 t}\int_{|x-y|=ct} h(y)\,dy$$



signal propagates outward along "light cone".

x only interacts w/ y at time t if separated by $|x-y| = ct$.

⚠ only true for dim > 2 !

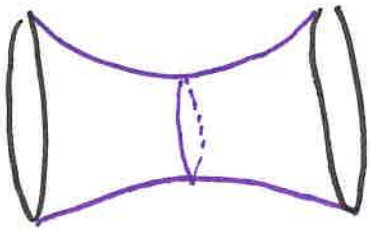
Lecture 35 Introduction to Nonlinear PDEs.

So far, we have considered exclusively linear PDEs. Many important equations are non-linear, and the goal is to use techniques to reduce these to linear problems.

35.1) Examples of Nonlinear PDEs

Ex 35.1 (Minimal Surface Equation)

Soap bubbles minimize surface area among possible surfaces w/ fixed boundary



$u : \mathbb{R}^2 \to \mathbb{R}$, the surface $\Sigma = (x, y, u(x,y))$ is minimal if u obeys

$$\nabla \cdot \left(\frac{\nabla u}{\sqrt{1+|\nabla u|^2}} \right) = 0 \quad \text{i.e.} \quad \left(1+\left(\frac{\partial u}{\partial y}\right)^2\right)\frac{\partial^2 u}{\partial x^2} + \left(1+\left(\frac{\partial u}{\partial x}\right)^2\right)\frac{\partial^2 u}{\partial y^2} - 2\frac{\partial u}{\partial x}\frac{\partial u}{\partial y}\frac{\partial^2 u}{\partial x \partial y} = 0.$$

This equation is called quasi-linear, and has the form

$$\sum_{ij} Q_{ij}(u, \nabla u)\, \partial_i \partial_j u + F(u, \nabla u) = 0.$$

↖ can view as linear eq where coefficients themselves depend on u.

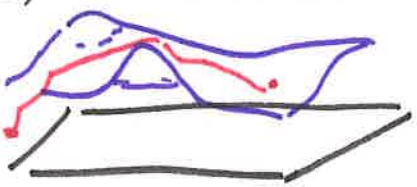
Ex 35.2 (Nonlinear Wave and Schrodinger Equations)

$$\partial_t^2 u - \Delta u + u(u^2 - 1) = 0$$

$$i\partial_t \psi + \Delta \psi + |\psi|^2 \psi = 0$$

Used to model phase transitions, classical field theories, more complex condensed matter systems.

Ex 35.3 (nonlinear ODE) On a surface



= (graph of f)

The shortest path between two points is called a geodesic. If $\gamma(t)$ is a geodesic

$$\frac{\partial^2 \gamma}{\partial t^2} + \Gamma\left(\nabla f(\gamma(t)), \partial_i \partial_j f(\gamma(t))\right) \frac{\partial \gamma}{\partial t} A \frac{\partial \gamma}{\partial t} = 0.$$

Nonlinear ODE.

- Planes fly along geodesics
- Objects in einsteins GR move along geodesics in spacetime.

35.ii) Minimization Principles

Each of the above equations can be expressed as the minimum/maximum of certain quantities

- Minimal Surfaces minimize area (scalar quantities)
- Geodesics minimize length (w/ constraints) ↓
- u, ψ in NL wave/Schrödinger minimize $\mathcal{L}$(total energy).)

Def 35.4 : The Calculus of Variations studies non-linear PDEs that can be characterized as the max/min of scalar functionals.

Each quantity is a nonlinear functional, eg.

$$\text{Area} : \{\text{surfaces}\} \longrightarrow \mathbb{R}$$
$$\text{graph } u \longrightarrow \text{Area}.$$

Key Idea 35.5 : Studying min/max requires generalizing maximization techniques from calculus to 'infinite-dimensions.

35.iii) Calculus in a way that generalizes

Recall the directional derivative in the direction v is

$$\lim_{\varepsilon \to 0} \frac{f(\vec{x} + \varepsilon v) - f(\vec{x})}{\varepsilon} = v \cdot \nabla f = \sum v_i (\partial_i f).$$

Lemma 35.6: The directional derivative at x in v direction is

$$\frac{d}{d\varepsilon}\Big|_{\varepsilon=0} f(x + \varepsilon v).$$

Proof:

$$\frac{d}{d\varepsilon}\Big|_{\varepsilon=0} f(x + \varepsilon v_1, y + \varepsilon v_2, z + \varepsilon v_3)$$

✓ chain rule

$$= \frac{\partial f}{\partial x} \cdot v_1 + \frac{\partial f}{\partial y} \cdot v_2 + \frac{\partial f}{\partial z} v_3$$

$$= v \cdot \nabla f = \sum v_i (\partial_i f).$$

Def 35.7: A function $f : \Omega \subseteq \mathbb{R}^d \to \mathbb{R}$ is differentiable at x if

$$\frac{d}{d\varepsilon}\Big|_{\varepsilon=0} f(x + \varepsilon v)$$

exists for every direction v. The linearization or total derivative or (Jacobian) is the map

$$Df \quad \Omega \to \left(\begin{array}{l} \text{Mat} \\ \mathbb{R}^d_v \to \mathbb{R} \end{array} \right).$$

$$\left[v \to \frac{d}{d\varepsilon}\Big|_{\varepsilon=0} f(x+\varepsilon v) \right] = (D_x f)(v)$$

Ex 35.8: $f(x,y) = \left(x^2 y + \sin(x) + x^3 y - y^2 \right)$.

at each (x,y) the linearization is the assignment

$$v \to v \cdot \nabla f = \left[2xy + \cos(x) + 3x^2 y \;,\; x^2 + x^3 - 2y \right] \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$$

so Linearization/Jacobian is the linear map w/ matrix given by

$$\nabla f = \left[\quad \right].$$

More generally $D_x f = \left[\frac{\partial f}{\partial x} \cdots \right] = \begin{bmatrix} \nabla f_1 \\ \nabla f_2 \\ \vdots \end{bmatrix}$ $\qquad f = (f_1 \ldots f_d)$.

Def 35.9: Define $x \in \Omega$ as a critical point of f if $(f: \Omega \to \mathbb{R})$

$$\Leftrightarrow D_x f = 0 \quad \text{(as a matrix)}$$
$$\nabla f = 0$$

i.e. if

$$\left.\frac{d}{d\varepsilon}\right|_{\varepsilon=0} f(x+\varepsilon v) = 0 \qquad \forall v \in \mathbb{R}^d$$

Ex 35.10: $F(x,y) = x^4 - 2x^2 + y^2$

Then $D_x f = 0 \iff (x,y)$ satisfies an algebraic equation(s).

$$0 = (4x^3 - 4x,\ 2y)$$

$$\begin{aligned} 4x^3 - 4x &= 0 \\ 2y &= 0 \end{aligned} \iff \begin{aligned} x &= 0 \text{ or } 1 \\ y &= 0 \end{aligned}$$

35.iv) Functionals

Def 35.11: A non-linear functional

$$\Sigma : (X, \|\ \|_x) \longrightarrow \mathbb{R}$$

on a Banach space is said to be differentiable at $u \in X$ if

$$\left.\frac{d}{d\varepsilon}\right|_{\varepsilon=0} \Sigma(x+\varepsilon\varphi)$$

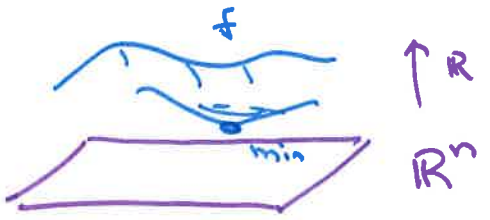
exists for every $\varphi \in X$. The assignment

$$D_x\Sigma : X \longrightarrow \mathbb{R}$$
$$\varphi \longmapsto \left.\frac{d}{d\varepsilon}\right|_{\varepsilon=0} \Sigma(x+\varepsilon\varphi)$$
$$= D_x\Sigma(\varphi)$$

is called the functional or Frechet derivative, or just the derivative by unpretentious people.

Lecture 36 | Variational PDEs.

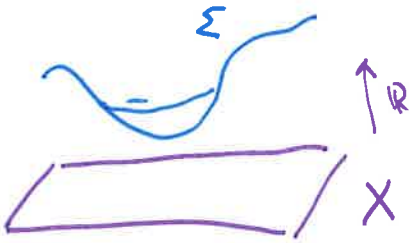
Recall we wish to minimize a functional $\mathcal{E}: (X, \|\ \|_X) \to \mathbb{R}$



In finite dimensions

$f: \mathbb{R}^n \to \mathbb{R}$ minimized at x $\iff$ $\nabla f = 0$ $\iff$ $\partial_v f = 0\ \forall v$.

$\iff$ algebraic eq.



Infinite dimensions

$\mathcal{E}: X \to \mathbb{R}$ minimized at u $\iff$ $D_u\mathcal{E} = 0$ $\iff$ $\frac{d}{d\varepsilon}\big|_{\varepsilon=0} \mathcal{E}(u+\varepsilon\varphi) = 0\ \forall \varphi$

Idea 36.1: In infinite dimensions, the algebraic eq. $\nabla f = 0$ becomes a PDE or ODE.

Function Spaces 36.2: $\Omega \subseteq \mathbb{R}^n$

$$C_0^2 = \{u \in C^2(\Omega:\mathbb{R}) \mid u|_{\partial\Omega} = 0\}$$ satisfy Dirichlet BC.

C^0: same.

Def 36.3: The Dirichlet Energy functional $\mathcal{E}: C_0^2(\Omega:\mathbb{R}) \to \mathbb{R}$. is

$$\mathcal{E}(u) = \frac{1}{2}\int_\Omega |\nabla u|^2\, dV.$$

Thm 36.4: $u \in C_0^2(\Omega:\mathbb{R})$ minimizes $\mathcal{E}$ $\iff$ $-\Delta u = 0$.

Proof: The functional is minimized if

Step 1 (calculate derivative)

$$0 = \frac{d}{d\varepsilon}\Big|_{\varepsilon=0} \mathcal{E}(u+\varepsilon\varphi)$$

$$= \frac{d}{d\varepsilon}\Big|_{\varepsilon=0} \frac{1}{2}\int_\Omega |\nabla(u+\varepsilon\varphi)|^2\, dV.$$

$$= \frac{d}{d\varepsilon}\Big|_{\varepsilon=0} \frac{1}{2}\int_\Omega \nabla(u+\varepsilon\varphi)\nabla(u+\varepsilon\varphi)\,dV$$

$$= \frac{d}{d\varepsilon}\Big|_{\varepsilon=0} \int_\Omega \tfrac{1}{2}|\nabla u|^2\,dV + \frac{d}{d\varepsilon}\Big|_{\varepsilon=0} \frac{1}{2}\int_\Omega 2\nabla u\cdot\varepsilon\nabla\varphi + \frac{d}{d\varepsilon}\Big|_{\varepsilon=0} \frac{\varepsilon^2}{2}\int |\nabla\varphi|^2\,dV.$$

$= 0$ as $\varepsilon = 0$.

$= \int_\Omega \nabla u\cdot\nabla\varphi\,dV$

$= \varepsilon \int_\Omega |\nabla\varphi|\,dV\Big|_{\varepsilon=0} = 0$

$$= \int_\Omega \nabla u\cdot\nabla\varphi\,dV.$$

Step 2
Isolate φ + invoke variations

$$= \int_\Omega -\Delta u\cdot\varphi\,dV \pm \int_{\partial\Omega} \partial_\nu u\cdot\varphi \pm \int_{\partial\Omega} u\cdot\partial_\nu\varphi$$

$= 0 + 0$. both $u, \varphi \in C^2_0$ so $\Big|_{\partial\Omega} = 0$

Minimum. $$0 = \int_\Omega -\Delta u\cdot\varphi\,dV \quad \forall \varphi \in C^2_0$$

In particular $$0 = \int_\Omega -\Delta u\cdot\varphi\,dV \quad \forall \varphi \in C^\infty_{cpct}$$

$\Rightarrow -\Delta u = 0$ by Fundamental lemma of calc. of variations.

Ex 36.5: Find the variational equation satisfied by the ~~min~~ critical points of

$$E_f(u) = \frac{1}{2}\int_\Omega |\nabla u|^2 - f\cdot u\,dV$$

$$0 = \frac{d}{d\varepsilon}\Big|_{\varepsilon=0} \frac{1}{2}\int_\Omega |\nabla(u+\varepsilon\varphi)|^2 - f\cdot(u+\varepsilon\varphi)\,dV.$$

$$= \frac{1}{2}\int_\Omega 2\nabla(u+\varepsilon\varphi)\cdot\nabla\varphi\Big|_{\varepsilon=0} - f\varphi\Big|_{\varepsilon=0}\,dV$$

$$= \int_\Omega \nabla u\cdot\nabla\varphi - f\varphi\,dV = \int_\Omega (-\Delta u - f)\varphi\,dV \quad \forall\varphi.$$

$$\Rightarrow -\Delta u - f = 0$$

$$\boxed{-\Delta u = f}$$

Ex 36.6 : Find the variational equation of

$$\mathcal{E}(u) = \int_\Omega \tfrac{1}{2}|\nabla u|^2 + \tfrac{1}{4}(u^4 - u^2)\,dV$$

$$0 = \frac{d}{d\varepsilon}\Big|_{\varepsilon=0} \int_\Omega \tfrac{1}{2}(\nabla(u+\varepsilon\varphi))^2 + \tfrac{1}{4}(u+\varepsilon\varphi)^4 - \tfrac{1}{4}(u+\varepsilon\varphi)^2\,dV$$

$$= \int_\Omega \tfrac{1}{2}\cdot 2(\nabla u + \varepsilon\nabla\varphi)\cdot\nabla\varphi\Big|_{\varepsilon=0} + (u+\varepsilon\varphi)^3\varphi - \tfrac{1}{2}(u+\varepsilon\varphi)\Big|_{\varepsilon=0} dV$$

$$= \int_\Omega \nabla u\cdot\nabla\varphi + u^3\varphi - \tfrac{1}{2}u\varphi\,dV$$

$\forall\varphi$

$$0 = \int_\Omega -\Delta u\cdot\varphi + u^3\varphi - \tfrac{1}{2}u\varphi\,dV$$

Fund. Lem (

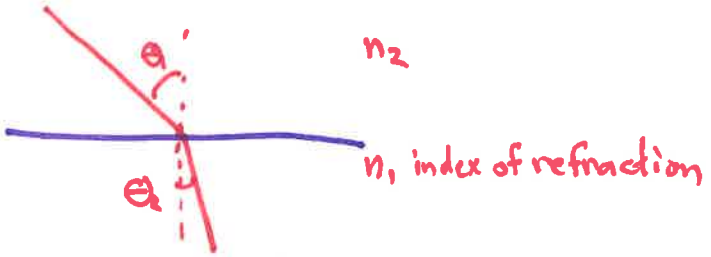
$$= \int_\Omega \left(-\Delta u + u^3 - \tfrac{1}{2}u\right)\varphi\,dV$$

$$\Leftrightarrow -\Delta u + u^3 \cdot \tfrac{1}{2}u = 0.$$

Lecture 37 Euler-Lagrange Equations in Physics

Another famous minimization principle is the first one

Principle 37.1 (Fermat's least time)
Light travels along the pathway that minimizes time of travel



Some calculus minimization derives Snell's law $n_1 \sin\theta_1 = n_2 \sin\theta_2$.

Newtonian Mechanics 37.2: Particle traces path $x(t): [0,T] \longrightarrow \mathbb{R}^n$, governed by Newtons law

$$ma = \text{Force} = \begin{cases} -\nabla V \text{ for } V: \mathbb{R}^n \to \mathbb{R} \text{ potential energy} \end{cases}.$$

$$mx''(t) = -\nabla V(x(t)).$$

Question 37.3: Can Newton's law be recovered as the minimum of some functional?

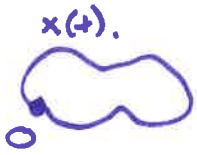
Definition 37.4: The Lagrangian $\mathcal{L}: \mathbb{R}^n \times \mathbb{R}^n \times \mathbb{R} \to \mathbb{R}$.

$$\mathcal{L}(x(t), x'(t), t) = KE - PE$$
$$= \frac{1}{2} m x'(t)^2 - V(x(t))$$

is the difference of Kinetic and potential energy

And the action functional

$$\mathcal{A}[x(t)] = \int_0^T \mathcal{L}(x(t), x'(t), t)\, dt$$



Function Spaces 37.5

$$C^2_0([0,T]; \mathbb{R}^n) = \{ x: [0,T] \to \mathbb{R}^n \mid x \text{ is } C^2, \ \begin{matrix} x(0)=0 \\ x(T)=0 \end{matrix} \}$$

- path is n C^2-functions
- We consider only loops, to simplify boundary conditions.

Thm 37.6: $x(t) \in C_0^2([0,T];\mathbb{R}^n)$ obeys Newton's law if and only if it is a critical point of $A[x(t)]$.

Proof:

$$0 = \frac{d}{d\varepsilon}\Big|_{\varepsilon=0} A[x(t)+\varepsilon\varphi(t)]$$

$$= \frac{d}{d\varepsilon}\Big|_{\varepsilon=0} \int_0^T \mathcal{L}(x(t)+\varepsilon\varphi, x'(t)+\varepsilon\varphi', t)\,dt \qquad L = L(\overset{x}{\cdot}, \overset{v}{\cdot}, t)$$

$$\nabla_x L = \frac{\partial}{\partial x} L$$

$$= \int_0^T \frac{d}{d\varepsilon}\Big|_{\varepsilon=0} \mathcal{L}(x+\varepsilon\varphi, x'+\varepsilon\varphi', t)\,dt \qquad \nabla_v L = \frac{\partial}{\partial v} L$$

$$= \int_0^T \nabla_x \mathcal{L}(x+\varepsilon\varphi, x'+\varepsilon\varphi', t)\cdot\varphi\Big|_{\varepsilon=0} + (\nabla_v L)(x+\varepsilon\varphi, x'+\varepsilon\varphi', t)\Big|_{\varepsilon=0}\cdot\varphi'(t)$$

Integrate by parts, use B.C.

$$= \int_0^T \left[(\nabla_x \mathcal{L})(x,x',t) - \frac{d}{dt}\left[\nabla_v L(x(t),x'(t),t)\right]\right]\varphi(t)\,dt.$$

$$\Rightarrow \quad (\nabla_x \mathcal{L})(x,x',t) - \frac{d}{dt}\left[\nabla_v \mathcal{L}(x,x,t)\right] = 0.$$

$\mathcal{L}(x,v,t) = \frac{1}{2}mv^2 - V(x)$

$\nabla_x \mathcal{L} = -\nabla V.$

$\nabla_v \mathcal{L} = mv.$

$$\Rightarrow 0 = -\nabla V(x,x',t) - \frac{d}{dt}[mv(t)]$$

$$0 = -\nabla V(x(t),x'(t),t) - m x''(t) \quad \checkmark.$$

Def 37.7: If $\mathcal{E}: X \to \mathbb{R}$ has the particular form

$$\mathcal{E} = \int_{\mathbb{R}\times[0,T]} F(u,\nabla u,t)\,dV\,dt$$

for some function $F: \mathbb{R}\times\mathbb{R}^n\times\mathbb{R}\to\mathbb{R}$, then it is called an action or Euler-Lagrange functional

In this case, the variational equation is the Euler-Lagrange Equation.

$$\partial_1 F(u, \nabla u, t) - \frac{d}{dt}\left[\nabla_2 F(u, \nabla u, t)\right] = 0.$$

$\nabla_2 F(u, \nabla u, t)$: 2nd order in u by chain rule.

37.2 : Lagrangian For the wave equation



$KE = \frac{1}{2}[\rho]\,(\partial_t u)^2$ (ρ: density)

$PE = \frac{1}{2}\tau\,[\partial_x u]^2$ (τ: tension)

$$\mathcal{A}[u(t,x)] = \int_0^T \int_a^b \frac{1}{2}\rho\, x'(t)^2 - \frac{1}{2}\tau x(t)^2 \, dx\, dt.$$

Step 1 :

$$0 = \frac{d}{d\varepsilon}\Big|_{\varepsilon=0} \mathcal{A}[u(t,x)]$$

$$= \frac{d}{d\varepsilon}\Big|_{\varepsilon=0} \int_0^T \int_a^b \frac{1}{2}\rho\left[\partial_t u + \varepsilon \partial_t \varphi\right]^2 - \frac{1}{2}\tau\left[\partial_x u + \varepsilon \partial_x \varphi\right]^2 dx\,dt$$

$$= \int_0^T \int_a^b \rho\left[\partial_t u\right]\cdot \partial_t \varphi - \tau\left[\partial_x u\right]\cdot \partial_x \varphi \, dx\,dt$$

Step 2

Int by parts both x, t w/ BC.

$$= \int_0^T \int_a^b \left\{\rho\left[-\partial_t^2 u\right] + \tau\left[\partial_x^2 u\right]\right\}\varphi \, dx\,dt$$

$$0 = \int_0^T \int_a^b \left[\rho\,\partial_t^2 u + \tau \Delta u\right]\varphi \, dx\,dt \quad \Rightarrow \quad -\rho\,\partial_t^2 u + \tau \Delta u = 0$$

wave eq.

Step 3 : Fundamental Lemma.

Rem 37.8 : Can also generalize Euler-Lagrange to functions of multiple variables to get same result.

Ex 37.8: The Einstein-Hilbert action for

$$g : \mathbb{R}^4 \longrightarrow \mathbb{R}^{10} = \{\text{Symmetric } 4\times 4 \text{ matrices}\}.$$

$$S[g] = \frac{c^4}{16\pi G} \int_{\text{space}\times\text{time}} \text{Tr}_g[\text{Ricci}(g)] \sqrt{\det g}\, dx\,dy\,dz\,dt.$$

yields Einstein's field equations for General Relativity.

And the Yang-Mills action for $A = (A_1, A_2, A_3, A_4)$, $A : \mathbb{R}^4 \longrightarrow \underbrace{SU(n) \times \cdots}_{\text{4 copies}}$ of skew self adjoint $n\times n$ $\mathbb{C}$-valued matrices.

$$Y[A] = -\frac{1}{4} \int_{\text{space}\times\text{time}} \sum_{i,j=1,\ldots 4} \text{Tr}\left[\partial_i A_j - \partial_j A_i - A_i A_j + A_j A_i\right]^2 dx\,dy\,dz\,dt$$

yields the Yang-Mills equations for gluons in the strong force/quantum chromodynamics.